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# DISTRIBUTION-INDEPENDENT TOUR-LENGTH ESTIMATION FOR THE TRAVELING SALESMAN PROBLEM ACROSS DIMENSIONS

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## ABSTRACT

We present Geometrically Assisted Regression Tree (GART) 2.0, a distribution-independent estimator of Traveling Salesman Problem (TSP) tour length whose feature set is defined for arbitrary dimension. Existing estimators either assume a known sampling density or depend on planar geometry that becomes unstable as dimension grows. GART 2.0 is a LightGBM regressor that predicts the instance-specific ratio  $\alpha = L_{\text{TSP}}/L_{\text{MST}}$  from minimum-spanning-tree topology and coordinate-range descriptors and returns  $\hat{L}_{\text{TSP}} = \hat{\alpha} L_{\text{MST}}$ ; no distribution label, no density, and no convex hull enters the computation. We train it on synthetic instances and it extrapolates to realistic ones, transferring to TSPLIB95 and, through a multidimensional-scaling embedding, to non-Euclidean instances given only as a distance matrix. Accuracy improves monotonically in both arguments the feature set was built to carry, growing instance size and growing dimension, out to a hundred dimensions and including one withheld from training entirely; on every stratum it is the most accurate of a seven-baseline roster. An estimator that sharpens exactly where an exact solve becomes infeasible is the one worth putting inside a routing or partitioning pipeline. We then score a certified Held–Karp 1-tree lower bound, which consumes no training corpus and proves its own output, and use it to fix the boundary of the method rather than to argue around it. On the corpus aggregate the bound is the better instrument, but that aggregate is mostly small instances, where the relaxation is nearly exact and nearly free. Read at fixed instance size the ordering turns over: above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the plane it is strictly better than the bound on both axes at every ascent budget we priced. The reason is structural rather than empirical: the bound pays a quadratic price in instance size for a fixed accuracy in every dimension, and the estimator does not. Below that size, and off the training grid in dimension, the bound is preferable; both halves of that rule are needed to use either method well. The same bound doubles as an audit instrument, refuting stored labels in our own corpus and exposing a distance-quantisation defect that we repair and verify.

**Keywords** Traveling Salesman Problem · Tour-length estimation · Distribution-independent · Multidimensional network

## 1 Introduction

The Traveling Salesman Problem (TSP) is a fundamental problem in combinatorial optimization, widely used for logistics operations planning, network design, and resource allocation [Chien, 1992, Applegate et al., 2006]. The TSP is NP-hard [Karp, 1972, Papadimitriou, 1977], so exact solution times grow prohibitive on large instances and rapid tour-cost estimates become essential for real-time decision-making in applications where a full solve is impractical. Tour-cost estimates also serve as inputs to other optimization problems where the TSP appears as a subproblem.

## 1.1 Background and Motivation

Tour-length estimation is a key input to strategic logistics decisions where solving the TSP exactly is unnecessary or infeasible. Last-mile delivery accounts for approximately 53% of total shipping costs [Finmile, 2025], and the UPS ORION system is reported to save roughly 100 million miles and \$300–400 million annually through route optimization [Holland et al., 2017]. In such systems recalculation is frequent, because orders arrive continuously and the number of customers on a route is large, so the cost of a full solve is paid many times over. A tour-length estimator replaces that solve wherever the decision depends on the relative cost of candidate routes rather than on their exact values.

The same estimation problem recurs wherever a TSP appears as a subproblem, and it rarely arrives in the plane. Radiation hybrid mapping in bioinformatics orders genetic markers by a TSP over an abstract similarity metric [Agarwala et al., 2000], and mobile-sink routing in wireless sensor networks contains a TSP through the data-collection locations [Yuan and Peng, 2010]. Large-scale routing methods partition an instance into smaller sub-TSPs and compare candidate partitions [Mariescu-Istodor and Fränti, 2021]; Pan et al. [2023] instead learn a hierarchical policy whose upper level repeatedly selects a subset of nodes and whose lower level extends a partial route through them. A planner choosing among node sets, route splits, or cluster assignments needs the cheaper option to be predicted as cheaper, so an estimator serving these tasks must be accurate and rank-preserving at once. We test both properties directly in Section 4.

Most classical tour-length estimators are nonetheless built for low-dimensional Euclidean inputs and depend on geometric quantities—most often convex-hull area—that become unstable or intractable as dimension grows; when dimension is variable, convex-hull output complexity alone can grow exponentially [Chazelle, 1993]. An estimator that serves the settings above must therefore be defined for arbitrary  $d$  from a feature set that never builds a hull. That requirement also opens the non-Euclidean case: instances given as a distance matrix admit an approximate Euclidean representation once enough dimensions are allowed [Bourgain, 1985], and an estimator that stays accurate at those dimensions can consume the embedding. Dimension generality is the design goal of this paper, and Section 4 shows the estimator meeting it: accuracy improves monotonically as both the instance size and the ambient dimension grow. Section 5 then fixes the boundary of the method by scoring it against a certified lower bound, and identifies the regime in which learned estimation is the right instrument.

## 1.2 Limitations of the Current Estimators

While the need for a general TSP estimator is clear, a persistent gap remains in the literature regarding high-dimensional and non-Euclidean instances. The BHH theorem [Beardwood et al., 1959] is stated for arbitrary dimension, but it is asymptotic in  $n$  and its calibrating constants and density integral are not recoverable from an arbitrary finite point set; published constants exist for  $d = 2$  [Johnson et al., 1996] and  $d = 3$  [Percus and Martin, 1996], and above that only a large- $d$  approximation [Bertsimas and van Ryzin, 1990]. The estimators of Chien [1992] are planar and depend on particular depot and covering-region definitions. A parallel line of work is tuned to the 2D Euclidean case: Daganzo [1984b] gives a strip-based construction, Kwon et al. [1995] fit regression and neural models on rectangular-region predictors, and Çavdar and Sokol [2015] propose a distribution-free estimator built from per-axis spread statistics about the covering rectangle. Space-filling heuristics such as the planar construction of Bartholdi and Platzman [1982] instead build a tour outright in  $O(n \log n)$ . Every member of this family consumes either a convex-hull area or an axis-aligned covering box, and both degrade with dimension, so the embedding route out of the non-Euclidean case returns them to the regime in which their geometric inputs are unstable. Section 4 scores BHH and Çavdar and Sokol [2015] numerically, both on the full benchmarks and on the subset where their published assumptions hold (Table 1). We survey Chien [1992], Daganzo [1984b] and Kwon et al. [1995] without scoring them, because we could not obtain their primary sources and decline to publish numbers produced by constants we have not read in the original.

A separate classical line does not estimate the tour at all: it bounds it, and it is the strongest comparator available to any tour-length estimator. Held and Karp [1970, 1971] relax the Hamiltonian-cycle constraint to a *1-tree*, a

spanning tree on all but one node together with the two cheapest edges at that node, and introduce node penalties  $\pi$  under which every tour’s cost shifts by the same constant  $2 \sum_i \pi_i$ . Maximising the concave piecewise-linear dual  $w(\pi) = L_{\text{tree}}(\pi) - 2 \sum_i \pi_i$  yields the Held–Karp bound, and because  $w(\pi) \leq L_{\text{TSP}}$  holds at *every*  $\pi$ , an ascent stopped anywhere returns a certificate rather than a prediction. Three properties follow. It consumes no training corpus and no distributional assumption, so it has no split to contaminate and no domain to leave. It never invokes the triangle inequality, so it is defined wherever a symmetric distance matrix is, including the non-Euclidean instances Section 6 reaches only through an embedding. And its cost is an explicit knob,  $\Theta(kn^2)$  for  $k$  ascent iterations, so accuracy is bought continuously rather than fixed by a closed form. The literature divides on the ascent: Held and Karp [1971] give the subgradient  $\deg_i - 2$ , Volgenant and Jonker [1982] supply a step schedule that consults no upper bound and Helsgaun [2000] adds direction smoothing to it, while Polyak [1969] instead sizes each step from the gap to a known feasible solution and converges faster wherever a good tour is cheap to build. We score both schedules as first-class rows on every benchmark in this paper (Table 1). Because the bound is certified, cheap in the plane only at small  $n$ , and quadratic in  $n$  everywhere, it is the comparator that fixes where a learned estimator is worth running; Section 5 draws that line.

Recent neural approaches address these limitations. Varol et al. [2024] report sub-1% deviation on standard distributions by combining an MLP with domain-knowledge features, but their pipeline requires solution-level features, which are computationally expensive and challenging to generalize across all instance sizes. Kou et al. [2022] use the standard deviation of random feasible tour lengths as a single-feature predictor, and evaluate it on both Euclidean and non-Euclidean instances. The predictor is distribution-agnostic but requires repeated tour sampling, which raises its per-instance cost. Kou et al. [2024] extend this randomization principle beyond the TSP, predicting optimal values for the VRP, MST, and matching problems from the standard deviation of random feasible solutions; like the original predictor it relies on repeated per-instance sampling, whose cost we exclude on the same grounds (Section 4.1). The most direct methodological neighbour is Kou et al. [2023], which pairs a Sammon-map embedding with a distribution-independent estimator on non-Euclidean TSPLIB95 instances; our pipeline differs by using classical metric MDS for the embedding while computing MST-topology features on the original distance matrix rather than the embedded coordinates.

### 1.3 Contribution

We present the Geometrically Assisted Regression Tree (GART 2.0), a LightGBM regressor [Ke et al., 2017] that predicts a per-instance scaling factor  $\hat{\alpha}$  rather than applying a fixed constant. It is distribution-independent: no distribution label and no parametric density is supplied at inference, although accuracy varies by distribution type. Distribution-independence is not by itself new, since Çavdar and Sokol [2015] and Kou et al. [2022, 2023] are also distribution-independent, but those estimators remain tied to 2D coordinate geometry or to per-instance sampling. The contribution of GART 2.0 is to be distribution-independent *and* dimension-scalable at once: all 31 features are defined for arbitrary  $d$  through MST topology, centroid dispersion and coordinate ranges, no feature requires a convex hull, and the same feature set applies unchanged as both dimension and instance size grow. Section 3.1 gives the feature set and Section 3.5 validates it against SHAP attributions on the held-out split.

We benchmark GART 2.0 against seven baselines: two classical estimators, three constant multiples of  $L_{\text{MST}}$  of which one is calibrated on the training split, the GART 1.0 predecessor [Feldman, 2025], and a space-filling-curve tour construction. Both classical estimators are scored on the full benchmarks and again on the i.i.d. uniform subset where their published assumptions hold, BHH there receiving the exact sampling region its theorem names. GART 2.0 has the lowest aggregate MAPE and standard deviation of percent error (SDPE) on all four strata, ahead of a calibrated per-cell lookup table on the multidimensional set, of Çavdar–Sokol on planar i.i.d. uniform instances, and of the asymptotic MST ratio on TSPLIB EUC\_2D. The behaviour that matters for deployment is that these gains are not concentrated at one operating point. Dispersion falls monotonically as instances grow, and again as the ambient dimension grows, holding at a dimension withheld from training entirely. An estimator whose accuracy improves in exactly the two directions that make an exact solve infeasible is the one worth putting into a routing or partitioning pipeline.

Section 5 then fixes the boundary of the method against a comparator no estimator can argue with. The Held–Karp 1-tree bound consumes no training corpus, proves its own output, and needs only a symmetric distance matrix, so it is defined on every instance in this paper and on some that GART 2.0 declines. On the corpus aggregate it is the better instrument, more accurate and cheaper at once. That aggregate is also mostly instances small enough for the relaxation to be very nearly exact and very nearly free, so it describes small instances more than it describes either method. Stratified by size as well as by dimension (Table 5), the ordering turns over above a few hundred nodes: the estimator is on the cost/accuracy front at every dimension it was trained on, and in the plane and just above it, in the two larger size bands, it is strictly better on both axes at every budget we priced. What the bound cannot do is scale, because it pays a quadratic price in instance size for a fixed accuracy in every dimension while the estimator does not; the measured planar cost exponents put the two families in different classes, and GART 2.0 predicts an instance many times larger than anything it was trained on in a fraction of a second. Below that size, and off the training grid in dimension, we recommend the bound instead. Stating both halves is what makes the recommendation usable.

Separately from the estimator, we contribute the audit that a certified bound makes possible. Because a certified lower bound  $B$  above a stored label  $L$  refutes that label with no tour entering the test, a corrupt tour cannot hide a corrupt cost behind it. Applied to our own corpus the test refuted released labels and exposed a distance-quantisation defect in a seventh of the multidimensional instances that four earlier tour-based audits had not (Section 5.7). Every table in this paper uses the repaired labels, and no estimator changes rank on any stratum as a result.

## 1.4 Paper Outline

The remainder of this paper is organised as follows. Section 2 establishes the multiplicative MST–TSP relationship and shows that  $\alpha$  is instance-dependent and concentrates with dimension, so that a fixed constant is biased. Section 3 describes the feature set, the synthetic corpus, the reference-tour provenance audit, model fitting, and inference cost. Section 4 declares the baseline roster and the three benchmark datasets, reports per-dataset results, re-evaluates each classical estimator on the subset where its published assumptions hold, withholds whole dimensions and size ranges from training to separate generalization from interpolation, diagnoses the degenerate-geometry failure, and closes on rank agreement and timing. Section 5 scores the certified Held–Karp 1-tree bound against GART 2.0 on cost and accuracy, reports the label audit that bound makes possible, and states the revised positioning. Section 6 extends GART 2.0 to non-Euclidean TSPLIB95 instances through an MDS embedding. Section 7 states what the paper establishes and what bounds it.

## 2 Theoretical Foundations

This paper predicts the ratio  $\alpha = L_{\text{TSP}}/L_{\text{MST}}$ , and this section establishes that the ratio is a property of the instance rather than a constant. Let  $\mathcal{G} = (V, E)$  be a complete undirected graph on the node set  $V = \{1, \dots, n\}$  with non-negative edge costs  $c_{ij}$ , typically the Euclidean distance between points in  $\mathbb{R}^d$ . Throughout,  $n$  is the number of nodes in an instance and  $d$  is the dimension of the space its coordinates occupy. The TSP asks for a minimum-cost Hamiltonian cycle. In the standard Dantzig–Fulkerson–Johnson formulation [Dantzig et al., 1954, Applegate et al., 2006] a binary  $x_{ij}$  indicates whether edge  $(i, j)$  is in the tour and  $\sum_{i < j} c_{ij} x_{ij}$  is minimised subject to a degree-two constraint at every vertex and a subtour-elimination constraint on every proper subset of  $V$ ; the problem is NP-hard [Karp, 1972, Papadimitriou, 1977]. What the formulation exposes, and what this paper exploits, is that any feasible tour is a 2-regular spanning subgraph, of which the minimum spanning tree (MST) is the cheapest connected relaxation.

The MST is solvable in polynomial time. Prim’s algorithm constructs one in  $O(n^2)$  on a complete graph with an adjacency-matrix representation [Prim, 1957]; Kruskal’s must sort  $\Theta(n^2)$  edges and so costs  $O(n^2 \log n)$  there [Kruskal, 1956]. For nodes in  $\mathbb{R}^d$  the pairwise distance computation scales as  $O(n^2 d)$ , linear in  $d$ , which is what makes the MST a computationally stable scaffold for high-dimensional inference. It also brackets the tour. Removing the heaviest edge from any tour leaves a spanning tree, so  $L_{\text{MST}} \leq L_{\text{TSP}}$ ; the double-tree heuristic [Rosenkrantz et al., 1977] doubles the

MST edges into an Eulerian multigraph and shortcuts repeated nodes to yield a feasible tour of length at most  $2 L_{\text{MST}}$  on any instance obeying the triangle inequality. Together these give

$$L_{\text{MST}} \leq L_{\text{TSP}} \leq 2 \cdot L_{\text{MST}}. \quad (1)$$

The upper constant cannot be reduced in general: for collinear point sets the optimal tour must traverse the line to its far end and return, so  $L_{\text{TSP}} = 2 L_{\text{MST}}$  exactly.

A separate and frequently conflated result concerns *algorithmic* approximation rather than this intrinsic ratio. The Christofides–Serdyukov algorithm [Christofides, 1976, Serdyukov, 1978] augments the MST with a minimum-weight perfect matching (MWPM) on the odd-degree nodes and returns a tour within a factor  $\frac{3}{2}$  of the *optimum*; Karlin et al. [2021] improve on that factor with a different construction, a randomized  $\frac{3}{2} - \epsilon$  approximation. Both are guarantees on an algorithm’s output relative to the optimal tour, and neither tightens the intrinsic ratio  $L_{\text{TSP}}/L_{\text{MST}}$ , which can still equal 2. The MWPM moreover costs  $O(n^3)$  [Edmonds, 1965, Gabow, 1990, Cook and Rohe, 1999], far above the  $O(n^2)$  MST construction, so the Christofides refinement is impractical as a fast estimation subroutine.

For symmetric metric instances we therefore treat  $\alpha \in [1, 2]$  as an intrinsic, instance-dependent quantity to be predicted. For uniformly distributed points the TSP constant itself varies with dimension [Percus and Martin, 1996], and no published MST constant covers every  $d$ , so we establish the behaviour of the ratio empirically: the standard deviation of  $\alpha$  over our training split falls monotonically from 0.1837 at  $d = 2$  to 0.0762 at  $d = 50$ . The relative difference in objective value, and not the computational complexity of either problem, decreases as  $d$  grows, so a static estimator calibrated in the plane is biased in higher dimensions. GART 2.0 predicts  $\alpha$  from the observed instance features instead.

### 3 Methodology: The Geometrically Assisted Regression Tree

GART 2.0 extends the estimator of Feldman [2025]. It predicts  $\alpha = L_{\text{TSP}}/L_{\text{MST}}$  with a LightGBM ensemble and returns  $\hat{L}_{\text{TSP}} = \hat{\alpha} L_{\text{MST}}$ . The following subsections describe its 31 input features, the synthetic corpus and the provenance audit we ran over it, model fitting and inference cost, and the feature-importance analysis. We constrain the regression target to  $\alpha \in [1, 2]$ , the metric-TSP interval implied by Eq. (1). Two raw corpus values fall outside that interval, one training row at 0.975 and one validation row at 2.060, and target preparation clips them.

#### 3.1 Feature Selection

We extract 31 features from each instance: eleven geometric and centroid descriptors of spatial spread and density, ten MST edge-weight statistics, nine MST topology and clustering proxies, and one constructive ratio. The count reflects the four aspects of instance structure the target responds to – spatial spread, edge-weight distribution, branching topology, and the gap between a constructed tour and the MST – the first three requiring several statistics apiece to cover the behaviours observed across distributions, the fourth a single ratio. MST edge weights and the degree sequence are invariant to rotation, translation and coordinate-system choice; the geometric descriptors are lightweight quantities such as bounding hypervolume and centroid distance that generalize to arbitrary  $d$  without a convex hull, supplying the scale information that the volume term of the BHH formula carries [Beardwood et al., 1959] and adapting their centroid statistics from Çavdar and Sokol [2015]. The dimension  $d$  enters as an explicit input rather than implicitly through coordinate geometry. Before any geometric descriptor is evaluated we rotate the coordinates into their principal-axis frame [Jolliffe, 2002], which costs  $O(nd^2 + d^3)$  and is dominated by dense MST construction whenever  $n$  is large relative to  $d$ ; the rotation leaves the rotation-invariant features untouched and makes the axis-aligned bounding-box descriptors independent of the input orientation. Appendix N lists the geometric group.

Nineteen features summarize the MST. Ten describe the edge-weight distribution: mean, standard deviation, skewness, kurtosis, maximum, and the 10/25/50/75/90 quantiles. Higher moments characterize heterogeneous edge-length distributions, including those produced by dense clusters separated by long connecting edges. Nine describe topology:

dominance, gaps, leaves, degrees, large edges, and the MST diameter. The MST’s total length is not among them, because it is the multiplicand  $L_{\text{MST}}$  and withholding it keeps the learned quantity a scale-free ratio. The thirty-first feature, `greedy_nn_over_mst`, is the length of an exact greedy nearest-neighbour tour divided by  $L_{\text{MST}}$ . It is the only input GART 2.0 adds to the 30-feature block it inherits, and it is a constructed upper bound on the target: a greedy tour is a feasible tour, so its length is at least  $L_{\text{TSP}}$  and the ratio is at least  $\alpha$ . The feature also gates coverage. On the non-Euclidean path of Section 6, where the metric is given but coordinates are not, the estimator returns a status row rather than a prediction for any instance whose ratio falls outside  $[1.035, 2.209]$ . That interval is the range the ratio spans over the full 106,272-row corpus, training, validation and test together, which is what the implemented gate uses; the training split alone spans the narrower  $[1.0465, 2.1295]$ . We state the gate as the corpus range rather than as a training range because that is the constant the code applies. Definitions and computational costs are reported in Appendix C.

### 3.2 Dataset Generation

To train a general model, we constructed a synthetic dataset spanning four one-dimensional generators: Uniform, Normal (Gaussian), Clustered ( $k$ -means centers), and Correlated (autoregressive across dimensions). Each axis of each instance draws its generator independently, producing anisotropic configurations whose topology varies by axis. The generator is selected from sixteen letter codes that map onto these four functions, and the map is not balanced: eleven codes resolve to the uniform draw, two to normal, two to clustered, and one to correlated. Measured over all 2,444,256 axes in the corpus, 68.7% are uniform, 12.5% clustered, 12.5% normal, and 6.2% correlated. The corpus is therefore predominantly uniform per axis and not balanced across the four classes, and the multidimensional results below should be read against that mix.

The full feature corpus comprises 106,272 solved instances with node counts  $n \in [5, 1000]$  and dimensions  $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$  (17 training-visible dimension values), plus a held-out extrapolation bucket at  $d = 100$ . Instances are partitioned into training (69,768), validation (19,584), and test (16,920) via a stratified split on  $(d, n, \text{grid size})$ ; all  $d = 100$  instances are locked into the test partition so the model never sees that dimension during training or validation. The full training grid is given in Appendix B (Table 11); extrapolation to  $d = 100$  is evaluated in Section 4.3.1.

For each instance, the generation pipeline ran Concorde [Applegate et al., 2006] and LKH-3 [Helsgaun, 2017] and retained the shorter tour. The canonical feature table stores the retained tour cost but not solver-specific costs or Concorde certificates; we therefore call these values reference tour costs and do not claim that every stored row is independently auditable as optimal. Coordinate scaling did *not* preserve floating-point precision uniformly. Two code paths with different integer scale factors were used, one on each of two consecutive generation days, and the coarser of the two returned a cost that is wrong on 14.22% of the corpus. Section 5.7 gives the mechanism, the census and the repair; every label behind every table in this paper is the repaired one.

### 3.3 Reference-tour provenance audit

The target clip above is a weak screen, so we audited the corpus directly. We recomputed the length of every stored tour on its own released coordinates for all 106,272 instances. The audit sorts each instance into one of four classes. For 100,883 instances (94.93%) the stored cost matches the recomputed tour length to within 0.1%. For 586 instances (0.55%) the stored cost is exactly the integer-rounded length of the stored tour, the TSPLIB EUC\_2D convention; 584 of these have  $n \leq 10$ , where per-edge rounding is largest relative to the tour, and the effect vanishes by  $n > 50$ . A further 4,619 instances (4.35%) sit between those thresholds, again concentrated at  $n \leq 10$ .

The remaining 184 instances (0.173%) disagree by more than 1% and are not explained by rounding. Their stored Concorde and LKH-3 lengths agree with each other on 166 of 184 rows while disagreeing with the released coordinates, so the solvers were given a coordinate file that the released instance no longer matches. The defect concentrates where

generation was most expensive: 151 of 184 have  $n > 500$ , and by dimension it clusters at  $d = 100$  (57),  $d = 35$  (38),  $d = 30$  (18), and  $d = 45$  (16). It spans all three partitions, 80 training, 30 validation, and 74 test.

We tested whether the stored costs are anomalous for their cell. The 184 rows fall in 29 ( $d, n$ ) cells, none of which retains fewer than 219 unaffected rows. We standardize the mean  $\alpha = L_{\text{TSP}}/L_{\text{MST}}$  of each cell’s affected rows against the mean and standard deviation of that cell’s unaffected rows; on that scale, 17 of the 29 cells depart by more than 1.6 standard deviations and those 17 hold 95 of the 184 rows; the largest departure is  $-15.93$ , at  $d = 25$ . The affected rows’  $\alpha$  is therefore far from the unaffected distribution of its own cell, which is what a coordinate swap predicts, because the stored cost and the recomputed  $L_{\text{MST}}$  then come from different coordinate files. That same mechanism means the per-cell comparison cannot be offered as evidence that the stored costs are sound; the solver agreement above and the sensitivity below carry that. Section 5.7 settles the question in the opposite direction: it certifies 40 stored costs across the whole corpus as strictly below a proven lower bound on the optimum of their own released coordinates, and 39 of those 40 are inside these 184 rows.

Before the labels were repaired, removing all 184 instances from the multidimensional benchmark moved GART 2.0 from 0.6201% to 0.6182% MAPE and every other estimator by less than 0.06 percentage points, and refitting the model on the cleaned training split moved test MAPE from 0.6182% to 0.6146%; the shipped model is the one fitted on the uncleaned split, and we report that refit as a sensitivity rather than adopting it. Neither the stored cost nor the stored tour describes the released point set for these rows, and no bound is tight enough to reconstruct the optimum from the coordinates alone. On the 33 of them with  $n \leq 500$ , a plain float-metric 2-opt descent on the released coordinates lands a median 0.81% above the stored label and finds a strictly shorter tour than it on 10, which is impossible if that label were the optimum of those coordinates; the descent is 1.5 orders of magnitude looser than the quantity being repaired, so it cannot supply a replacement either. All 184 are therefore *quarantined* rather than disclosed: they carry an empty reference cost and the status `quarantined_label_unrecoverable` in every released results CSV, so an aggregate that ignores the status column yields NaN and never a plausible number, and the scoring harness raises on any attempt to score one. 74 of them fall in the multidimensional test partition, which is scored on the remaining 16,846 instances throughout. Quarantining all 184 moves GART 2.0 from 0.6201% to 0.6177% MAPE, jointly with the repair of Section 5.7, and every other estimator by less than 0.14 percentage points. The target clip flags two of the 184, so range screening alone is not an adequate provenance check.

### 3.4 Model Architecture and Training

The estimator for  $\alpha$  is a *gradient-boosted ensemble* of regression trees. A single regression tree is easy to interpret but high-variance, so combining many trees yields a far more stable estimate. Random forests [Breiman, 2001] average trees grown independently on bootstrap samples; gradient-boosted decision trees [Friedman, 2001] instead grow the trees *sequentially*, each new tree fit to the error left over by the trees before it, so the ensemble keeps improving where it is currently weakest. We use LightGBM [Ke et al., 2017], a fast and memory-efficient GBDT implementation whose leaf-wise growth splits the leaf with the largest estimated loss reduction instead of expanding every depth level uniformly; its final prediction  $\hat{\alpha}$  is the sum of all tree outputs.

The learner uses squared-error regression on the logit-transformed target  $z = \log((\alpha - 1)/(2 - \alpha))$ , and predictions are returned through its inverse  $\alpha = 1 + (1 + e^{-z})^{-1}$ . The bound  $\alpha \in [1, 2]$  is therefore structural: the inverse transform cannot leave the interval, and no clip is applied at inference. On the 16,920 multidimensional test predictions  $\hat{\alpha}$  fell in  $[1.029, 1.922]$ , well inside the bound, so the transform is not saturating at either end. The hyperparameters are inherited and frozen; no search was run for this model, so Section 4 compares a tuned field against an untuned entrant. We ran the search and rejected it. Optuna TPE [Akiba et al., 2019] against in-distribution validation improves multidimensional accuracy and loses more than it gains on TSPLIB, and 200 trials, 60 of them completed and 140 pruned, moved multidimensional MAPE by 0.011 percentage points while costing 0.16 points of TSPLIB MAPE. The tuned booster reaches 0.6112% on the multidimensional test split against 0.6226% for the same booster on the frozen

hyperparameters, and 2.6364% against 2.4725% on TSPLIB EUC\_2D; both are unconstrained and both are on the logit target, so the monotone constraint of the shipped model does not enter the difference, and scoring the tuned booster against the shipped model instead would confound the search with that constraint. Early stopping used cost-level MAPE on the validation split with a patience of 100 rounds. The model was fit at seed 42 on the training split only, with no train-validation refit, and the test split was scored once after that fit. The resulting ensemble contains 1,118 trees with 148 leaves per tree as the growth cap and 147.80 realized on average; its configuration is reported in Appendix O.

Two inputs carry non-increasing monotone constraints,  $n$  and  $d$ , encoding that  $\alpha$  falls towards 1 as an instance grows or the ambient dimension rises. The constraints are enforced inside the tree builder, so a sum of such trees cannot increase along a ceteris-paribus sweep of either variable; a probe over 1,000 held-out instances accordingly returns 100% non-increasing sweeps on both axes with zero violations. Those rows are drawn from the multidimensional test split at seed 42 and swept on log-spaced grids of 24 points,  $d$  from 2 to 200 and  $n$  from 5 to 4,000, against a tolerance of  $10^{-9}$  in  $\alpha$  units below which a difference is float noise. Both grids deliberately overshoot the synthetic evaluation range, because the guarantee is wanted outside that range and not only inside it. The probe is ceteris paribus in the strict sense, which bounds what it establishes: it overwrites one column of a real held-out instance and holds the other thirty at that instance’s measured values, so a swept row corresponds to no realisable instance. What it measures is the shape of the learned function in one argument, which is what a structural guarantee is a claim about, and it says nothing about accuracy at those  $n$  and  $d$ . That figure also verifies only that the constraint was applied, and is not evidence about the model’s inductive bias. A constant multiple of  $L_{\text{MST}}$  is trivially non-increasing on both axes, so the probe does not compare this model against that family at all.

The informative control is the same fit with the constraints removed, which holds monotonicity on 8.4% of the dimension sweeps and 19.0% of the size sweeps. The constraint is therefore doing the work: without it the same feature block and the same data do not produce a monotone response in either argument.

Inference cost is dominated by feature extraction, not tree traversal. The trained model performs  $K = 1,118$  traversals with empirical mean root-to-leaf depth  $\bar{D} \approx 11.83$  (maximum 40). Because the trees are unbalanced, traversal cost is better represented by  $\bar{D}$  than by  $\lceil \log_2 148 \rceil$ ; the ensemble therefore performs approximately  $K\bar{D} = 1.3 \times 10^4$  comparisons per prediction, independent of  $n$ . With a dense distance matrix, the pipeline costs  $O(n^2d + nd^2 + d^3)$ , combining pairwise distances and MST construction with the PCA rotation. In the plane the Euclidean MST is a subgraph of the Delaunay triangulation [Shamos and Hoey, 1975], so it can be built in  $O(n \log n)$ ; sorting MST edges adds  $O(n \log n)$ , and the remaining topological statistics cost  $O(n)$ . The greedy nearest-neighbour tour behind `greedy_nn_over_mst` does not take that path: it scans a dense distance matrix in  $O(n^2)$  for  $n \leq 3000$  and switches to precomputed 16-neighbour candidate lists with an exact nearest-unvisited fallback above that threshold. Below  $n = 3000$  the greedy tour, not the MST, is therefore the leading term of 2D extraction.

### 3.5 Feature importance and design validation

We assess the 31-feature set with mean absolute SHAP magnitudes [Lundberg and Lee, 2017] against the production booster, on a 5,000-row sample of the 16,920-row held-out test split. The MST dominance ratio contributes 26.5% of the total magnitude and `greedy_nn_over_mst` 23.9%; those two features carry 50.4% of it between them. Four of the top ten features are MST-derived, five are geometric, and the greedy ratio is the remaining one. Node count and dimension jointly contribute 14.1%, the two bounding-hypervolume features 7.2%, and the four centroid-distance descriptors 3.0%. The greedy ratio ranking second on its first appearance is the case for adding it; it is also the input that lets the estimator decline an instance, since the coverage gate of Section 6 is a range test on this feature and no other. Appendix D reports the complete ranking.



## 4 Benchmarking

This section declares the roster, the datasets, and the metrics, then reports per-dataset results, a matched-domain re-evaluation of every classical estimator, and two tests that withhold whole dimensions and size ranges from training. Section 4.2 describes the three benchmarks; the  $d = 100$  and near-linear cases in them lie outside the training distribution and serve as extrapolation tests.

We report four error metrics because no one of them ranks estimators correctly on its own. Mean Absolute Percentage Error (MAPE) reports average magnitude. The Standard Deviation of Percent Error (SDPE) occupies the first column of every results table and measures precision, but it is blind to bias: an estimator that is uniformly 20% high scores an SDPE of zero. We therefore complement it with MAPE, with the median absolute percentage error as an outlier-robust companion, and with the Mean Signed Percent Error (MSPE) so that a systematic offset stays visible. The distinction is not hypothetical here. BHH given the exact sampling region carries a  $-8.65\%$  offset on uniform instances with only  $7.76\%$  SDPE, and the custom Hilbert sort has  $30.95\%$  MAPE against  $14.19\%$  SDPE; reading either row on SDPE alone would misdescribe it. SDPE is defined as:

$$\text{SDPE} = 100 \cdot \sqrt{\frac{1}{N-1} \sum_{i=1}^N (e_i - \bar{e})^2}, \quad \bar{e} = \frac{1}{N} \sum_{i=1}^N e_i.$$

Here  $e_i = (\hat{L}_i - L_i)/L_i$  is the signed relative error and  $N$  is the number of scored instances; the leading factor of 100 expresses SDPE in percentage points. In every results table, bold marks GART 2.0, the model proposed here; it does not mark the best value in a column, and several baselines beat it in individual cells. We accompany each SDPE with a 95% bootstrap confidence interval over 1,000 resamples, seeded at 42, and suppress the interval where a group holds fewer than three instances. Those intervals are computed separately for each estimator and therefore ignore the pairing, so overlapping intervals establish nothing on their own; Appendix I supplies paired tests of the difference. A paired test that fails to reject is not evidence of equivalence, which would require an explicit equivalence margin we do not set.

### 4.1 Benchmark Roster

We compare GART 2.0 against seven baselines in four families: classical region-based estimators, constant-ratio references, the published GART 1.0 predecessor, and a constructive heuristic. A source-to-code audit found that the repository had been feeding the classical estimators the convex hull of the realized sample in place of the quantity their sources require, and that the substitution rather than the estimators produced their reported errors. Where we could read the primary we repaired the implementation instead of dropping the comparison, and we state for every row which inputs it receives. Where we could not, we withdrew the estimator: the same audit established that our Daganzo, Chien, and Kwon–Golden–Wasil coefficients had been transcribed from a secondary source, so those three are surveyed in Section 1.2 and scored nowhere in this paper.

**Classical region-based estimators.** Each estimator now receives the quantity its source defines and is gated to the domain its source covers. The BHH form is  $\hat{L} = \beta_d n^{(d-1)/d} \mathcal{V}^{1/d}$ , where  $\mathcal{V}$  is the measure of the sampling region [Beardwood et al., 1959] and  $\beta_2 = 0.7124$  [Johnson et al., 1996]; for  $d = 3$  we use  $\beta_3 = 0.6979$  [Percus and Martin, 1996] and for  $d \geq 4$  the conjectured large- $d$  asymptotic  $\beta_d \approx \sqrt{d/(2\pi e)}$  [Bertsimas and van Ryzin, 1990]. Çavdar and Sokol [2015] gives  $\hat{L} = 2.791 \sqrt{n \text{cstdev}_x \text{cstdev}_y} + 0.2669 \sqrt{n \text{stdev}_x \text{stdev}_y} A / (\bar{c}_x \bar{c}_y)$ , where  $A$  is the area of the rectangle covering the nodes,  $\text{stdev}_j$  is the standard deviation of the raw coordinates on axis  $j$ , and  $\bar{c}_j$  and  $\text{cstdev}_j$  are the mean and standard deviation of the absolute distances from the nodes to that rectangle’s midpoint line on axis  $j$ . The Çavdar–Sokol estimate is divided by the finite- $n$  adjustment factor its source fits, held fixed outside the node range that fit covers. Both forms are transcribed from primary documents we read: the BHH constants from the sources cited above, and the Çavdar–Sokol estimator together with both of its constant sets and its adjustment factor from Çavdar

[2014]. Appendix E records what each estimator receives, and why three further classical estimators are surveyed but not scored.

**Constant-ratio references.** Three rows bound what is achievable without learning, and each answers a different question. The  $\alpha = 1$  floor returns  $L_{\text{MST}}$  itself: it is the certified lower bound of Eq. (1), the only constant defined at every  $d$ , and the reference  $R_\alpha^2$  is measured against, so it isolates what the learned multiplier adds to the scaffold it multiplies. The asymptotic ratio uses the published constants  $\beta_{\text{TSP}}/\beta_{\text{MST}} \approx 1.1253$  [Johnson et al., 1996, Cortina-Borja and Robinson, 2000] and is defined at  $d = 2$  only, the one constant here taken from the literature instead of fitted on our data. The calibrated row uses  $\hat{\rho}(d, n)$ , the mean of  $\alpha$  within each dimension–size cell of the training split, which is the ceiling of this family: the best constant obtainable from the training split conditioned on everything cheap that is known about an instance. Two further constants stay in the released tidy tables and are not printed as baselines: an author-chosen schedule whose  $d \geq 3$  values sit 8 to 15 percentage points below the training-split mean of  $\alpha$ , and the per-dimension mean  $\hat{\rho}(d)$ , which  $\hat{\rho}(d, n)$  dominates everywhere and which the  $\alpha = 1$  floor itself beats on TSPLIB EUC\_2D. The  $\alpha = 1$  and asymptotic rows are one estimator at two values of  $c$  in  $\hat{L} = cL_{\text{MST}}$ , so their dispersion is related exactly by  $\text{SDPE}(c) = c\text{SDPE}(1)$ ; we keep both because that identity does not extend to MAPE.

**Prior model.** GART 1.0 [Feldman, 2025] is the published predecessor of this work and the one learned estimator in the baseline count. It is quadratic in construction and planar only, which is what this paper set out to remove: it appears only on the 2D synthetic and TSPLIB EUC\_2D benchmarks (Tables 22, 23). Its artifact-driven extractor uses a dense pairwise-distance matrix, so its effective time and memory cost is  $O(n^2)$  rather than  $O(n \log n)$ , and it treats the lexicographically first unique coordinate as a depot-like reference for two features although no depot is defined in these datasets. It differs from GART 2.0 in data, features, and training, so the comparison isolates no single causal change.

**Constructive heuristic.** The Hilbert row builds a tour from a bare  $d$ -dimensional Hilbert ordering after independent per-axis min–max normalisation to a 16-bit grid. It constructs a tour rather than estimating one, and the planar guarantee of Bartholdi and Platzman [1982] does not transfer to this multidimensional variant.

**Certified lower bound.** One further row is not an estimator at all. The Held–Karp 1-tree bound [Held and Karp, 1970, 1971] returns a proven lower bound on the optimal tour rather than a prediction of it, at a cost fixed by an explicit ascent budget  $k$ . Two rows carry it, differing only in the step rule of the subgradient ascent: the Volgenant–Jonker schedule [Volgenant and Jonker, 1982], which consults no upper bound, and the Polyak step [Polyak, 1969], which sizes each move from the gap to a constructively built tour. Both are scored on all four benchmarks – the 2D diverse set, the multidimensional set, TSPLIB EUC\_2D, and the non-EUC\_2D TSPLIB instances – both raw and scaled by a single constant fitted per budget on the training split, and Section 5 reports them in full. We report the bound outside the seven-baseline count because it answers a different question than an estimator does. It is nevertheless the comparator that decides this paper.

**Region measure and matched domains.** A *matched domain* is the subset of a benchmark on which a classical estimator’s own published assumptions hold, so that scoring it there measures the method rather than the mismatch. Every synthetic generator draws coordinates inside  $[0, G]^d$  for a coordinate-grid side length  $G$ , so the sampling region measure is  $G^d$  exactly. BHH is the only estimator here whose source names that measure, and on the synthetic benchmarks we report it as two rows, one given  $G^d$  and one given the convex hull of the realized sample. Both are reported because the region measure is the source-faithful input only where the generator is uniform on that region: a near-collinear point set has one-dimensional effective support, and handing an area-based estimator  $G^2$  inflates its prediction by an amount that describes the data and not the estimator. TSPLIB defines no sampling region at all, so there BHH receives the convex hull as a stated plug-in. Çavdar–Sokol takes no region input: its  $A$  is the area of the rectangle covering the nodes, a statistic of the sample, so it has a single row on every benchmark and supplying it  $G^2$  would be our construction rather than its authors’. Section 4.4 reports both estimators on the subset where their

published assumptions hold. Neither assignment is a free parameter: the region measure is  $G^d$  by construction of the generator, and the matched domain is pinned by the source itself at i.i.d. uniform sampling, so a reader can reconstruct the subset without reference to our results.

Table 1: Benchmark roster. Every row is evaluated numerically. “Domain” states the range each source fits or derives; rows outside it are reported separately as extrapolations.

| Estimator                                     | Formula                                                                                                                        | Domain                                          | Cost             | Source                                                                   |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------|--------------------------------------------------------------------------|
| <i>Classical region-based estimators</i>      |                                                                                                                                |                                                 |                  |                                                                          |
| BHH                                           | $\beta_d n^{(d-1)/d} \mathcal{V}^{1/d}$ , $\mathcal{V}$ the sampling-region measure                                            | i.i.d. uniform, $n \rightarrow \infty$          | $O(nd)$          | Beardwood et al. [1959], Johnson et al. [1996]                           |
| Çavdar–Sokol                                  | $2.791 \sqrt{n \text{ cstdev}_x \text{ cstdev}_y} + 0.2669 \sqrt{n \text{ stdev}_x \text{ stdev}_y A / (\bar{c}_x \bar{c}_y)}$ | Minimum-area rectangle; $n \geq 100$ correction | $O(nd)$          | Çavdar [2014]                                                            |
| <i>Constant-ratio references</i>              |                                                                                                                                |                                                 |                  |                                                                          |
| $L_{\text{MST}} (\alpha = 1)$                 | $L_{\text{MST}}$                                                                                                               | All                                             | $O(n^2 d)$       | Lower bound of Eq. (1)                                                   |
| Asymptotic MST ratio                          | $(\beta_{\text{TSP}} / \beta_{\text{MST}}) L_{\text{MST}} \approx 1.1253 L_{\text{MST}}$                                       | $d = 2$ , uniform, $n \rightarrow \infty$       | $O(n^2 d)$       | Johnson et al. [1996], Cortina-Borja and Robinson [2000]                 |
| Calibrated $\hat{\rho}(d, n)$                 | $\hat{\rho}(d, n) L_{\text{MST}}$ , per dimension–size cell                                                                    | Trained cells                                   | $O(n^2 d)$       | This work                                                                |
| <i>Prior model and constructive heuristic</i> |                                                                                                                                |                                                 |                  |                                                                          |
| GART 1.0                                      | 2D-only feature set, GBDT regression on $\alpha$                                                                               | $d = 2$                                         | $O(n^2)$         | Feldman [2025]                                                           |
| Custom Hilbert sort                           | Bare $d$ -dimensional ordering, per-axis normalised, $h = 16$                                                                  | Constructs a tour                               | $O(n \log n)$    | After Bartholdi and Platzman [1982]                                      |
| <i>Certified lower bound</i>                  |                                                                                                                                |                                                 |                  |                                                                          |
| Held–Karp 1-tree, V&J step, budget $k$        | $L_{1\text{tree}}(\pi_k) - 2 \sum_i \pi_i$ after $k$ ascent steps; step schedule consults no upper bound                       | Any symmetric distances, any $d$ ; certificate  | $\Theta(kn^2 d)$ | Held and Karp [1970, 1971], Volgenant and Jonker [1982], Helsgaun [2000] |
| Held–Karp 1-tree, Polyak step, budget $k$     | The same bound; step $\gamma(\text{UB} - w) / \ g\ ^2$ against a constructive tour                                             | Any symmetric distances, any $d$ ; certificate  | $\Theta(kn^2 d)$ | Held and Karp [1970], Polyak [1969]                                      |
| Calibrated 1-tree                             | $c_k$ times either bound, $c_k$ fitted per budget on the training split                                                        | As trained                                      | $\Theta(kn^2 d)$ | This work                                                                |

## 4.2 Datasets

We evaluate on a multidimensional synthetic set, a diverse 2D synthetic set, and TSPLIB95. The synthetic sets are held-out splits from our generation pipeline (Appendix B). The evaluation assumes symmetric metric distances; Section 6 lists the ten EXPLICIT TSPLIB matrices whose structural triangle-inequality violations put them outside every metric-dependent aggregate here.

#### 4.2.1 Multidimensional synthetic

The benchmark contains 16,920 held-out instances with  $n \in [5, 1000]$  and  $d$  drawn from the 17 training-visible values plus  $d = 100$ , sampled by stratified dimension–size–grid buckets from the same four per-axis generators as the training set: uniform, normal, clustered ( $k$ -means [Lloyd, 1982]), and autoregressive-correlated. Stratification enforces zero row overlap with the training split, but the generator families and axis mix of §3.2 are shared, so this benchmark measures held-out sampling and not held-out geometry. Section 4.5 withholds whole dimensions and whole size ranges from training to separate the two.  $d = 100$  is outside the training range and serves as the dimension-extrapolation test. The full  $(d, n)$  count grid is given in Appendix B.3 (Table 10).

#### 4.2.2 2D diverse synthetic

The benchmark contains 2,580 instances with  $d = 2$ ,  $n \in [5, 1000]$ , and coordinate-grid side length  $G \in [1000, 10000]$ . It was generated by the separate 2D benchmark pipeline; an identifier-level audit found zero overlap with the 106,272-row training/validation/test corpus. Five distribution classes (Table 2) each probe a different assumption made by classical 2D estimators. Two of the thirteen sub-generators test geometry the training corpus does not contain: Line Noise, which is near-collinear, and grid, a jittered square lattice. Near-collinear and piecewise-linear point sets are a standard degenerate TSP test case [Smith-Miles et al., 2010, Bossek et al., 2019]. Appendix B gives the construction of both classes, and Table 17 reports grid as its own row instead of inside its class aggregate.

Table 2: 2D benchmark dataset composition (2,580 instances).

| Class                                                        | Generators                                                                 | Count        | Stress target                                       |
|--------------------------------------------------------------|----------------------------------------------------------------------------|--------------|-----------------------------------------------------|
| Isotropic [Çavdar and Sokol, 2015, Golden and Stewart, 1985] | Uniform, Normal, Triangular, Truncated Exp.                                | 840          | Control: $A_{\text{hull}} \approx A_{\text{box}}$ . |
| Biased / Skewed [Çavdar and Sokol, 2015]                     | Squeezed, Uni-Tri, Tri-Squeezed, Correlated                                | 840          | Aspect-ratio distortion: $\sigma_x \neq \sigma_y$ . |
| Geometric Struct. [Reinelt, 1991, Bossek et al., 2019]       | Grid <sup>†</sup> , Boundary (hollow), X-Central                           | 630          | Lattice regularity, hollow voids.                   |
| Clustered [Golden and Stewart, 1985, Bossek et al., 2019]    | $k$ -Means ( $k \in [5, 20]$ )                                             | 60           | Bimodal edge-weight distribution.                   |
| Line Noise <sup>†</sup>                                      | Boundary-clipped $y = mx + b + \epsilon$ ; realized as a 2–4 segment union | 210          | Locally 1D structure; hull/box divergence.          |
| <b>Total</b>                                                 |                                                                            | <b>2,580</b> |                                                     |

<sup>†</sup> Not represented in training; included to measure generalization to unseen geometries.

Grid is one of the three Geometric Struct. generators; Table 17 reports it as its own row.

#### 4.2.3 TSPLIB95 EUC\_2D common subset

The common-coverage benchmark contains 78 EUC\_2D instances with  $n$  from 51 (e1151) to 18,512 (d18512) [Reinelt, 1991]. Every numerically evaluated 2D baseline is defined on this complete subset. The 23 retained non-EUC\_2D instances are evaluated separately in Section 6. For large coordinate instances, a Delaunay-based MST construction [Delaunay, 1934] avoids materializing the dense distance matrix.

### 4.3 Results

The detailed tables in the appendices report SDPE with a 95% bootstrap interval (1,000 resamples), MAPE, median absolute percentage error, MSPE,  $R_\alpha^2$  for estimators that predict a multiple of  $L_{\text{MST}}$ , and median prediction time. Writing  $e_i = (\hat{L}_i - L_i)/L_i$ , MAPE is  $100 \overline{|e_i|}$  and MSPE is  $100 \bar{e}$ . We define  $R_\alpha^2 = 1 - \sum_i (\hat{\alpha}_i - \alpha_i)^2 / \sum_i (\alpha_i - \bar{\alpha})^2$  with  $\bar{\alpha}$  the mean over the reported bucket, so a negative value means the estimator predicts  $\alpha$  worse than that bucket mean does. Each estimator in a bucket uses the stated common instance set. SDPE is the primary precision metric;  $R_\alpha^2$  measures the learned multiplicative correction beyond MST length.

For synthetic tables, the separate reference-tour row reports the median generation time stored in the canonical benchmark results. The logs contain one timed estimator invocation per instance, with no recorded warm-up repetitions or within-instance dispersion; feature and inference time exclude file I/O and reference-tour generation. All timings are therefore descriptive single-machine measurements, and none supports an inferential comparison. We omit TSPLIB reference-generation timing because the available logs have incomplete instance coverage and mix hardware sources.

#### 4.3.1 Multidimensional benchmark

The multidimensional benchmark is 16,920 held-out instances, of which the 16,846 with a recoverable reference cost are scored (Section 5.7). We report it in two orthogonal slices of the same instances, by size (Table 21) and by dimension (Table 20), with the error distributions in Figure 1.

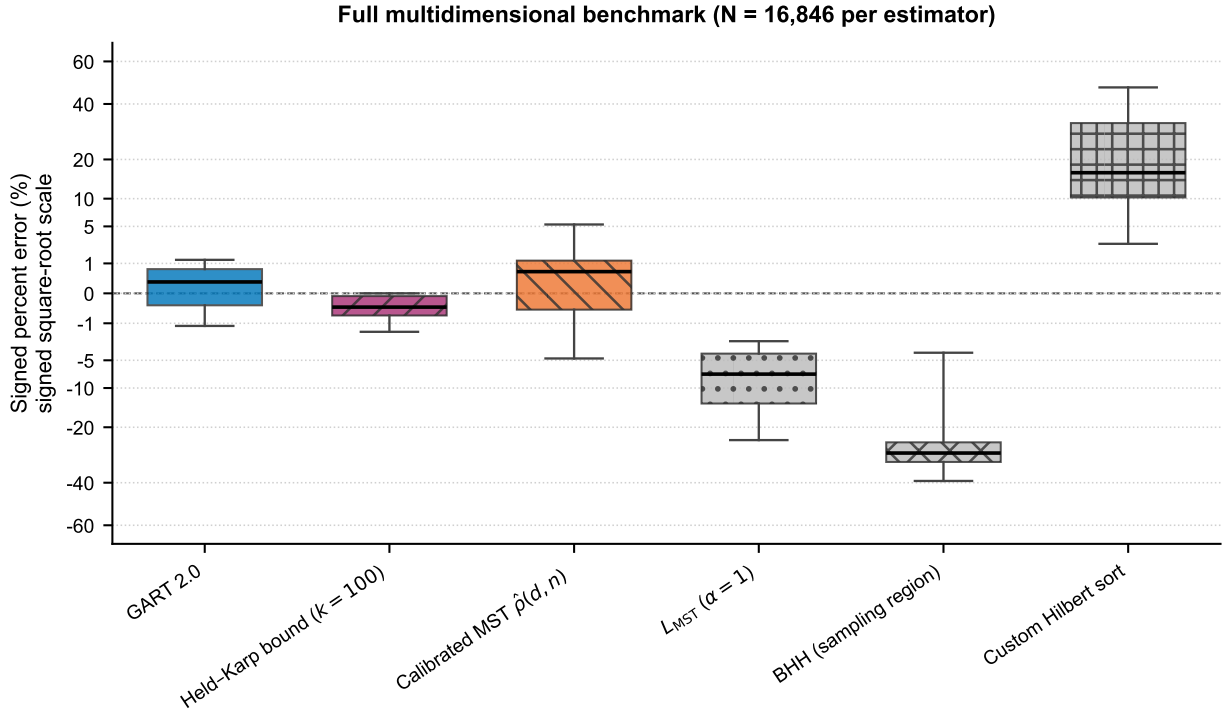


Figure 1: Signed percent error on the multidimensional benchmark ( $N = 16,846$  per estimator). Boxes show the interquartile range and median; whiskers span the 5th–95th percentiles, and outliers are suppressed for legibility. The roster is that of Table 20 plus the certified Held–Karp 1-tree bound of Section 5, drawn at the ascent budget  $k = 100$  at which Table 4 first has it overtake GART 2.0. The bound is one-sided by construction, so its box sits at or below zero with its upper quartile on the zero line, and its box is the only one in the figure that carries a guarantee. The vertical axis is a signed square root,  $\text{sign}(x)\sqrt{|x|}$ , with ticks labelled at true percentages and the same limits on all three panels; nothing is clipped. Neither obvious alternative carries these rosters. A linear axis collapses the tightest boxes to lines, because the interquartile ranges across this roster differ by more than an order of magnitude. A logarithmic or symmetric-log axis fails differently: height on a log arm is a ratio rather than a dispersion, so it reorders the boxes, and any box straddling zero stretches without bound.

GART 2.0 obtains 0.62% MAPE and 0.99% SDPE overall. The strongest baseline in the roster of Section 4.1 that uses no learned model is the calibrated ratio  $\hat{p}(d, n)$  at 1.82%/2.94%, so the learned  $\alpha$  improves on a lookup table of 102 constants fitted on the same training split by a factor of 2.9 on MAPE and 3.0 on SDPE. The  $\alpha = 1$  floor reaches 9.71%/7.26%, the custom Hilbert sort 21.21%/14.40%, and BHH given the exact sampling region 28.01%/13.57%. Figure 1 draws the certified Held–Karp 1-tree bound beside them, and the shape of its box is a difference in kind rather than in degree: it lies wholly at or below zero because the bound cannot exceed the optimum, so the sign of its error is

known before the instance is seen. Several estimator boxes also fall on one side of the line, but only as a fitted tendency observed after the fact; none of them can promise a side on the next instance. Every number in this subsection ranks the estimator roster against itself; Section 5.3 scores that bound against GART 2.0 on cost and accuracy together.

SDPE decreases monotonically across all six size buckets, from 1.56% at  $n \leq 10$  to 0.46% at  $n \in [201, 500]$  and 0.45% at  $n \in [501, 1000]$ . Against the as-published labels the last step ran the other way, by 0.01 points; the label repair of Section 5.7 removed the only non-monotone step this benchmark had. Across dimension groups it falls from 2.37% at  $d = 2$  to 0.48% at  $d \in [30, 50]$ , and falls again, to 0.44%, at the unseen  $d = 100$ , while MAPE rises there from 0.29% to 0.59%. The standard deviation of  $\alpha$  over the training split falls monotonically with dimension, from 0.1837 at  $d = 2$  to 0.0762 at  $d = 50$ , so the target concentrates as  $d$  grows and the prediction problem becomes easier in exactly the direction the extrapolation runs. The  $d = 100$  result is therefore favourable-direction extrapolation, and Section 4.5 supplies the unfavourable-direction tests.

BHH’s 28.01% error deserves its own explanation, because it is not a defect of the reimplementation. Two of its premises fail here: only 68.7% of axes are uniform, so the region measure  $G^d$  exceeds the effective support of the rest, and  $n \leq 1000$  with  $d \geq 4$  is far from the asymptotic regime the theorem describes, where the sample must fill the region. Supplying the exact region rather than the convex hull nonetheless cuts its MAPE from 39.07% to the 28.01% in Table 20, which isolates how much of the previously reported error was an input substitution; the convex-hull row itself is in the released tidy tables rather than here, because at  $d > 3$  that hull is replaced by the coordinate bounding box (Appendix E) and the row would mix two region definitions across dimensions.

#### 4.3.2 2D diverse benchmark

The 2D benchmark is 2,580 instances in five distribution classes, each probing a different geometric regime; Figure 2 shows the common-coverage error distributions and Appendix L disaggregates by size.

GART 2.0 obtains 2.89% MAPE and 4.68% SDPE overall, against 5.76%/9.38% for the calibrated ratio  $\hat{\rho}(d, n)$ , 7.30%/7.90% for GART 1.0, 9.20%/11.55% for the asymptotic ratio, and 17.95%/10.27% for the  $\alpha = 1$  floor. Its SDPE stays below 5.9% in every size bucket and falls to 2.75% in the [501, 1000] bucket, and its aggregate is less than half the asymptotic ratio’s. The custom Hilbert sort has 30.95% MAPE against 14.19% SDPE, a gap that identifies its error as a near-constant positive offset and not scatter, which is what a space-filling-curve tour should produce. The certified bound plotted beside them in Figure 2 is the one row there that is not an estimator, and it reads that way: its interquartile range is the narrowest in the figure and it never crosses zero, which is what a one-sided guarantee looks like when it is drawn next to two-sided estimates.

Table 17 disaggregates the five stress classes, which the aggregate hides. GART 2.0 ranges from 1.47% MAPE on the isotropic class to 10.82% on Line Noise, and every estimator degrades on Line Noise in the same direction: the calibrated ratio moves from 3.31% to 20.45%, the asymptotic ratio from 7.12% to 25.88%, and the  $\alpha = 1$  floor from 17.15% to 34.13%. Near-collinear point sets drive  $\alpha$  toward its upper bound of 2, and no estimator anchored on a typical ratio tracks that. Line Noise is not the only unrepresented geometry, and the other one fails in the opposite direction. On the 210 grid instances GART 2.0 reads 7.07% MAPE against 1.96% on the 420 Geometric Struct. instances that are represented, and its MSPE equals that MAPE exactly, so every one of those predictions is high. Realized  $\alpha$  averages 1.05 on grid against 1.56 on Line Noise, so the two unrepresented generators sit at opposite ends of the target range and the estimator errs toward the middle on both. Both therefore confound unseen geometry with an extreme target value.

#### 4.3.3 TSPLIB95 EUC\_2D benchmark

On the full EUC\_2D benchmark GART 2.0 obtains 2.53% MAPE and 2.92% SDPE, against 3.74%/4.48% for the calibrated ratio  $\hat{\rho}(d, n)$ , 3.53%/4.74% for the asymptotic ratio, and 8.43%/7.45% for GART 1.0. Table 23 reports all 78 instances, spanning  $n = 51$  to 18,512, in three prespecified size buckets (23/16/39 instances), and Figure 3 shows

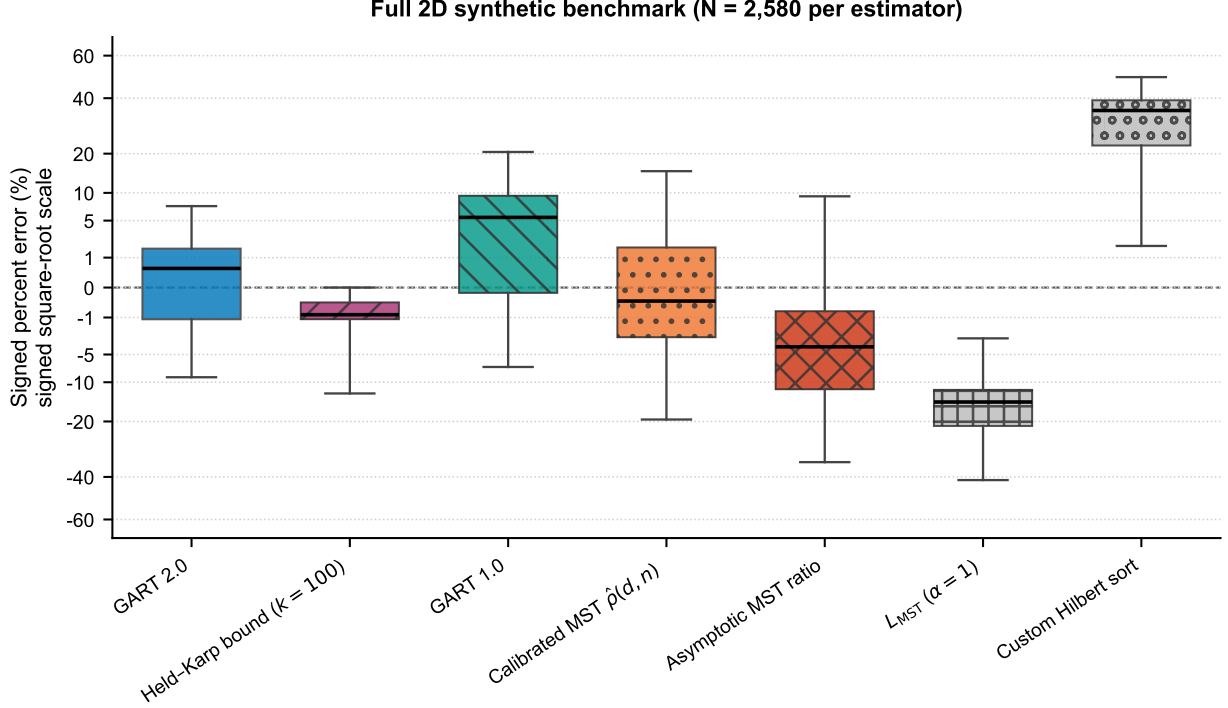


Figure 2: Signed percent error on the diverse 2D benchmark ( $N = 2,580$  per estimator). Boxes show the interquartile range and median; whiskers span the 5th–95th percentiles, and outliers are suppressed for legibility. The roster is that of Table 22 plus the certified Held–Karp 1-tree bound of Section 5, drawn at the same ascent budget  $k = 100$  as in Figures 1 and 3; being one-sided it sits at or below zero. The vertical axis is the signed square root of Figure 1, on the same limits. The classical estimators are omitted from this figure only for legibility; their errors are an order of magnitude larger and are reported in Table 16.

the error distributions; the coarse  $n > 400$  bucket should not be read as a homogeneous asymptotic regime. This is the narrowest margin in the paper, and it is narrow for a structural reason:  $\alpha$  on these 78 instances has mean 1.1306 and standard deviation 0.0558, so a single well-chosen constant is already close. The MAPE-minimising constant on these instances is 1.1275 and reaches 3.52%, so no constant multiplier does better on MAPE, and GART 2.0’s 2.53% improves on that bound by 0.99 percentage points. The dispersion gap is the larger one, 2.92% SDPE against 4.74%, but SDPE alone cannot carry that comparison: for  $\hat{L} = cL_{\text{MST}}$  the signed error is  $c/\alpha - 1$ , so SDPE is proportional to  $c$  and shrinking  $c$  drives it to zero at ruinous MAPE. The comparison holds because GART 2.0 attains the lower SDPE at the lower MAPE. Figure 3 puts the certified bound on the same axis, and the two interquartile ranges fall on opposite sides of it: GART 2.0’s lies wholly above the zero line, the bound’s wholly below. On the stratum where GART 2.0 is strongest the two are therefore comparable in magnitude and opposite in direction, and only one of them fixes its direction in advance.

The advantage is not uniform across size. In the coarse  $n > 400$  bucket the GART and asymptotic-ratio SDPE intervals overlap, on  $[3.20, 3.62]$ . In the post hoc  $n > 10,000$  slice of five instances GART 2.0 is lower on both metrics, 0.57% MAPE and 0.72% SDPE against the asymptotic ratio’s 2.74% and 1.46%. The largest instance, d18512, is more than  $18\times$  the training-size cap. For i.i.d. points of bounded density, normalised TSP and MST lengths each converge to dimension-dependent constants [Beardwood et al., 1959, Steele, 1988], which motivates a fixed-ratio baseline but does not establish that the heterogeneous  $n > 400$  bucket is asymptotic; the advantage at extreme sizes is an observation on five instances and not proof of a convergence mechanism.

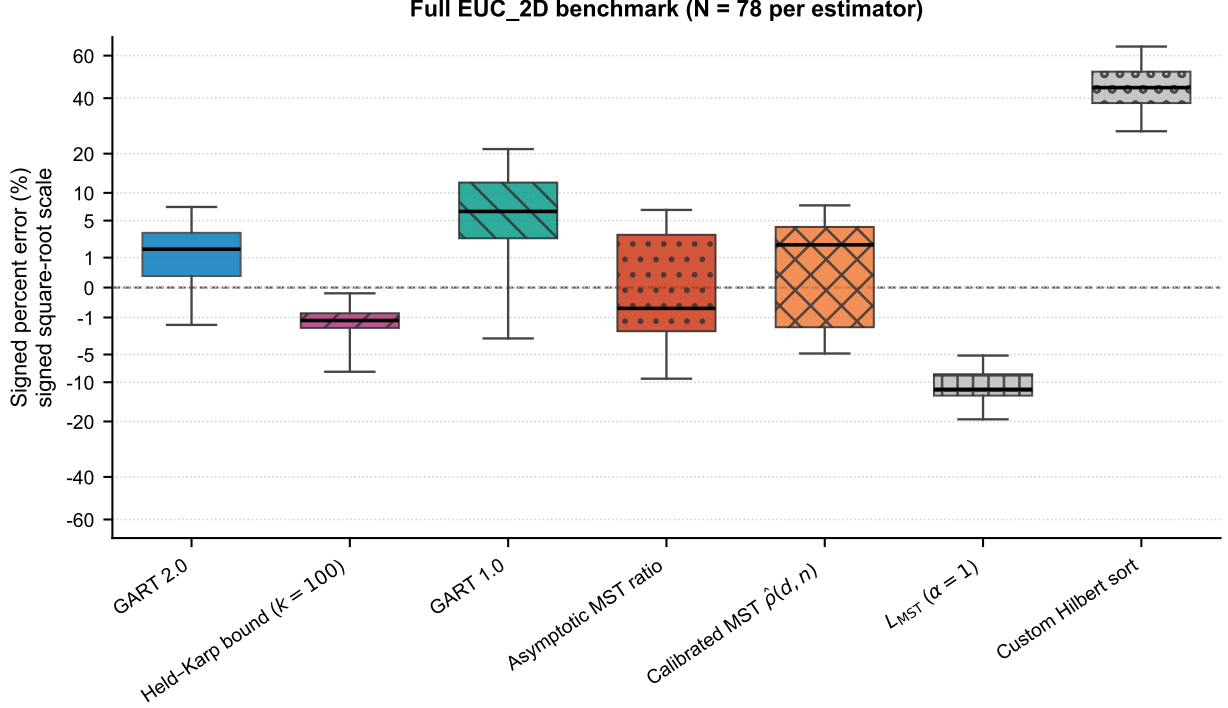


Figure 3: Signed percent error on the TSPLIB95 EUC\_2D benchmark ( $N = 78$  per estimator). Boxes show the interquartile range and median; whiskers span the 5th–95th percentiles, and outliers are suppressed for legibility. The roster is that of Table 23 plus the certified Held–Karp 1-tree bound of Section 5 at ascent budget  $k = 100$ . This is the stratum on which GART 2.0 is closest to the bound, and the two interquartile ranges sit on opposite sides of the zero line, GART 2.0’s just above it and the bound’s just below: the two err in opposite directions, and only one of them can be trusted not to overshoot. The vertical axis is the signed square root of Figure 1, on the same limits. The bound is scored here on all EUC\_2D instances, including the one Table 3 excludes for cost, and on the released reference costs rather than the integer-rounded metric, which is why a few instances place it marginally above the reference.

Both classical estimators are far behind on this benchmark, and the reason is the same one that motivates the paper. Çavdar–Sokol obtains 23.87% MAPE and BHH 25.14%. TSPLIB instances are drawn from real geography and circuit layouts rather than from a uniform density on a known region, so an area-based estimator is applied outside the conditions it was derived under: BHH receives a convex hull in place of a sampling region that does not exist, and Çavdar–Sokol is evaluated on graphs that do not carry a node at the corners of their covering rectangle, which every one of its training graphs did. Both overpredict, by +13.66% and +20.07% respectively. These rows measure the cost of that mismatch, not the quality of the cited methods, which is why Section 4.4 evaluates both where their assumptions hold.

#### 4.4 Matched-domain comparison

The full-benchmark rows above apply both classical estimators outside the conditions they were derived under, so they cannot support a claim about the methods themselves. This section evaluates each one where its published assumptions hold. The 210 instances of the random generator, the Uniform member of the Isotropic class of Table 2, are drawn i.i.d. uniformly on  $[0, G]^2$ , which is exactly the sampling model BHH assumes, and the region measure  $A = G^2$  is known exactly. Çavdar–Sokol carries the same single row here that it carries above, because it consumes no region: between the two panels its instance set changes and nothing else, which is what makes the pair readable as a domain effect. Table 16 reports both panels.



Both estimators’ errors fall substantially on the matched domain. BHH falls from 23.80% MAPE on the full 2D benchmark to 8.91% here. Two things change at once between its two panels, the region input and the instance set, so that figure is the combined effect and not the isolated value of the region input. What is left is a systematic underprediction,  $-8.65\%$  signed against  $7.76\%$  SDPE, which is what an asymptotic constant evaluated at  $n$  as low as 5 should do. Çavdar–Sokol falls from 18.24% to 8.16%, and here only the instance set changed, so that figure is a clean domain effect. It has the lower MAPE of the two classical estimators on every panel of Table 16, and the lower median absolute error here, but not the lower dispersion, and it is also the least even: at 14.28% SDPE its dispersion is the widest in the panel, and its median absolute error of 2.84% sits at just over a third of its mean, so it is close on most uniform instances and badly wrong on a few.

GART 2.0 beats both classical estimators on the matched domain, on both metrics, by a smaller and more credible margin than the full-benchmark tables suggest. On the 210 uniform instances it obtains 1.31% MAPE against Çavdar–Sokol’s 8.16%, BHH’s 8.91% and the  $\alpha = 1$  floor’s 15.60%. A factor of 6.2 over Çavdar–Sokol on i.i.d. uniform draws states what this model adds over the classical literature in the plane; the wider gaps in the full-benchmark table describe domain mismatch instead.

#### 4.5 Generalization beyond the training grid

The multidimensional test split is held out by row, not by geometry: it shares the four generators and the 68.7% uniform axis mix of the training split, and it is stratified on the same  $(d, n, \text{grid size})$  cells. Its 0.62% MAPE therefore measures held-out sampling, and a reader is entitled to ask how much of that is interpolation within a seen stratum. We answer with three refits that hold the hyperparameters and seed fixed and vary only the training data, so no run-to-run noise confounds the comparison. A baseline refit reproduces the released model bit-for-bit at 1,118 trees, reading 0.6177% MAPE over the 16,846 scored rows of the repaired test split and 0.6201% over all 16,920 as published; the refit script enforces this, requiring an identical serialized booster and a maximum absolute prediction difference of zero over all 106,272 corpus rows before it will report anything else. The pipeline is therefore deterministic under the fixed seed on this machine, and every difference below is attributable to the training data and not to refit noise. Scoring the same model against the as-published labels on those same 16,846 rows gave 0.6182%, so 0.0005 percentage points of the change is the label repair and the rest is the quarantine.

Withholding both  $d = 15$  and  $d = 25$  from training, 11.8% of the training rows, raises test MAPE at  $d = 15$  from 0.50% to 0.58% and at  $d = 25$  from 0.35% to 0.38%, while the neighbouring trained dimensions  $d \in \{10, 20, 30\}$  do not degrade at all, moving from 0.43% to 0.42%. The loss is specific to the withheld strata rather than to the reduced training volume, and it takes the form of a bias shift: mean signed error at  $d = 15$  moves from  $-0.008\%$  to  $+0.134\%$ . Training only on  $n \leq 200$  and testing on  $n \in (200, 1000]$  raises MAPE from 0.41% to 0.59%, again almost entirely as a uniform offset, since SDPE moves only from 0.47% to 0.56% while mean signed error rises from  $+0.28\%$  to  $+0.51\%$ , a factor of 1.8.

Both held-out regimes therefore land at 0.58–0.59% MAPE, below the 0.62% the released model achieves on the full test split. The estimator does exploit having seen a dimension’s stratum, worth up to 0.08 MAPE points; off the grid it acquires a systematic offset and keeps its structure. We demonstrate no correction for that offset. The Line Noise class of the 2D benchmark supplies the complementary test on geometry rather than on scale, and Section 4.3.2 shows the cost there is larger. Section 4.6 decomposes that cost and finds it is a coverage gap and a feature gap at once.

#### 4.6 Degenerate Geometry: Coverage and Feature Gaps

Section 4.3.2 reports 10.82% MAPE on the near-collinear Line Noise class against 1.47% on the isotropic class, and the error is almost entirely one-sided: MSPE is  $-10.82$ , so the model under-predicts. Two explanations compete: the training split may never show the model this regime, or the feature set may be unable to represent it. We separate them below, and both hold.

The support gap is exact. The training split contains zero instances with  $\alpha > 1.5$  at  $n \geq 200$ , and every training instance with  $\alpha > 1.7$  has  $n \leq 10$ . Its  $\alpha$  distribution has a 99th percentile of 1.544, and 54% of the 210 benchmark Line Noise instances lie above that percentile. No tree can map large  $n$  at high  $\alpha$  onto a high  $\alpha$  when the fit never contains the combination. The symptom matches: regressing predicted  $\alpha$  on true  $\alpha$  over the 90 Line Noise instances at  $n \geq 200$  gives a slope of 0.36 against a correlation of 0.95, so the model ranks those instances almost perfectly and declines to move its  $\alpha$  level.

A  $2 \times 2$  ablation separates the two causes. Holding hyperparameters and seed fixed, we vary the feature set, extending the 30-feature baseline with five degeneracy, local-intrinsic-dimension and MST-topology descriptors, and we vary the training support, adding 874 newly generated and solved instances at large  $n$  and high  $\alpha$  to the training split only. Measured by the Line Noise slope at  $n \geq 200$ , features alone move the baseline from 0.294 to 0.569, support alone moves it to 0.332, and the two together reach 0.798. Neither intervention approaches sufficiency alone and the two are not additive, so the deficit is joint. The ladder is anchored on the 30-feature model, whose slope on this slice is 0.294; the released 31-feature model reaches 0.362 on the same slice, so these figures compare the two interventions against each other rather than against the shipped estimator.

We attempted the repair twice against criteria fixed before either candidate was scored, and rejected both. Adding all 874 instances to the training split clears every one of nine gates, moving 2D MAPE from 2.89% to 2.04%, Line Noise MAPE from 10.82% to 5.83% and the slope from 0.366 to 0.838, but two defects disqualify it. The slope is not a property of the model: eight permutations of the row order of the identical training rows, at one seed, span  $[0.637, 0.803]$ , and the reported 0.838 lies above that entire band. And the gain is not a cross-family result, because the augmentation’s  $d = 2$  lattice generator and the benchmark’s grid generator place their sites, spacing, cell centres and  $\pm 5\%$  jitter identically, so half of the 2D gain is measured on the candidate’s own training geometry. A de-contaminated second candidate, re-run over seven seeds fixed in advance, removes that overlap and clears ten of eleven gates at a median slope of 0.735, but forfeits the paper’s one distribution-free dispersion claim to do so: its Holm-adjusted advantage over the asymptotic ratio on TSPLIB reaches only  $p = 0.065$  at the median against 0.0048 for the released model. We did not make that trade. A per-family multiplicative constant, an oracle a deployed estimator does not have, recovers 97.9% of the first candidate’s Line Noise gain by itself, so most of what either arm buys is a per-family offset rather than a repaired response to geometry. The degenerate-geometry failure is therefore diagnosed and not repaired, the released model is unchanged, and the full criteria, seed bands and per-arm statistics are released with the results.

## 4.7 Discussion

Everything above judges GART 2.0 on pointwise accuracy. Many of the downstream uses motivated in Section 1 never consume a single cost in isolation: they compare instances and act on the comparison, asking which candidate node set is cheaper or which route split to keep. For those uses the estimator must *order* instances the way the stored references do even when its absolute numbers are off, so we measure order agreement directly. Spearman’s  $\rho$  correlates the cost *ranks* and Kendall’s  $\tau$  is the concordant minus discordant fraction of instance pairs; both equal 1 for perfectly matching order and 0 for an unrelated one, while the close-pair statistic below is a plain concordant fraction and sits at 0.5 under an unrelated ordering. Global rank correlation is close to 1 for every estimator here, so it separates almost nothing. Table 18 reports both metrics with  $L_{\text{MST}}$  itself as the control. GART 2.0 obtains Spearman  $\rho$ /Kendall  $\tau$  of 0.9994/0.983 on 2D, 1.0000/0.998 on the multidimensional set, and 0.9993/0.987 on TSPLIB EUC\_2D; the  $\alpha = 1$  control obtains 0.9961/0.962, 0.9996/0.986, and 0.9986/0.979. Cross-instance cost varies over orders of magnitude and  $L_{\text{MST}}$  already tracks that variation, so a high  $\rho$  demonstrates nothing about the learned ratio. Every constant multiplier is a monotone transform of  $L_{\text{MST}}$  and therefore scores identically to it, which is why the asymptotic-ratio and  $\alpha = 1$  rows coincide exactly on 2D and TSPLIB.

Close-pair ordering is the discriminating measure. For pairs satisfying  $|L_A - L_B| / \max(L_A, L_B) < 5\%$ , GART 2.0 orders 74.5% of 74,826 2D pairs, 92.4% of 1,647,719 multidimensional pairs, and 72.2% of 54 TSPLIB pairs correctly,

against 71.0%, 69.1%, and 61.1% for the  $\alpha = 1$  control. The multidimensional gain of 23.4 percentage points is the substantive one; the 2D gain of 3.4 points is not, and on TSPLIB 54 pairs cannot support an 11-point claim. At a 10% threshold GART obtains 83.6%, 96.1%, and 81.0%. Every pair set is enumerated exhaustively rather than sampled (Appendix H), and the percentages remain descriptive because the pairs share instances and we compute no block-bootstrap interval over them. GART 1.0 led GART 2.0 on this statistic before the label repair of Section 5.7 and trails it after, at 70.4%, the only ordering change the repair produced anywhere in this paper. Cost accuracy and pair ordering are distinct properties, and 54 pairs resolve none of these gaps.

Two calibration caveats bound the cost-level results. The  $n > 400$  TSPLIB bucket has  $R_\alpha^2 = -0.071$ , verified with both a library implementation and the sum-of-squares definition computed directly; a negative value means the model predicts  $\alpha$  worse than the bucket mean does. That bucket has low target variance and an upward mean prediction offset while its tour-cost MAPE remains 2.89%, so the cost-level result does not remove the ratio-calibration failure. The aggregate margins are also not uniformly significant. We test each comparison on the instances where both estimators produce a finite prediction, using the paired difference in absolute percent error with a 1,000-resample paired bootstrap and a Wilcoxon signed-rank test (Appendix I). On TSPLIB EUC\_2D the difference between GART 2.0 and the asymptotic MST ratio is  $-1.00$  percentage points with a 95% interval of  $[-1.77, -0.22]$  and  $p = 0.0049$ : on that benchmark GART 2.0’s MAPE advantage is small but resolvable, and its larger gain is dispersion, 2.92% SDPE against 4.74%.

Accuracy varies more across geometry than across any other cut of the 2D benchmark. GART 2.0’s MAPE is 1.47% on the isotropic class, 1.79% on the biased class, 1.96% on the two represented geometric-structure generators, 2.24% on the clustered class, 7.07% on the grid generator, and 10.82% on Line Noise. The two worst rows are the two whose geometry the training corpus does not contain, and no represented row exceeds 2.24%, which locates the model’s weakness in unseen geometry, not in unseen size or dimension. On grid the failure is almost pure bias: SDPE is 1.92, the lowest GART 2.0 records on any row of Table 17, while the signed error carries the whole of that MAPE and is positive on every instance. One of the five other rows in that table is more accurate on it than GART 2.0 is, the  $\alpha = 1$  floor at 4.54%, and grid is the only row anywhere in this paper on which the  $\alpha = 1$  floor beats the shipped model.

Cost is dominated by feature extraction, not by the ensemble. In the TSPLIB timing log GART 2.0 spends 92.90% of its wall time on feature extraction and MST construction against 5.03% on LightGBM inference, so the tree traversal is not the target for any speed work and the ensemble size is not what the cost claim turns on. The Time column of Table 23 puts every estimator on one protocol: one estimator per process, a single thread, the median of 11 repeats. On that protocol GART 2.0 costs 6.12 ms over the full EUC\_2D set, matching GART 1.0, roughly twice the 2.57–3.08 ms of the rows that build an MST or a Hilbert sort, and roughly an order of magnitude over the 0.645–0.880 ms of the two classical closed forms, which compute a convex hull and then evaluate an algebraic expression. That ordering is the price of the feature set and it is stable across the size range; at  $n > 400$  GART 2.0 is the most expensive row in the table at 20.54 ms.

Those multiples are the wrong comparison for the deployment case, because the alternative in that case is a solve. On the multidimensional benchmark the median prediction and the median reference-tour generation cost the same, 121 ms against 122 ms, since most of those instances are small enough to solve outright and an estimator buys nothing. The argument holds only where the solve is expensive, and there it holds decisively: at  $n \in [501, 1000]$  the reference tour takes 3,799 ms against GART 2.0’s 460 ms, and on TSPLIB d18512, at  $n = 18,512$  and more than  $18\times$  the largest training instance, GART 2.0 predicts in 239 ms. Every row in this accounting is an estimator, and none of them certifies anything. Section 5 runs the accounting again against a comparator that does.

## 5 Comparison Against a Certified Lower Bound

Every baseline of Section 4.1 is an estimator: it returns a number that carries no guarantee and is scored only by how close that number lands. This section admits a comparator of a different kind, the Held–Karp 1-tree lower bound,

and uses it to establish where a learned estimator is the right instrument and where it is not. The bound is the hardest available test, because it consumes no training corpus, certifies its own output, and is defined on every instance in this paper. We score it on all four benchmarks, on cost and accuracy together. On the aggregate of every one of them the bound is both cheaper and more accurate than GART 2.0, on TSPLIB EUC\_2D once scaled by a single training-split constant, so the estimator is off the cost/accuracy Pareto front of all four and no aggregate in this paper argues otherwise. What the aggregates conceal is the one property that separates the two families, which is how each scales in  $n$ : the ascent is quadratic and the planar estimator is not. That difference, and not any error figure, fixes the operating regime we recommend at the end of this section.

### 5.1 The Held–Karp 1-Tree Bound

Two properties separate this row from every other row in this paper, and both follow from the same relaxation. Fix a node  $s$ . A *1-tree* is a spanning tree on  $V \setminus \{s\}$  together with any two edges incident to  $s$ ; it has  $n$  edges and total degree  $2n$ , and every Hamiltonian tour is a 1-tree, so the minimum 1-tree — obtained by taking the two cheapest edges at  $s$  — is a relaxation of the tour problem. Introduce node potentials  $\pi \in \mathbb{R}^n$  and transformed costs  $c_{ij}^\pi = c_{ij} + \pi_i + \pi_j$ . Under  $c^\pi$  every tour costs exactly  $L_{\text{TSP}} + 2 \sum_i \pi_i$ , because every node has degree two, so for *any* real  $\pi$

$$w(\pi) = L_{\text{1tree}}(\pi) - 2 \sum_{i \in V} \pi_i \leq L_{\text{TSP}}, \quad (2)$$

which is the Held–Karp bound [Held and Karp, 1970, 1971]. The map  $w$  is concave and piecewise linear, a subgradient at  $\pi$  is  $g_i = \deg_i(\pi) - 2$ , and  $\max_\pi w(\pi)$  is approached by subgradient ascent. We write  $w(\pi_k)$  for the bound after  $k$  ascent iterations and treat  $k$  as an explicit knob trading cost against tightness. Validity needs neither  $\pi \geq 0$  nor  $c^\pi \geq 0$ : the argument above is sign-free, and the spanning tree is built on the perturbed costs directly.

The first property is the certificate. Eq. (2) holds at every  $\pi$ , so  $w(\pi_k)$  is a *certificate* whatever the ascent did: the printed number is a proven lower bound on that instance’s optimal tour, and a reader can verify it from the 1-tree alone. No other row in this paper certifies anything about its own output, GART 2.0 included. The second is that the bound consumes no training corpus, so no question of coverage, distribution shift, or leakage arises about it at all.

The certificate is paid for with one-sidedness. Because  $w(\pi_k)$  never exceeds the optimum, its error against a reference tour is a duality gap plus whatever that reference is itself above the optimum, and not the two-sided spread of an estimator. Turning the bound into an estimator requires one fitted scalar, and we report that row too: the *calibrated* bound  $c_k w(\pi_k)$ , where  $c_k$  is a single scalar fitted per budget on the planar training split and never on any scored instance. That row is not a certificate, since  $c_k > 1$  pushes some predictions above the optimum, and it is not training-free. It is the like-for-like comparator against a trained estimator, and it is the stronger of the two rows on every corpus scored below.

Both ascents run on all four benchmarks, and each headline table reports the arm that leads on its own corpus. The planar tables use the Volgenant–Jonker step schedule with Helsgaun’s direction smoothing [Volgenant and Jonker, 1982, Helsgaun, 2000]. The multidimensional tables use a Polyak step [Polyak, 1969] against a constructive upper bound, a nearest-neighbour tour improved by 2-opt and built from coordinates alone, with the stored label never read during the ascent. Neither ascent is proved to attain  $\max_\pi w(\pi)$ , so every accuracy figure below is a floor on what this family reaches and not a ceiling. The two arms disagree: on the multidimensional split the Volgenant–Jonker arm returns the higher bound on 4.30% of instances, and taking the per-instance maximum of both, at the cost of running both, improves the converged multidimensional error from 0.0518% to 0.0486%. Validity was checked against exhaustively enumerated optima on small instances and against a second exact solver, with no violation of Eq. (2) anywhere.

## 5.2 Results on TSPLIB95 EUC\_2D

Cost here is measured on the protocol of Table 23, one estimator per process inside a single quiet window, and both arms are compared on the 77 of the 78 EUC\_2D instances that the bound’s own dense kernel covers. *Corpus median* names one statistic wherever this paper uses it: the cost of the typical instance, the median over the corpus of each instance’s own median over its repeats, which is what the Time column of Table 23 and of every ladder in this section carries. The accuracy printed beside it is a MAPE and therefore a mean, so the word describes the cost column and not the error. Table 3 gives the ladder. The one excluded instance is reported at the end of this subsection.

Read on the corpus median, the raw certified bound first matches GART 2.0’s MAPE at an ascent budget of 50, where it costs 1.46 times GART 2.0 for a margin of 0.0041 percentage points, and where the paired win rate is exactly 50.0%. The calibrated row settles what the raw row only balances. At a budget of 25 it costs 0.90 times GART 2.0 and reaches 2.00% MAPE against 2.56%: strict domination on both axes, on the same instances and the same protocol. That multiple is a ratio of typical-instance costs, and a median of medians does not compose: the second aggregation that does, the sum of those same per-instance medians over the corpus, puts the same pair at 24.94 instead, because the bound is cheap on the small instances that set the median and dear on the large ones that set the total. The first statistic is what a caller holding one instance faces and the second is the cost of running the benchmark end to end; the ladders in this section report the first. GART 2.0 is therefore not on the cost/accuracy Pareto front of this corpus median once the family is admitted.

Splitting by size shows what the median hides, and the split rather than the median is this section’s useful output. On the 23 smallest instances the calibrated bound at that same budget costs 0.16 times GART 2.0 at 1.69% MAPE against 2.01%, and even the raw uncalibrated certificate reaches 22.2% lower error at 0.46 times the cost. In the middle bucket nothing dominates, and the raw bound needs the top of the ladder to match at all. In the largest bucket the raw bound matches at 6.65 times the cost and again nothing dominates. The reversal is monotone in  $n$ , which is why the output of this comparison is a rule keyed to instance size. Table 3 carries that reversal bucket by bucket: the bound is cheaper and tighter on the smallest instances, neither row dominates in the middle bucket, and on the largest the bound’s accuracy is only ever paid for in time. A reader deciding between the two methods should locate their own instance size on that axis before reading any aggregate.

The cost column carries two disclosures. The bound is far more load-sensitive than GART 2.0, its median rising by a factor of 1.25 between a quiet and a noisy window at the crossing budget against 1.05 for GART 2.0, so every published repeat is from one quiet window and the noisy repeats are retained only as the control; mixing the two would have overstated the bound’s cost and flattered GART 2.0. And the excluded instance is real cost rather than a hidden exclusion: on d18512, whose node count is above the bound’s own dense kernel switch, the bound costs 167 times GART 2.0 at the crossing budget and 2,078 times at the top of the ladder, where that single instance takes 493 seconds against GART 2.0’s 237 milliseconds on those same three repeats; the 239 ms quoted for this instance in the Discussion above is the published harness figure for the same work under a different protocol.

## 5.3 Results on the Multidimensional Benchmark

On the multidimensional benchmark the result is not a trade-off. Table 4 reports the ladder by dimension group. At an ascent budget of 200 the bound reaches 0.111% MAPE against GART 2.0’s 0.618%, a factor of 5.56, at 0.71 times the cost, and it wins the paired comparison on 82.9% of instances. It first overtakes GART 2.0 at a budget of 100, where it costs 0.40 times as much. Those ratios are medians over a serial single-thread timing sample; weighted by the corpus composition instead, the cost ratio at a budget of 200 is 0.54.

The domination is not an artefact of one dimension group. It holds at  $d \in [4, 10]$ , at  $d \in [15, 50]$  and at  $d = 100$ , and it is widest exactly where the estimator was never trained: at  $d = 100$  a budget of 500 costs 0.35 times GART 2.0 and is

Table 3: TSPLIB95 EUC\_2D cost/accuracy ladder, 77 instances matched between both arms. Time is the median per-instance time in milliseconds on the solo protocol of Table 23;  $\times$  is that time over GART 2.0’s in the same bucket. Raw is the certified bound  $w(\pi_k)$ ; Cal. is  $c_k w(\pi_k)$  with  $c_k$  fitted per budget on the planar training split. Bold marks every cell where the bound family is both cheaper and more accurate than GART 2.0 in that bucket.

| Row       | $n \in [51, 150], N = 23$ |          |             |             | $n \in [151, 400], N = 16$ |          |       |      | $n > 400, N = 38$ |          |      |      |
|-----------|---------------------------|----------|-------------|-------------|----------------------------|----------|-------|------|-------------------|----------|------|------|
|           | ms                        | $\times$ | Raw         | Cal.        | ms                         | $\times$ | Raw   | Cal. | ms                | $\times$ | Raw  | Cal. |
| GART 2.0  | 3.73                      | 1.00     | 2.01        |             | 4.55                       | 1.00     | 2.38  |      | 18.29             | 1.00     | 2.97 |      |
| $k = 0$   | 0.19                      | 0.05     | 11.83       | 3.12        | 0.45                       | 0.10     | 11.31 | 3.59 | 24.49             | 1.34     | 9.13 | 4.01 |
| $k = 10$  | 0.37                      | 0.10     | 8.47        | 2.04        | 0.91                       | 0.20     | 9.85  | 3.76 | 43.93             | 2.40     | 7.63 | 2.53 |
| $k = 25$  | 0.59                      | 0.16     | 2.91        | <b>1.69</b> | 1.59                       | 0.35     | 4.84  | 3.09 | 72.52             | 3.97     | 3.48 | 1.72 |
| $k = 50$  | 0.96                      | 0.26     | 2.09        | <b>1.68</b> | 2.79                       | 0.61     | 3.82  | 3.03 | 121.59            | 6.65     | 2.31 | 1.56 |
| $k = 100$ | 1.72                      | 0.46     | <b>1.56</b> | <b>1.20</b> | 5.16                       | 1.13     | 2.97  | 2.29 | 222.22            | 12.15    | 1.96 | 1.33 |
| $k = 200$ | 3.22                      | 0.86     | <b>1.57</b> | <b>1.22</b> | 10.10                      | 2.22     | 2.96  | 2.28 | 427.85            | 23.40    | 1.88 | 1.26 |
| $k = 500$ | 7.88                      | 2.11     | 0.97        | 0.74        | 25.65                      | 5.64     | 1.93  | 1.42 | 1120.37           | 61.27    | 1.62 | 1.12 |

140.8 times more accurate. The one group GART 2.0 holds is  $d \in \{2, 3\}$ , where matching its accuracy costs the bound 1.27 times as much.

It is, however, an artefact of instance size, and the aggregate above conceals that entirely. Of the 16,846 scored instances 10,569, or 62.74%, carry at most 100 nodes, a regime in which the relaxation is very nearly exact and very nearly free; a corpus mean over that composition is largely a statement about small instances rather than about the estimator. Stratifying by size as well as by dimension separates the two. We draw 20 instances per (dimension, size band) cell, matched on size across dimensions so that the cost ratio reports a dimension effect and not a size effect, and price the ladder on them under the protocol of Table 3; accuracy in each cell is measured on every instance both methods score, between 108 and 2,209 of them. Table 5 reports the result. GART 2.0 is undominated in 10 of the 18 cells, and the eight in which the bound is both cheaper and more accurate are the six at  $n \in [20, 100]$  together with the two at  $d = 100$ . At  $n \geq 200$  the estimator is on the cost/accuracy front at every dimension it was trained on, and it is dominated only at  $d = 100$ , the dimension withheld from training, where its own accuracy degrades to 0.824% against 0.062% at  $d = 50$ .

At deployment scale the margin is not narrow. In the  $n \in [600, 1000]$  band GART 2.0 reaches 0.429% MAPE at 10.0 ms at  $d = 2$ , where the tightest rung we priced,  $k = 500$ , is both dearer and looser at 0.784% and 246.1 ms; 0.332% at 25.7 ms at  $d = 3$ , against 0.450% at 143.9 ms for  $k = 200$ ; and 0.251% at 58.2 ms at  $d = 4$ , against 0.291% at 86.2 ms for the same budget. At  $d = 10$  and  $d = 50$  no rung dominates it in either direction and the cell is a genuine trade-off. The mechanism is the one Section 5.5 states: the bound is  $\Theta(kn^2)$  in every dimension, so the cost of matching a fixed accuracy grows quadratically while the estimator’s does not, and the size at which the ordering reverses is what moves with  $d$  — roughly 200 nodes in the plane, roughly 600 at  $d = 3$ .

Three checks support the finding. The ascent never reads the stored label, so the comparison is not circular: its upper bound is a nearest-neighbour tour improved by 2-opt from coordinates alone, and supplying the released optimum in its place converges faster early and to the same value. On the 3,069 instances whose label came from an exact solver, where the target is a proven optimum rather than a heuristic tour, GART 2.0 scores 1.03% against the bound’s 0.046% at a budget of 200. And the relaxation closes exactly, returning the optimum rather than a bound, on 37.2% of the split. That last figure is also the mechanism: pairwise distances concentrate as  $d$  grows, so the integrality gap this relaxation is usually described by is a planar fact rather than a general one.

Roughly one instance in seven exhausts the largest budget we ran, so at the top of the size range the reported accuracy is a floor and not a converged value. Two limits bound the size-stratified reading in particular. The cost figure in each

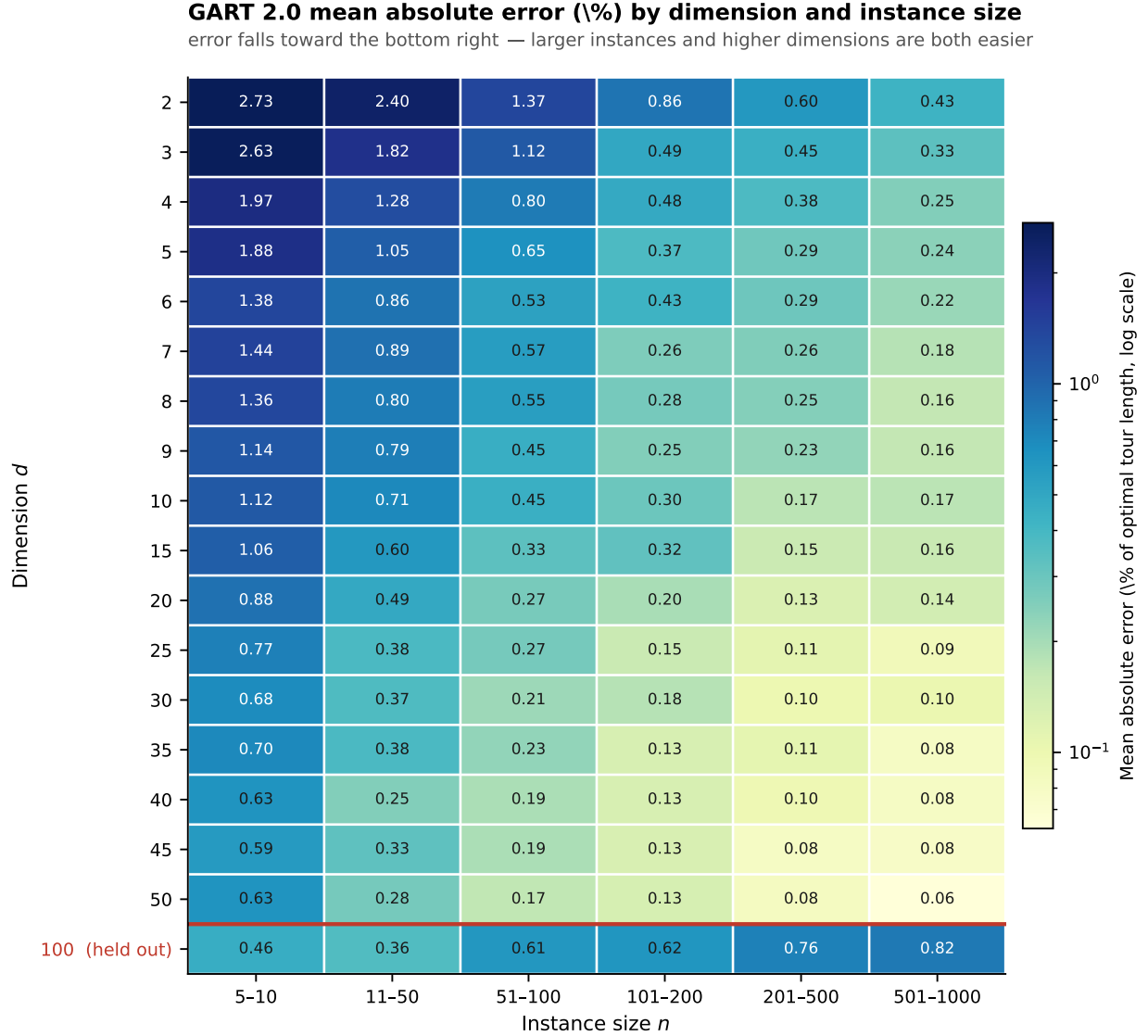


Figure 4: GART 2.0’s mean absolute error on the multidimensional benchmark, as a grid in the two arguments the feature set was built to carry. Each cell is the mean over every scored instance of that dimension and size band; colour is on a logarithmic scale. The field runs in one direction: error falls as instances grow and again as dimension grows, so the corner where an exact solve is least affordable is the corner the estimator predicts best. The bottom row is  $d = 100$ , withheld from training entirely, and it is the one row that breaks the pattern, rising with instance size instead of falling. That row is the boundary Section 5.8 states, drawn rather than argued.

cell is a median over 20 timed instances against an accuracy figure over the whole cell, so the cost column is the weaker of the two and a cell whose verdict turns on a few milliseconds should be read as a trade-off rather than as a ranking. And the estimator’s own asymptotics change with dimension: `compute_mst` dispatches to the Delaunay construction only for  $d \leq 3$  and to a dense kernel from  $d = 4$  upward, so above three dimensions both families are quadratic in  $n$  and the separation is a constant rather than an exponent. That constant favours the estimator at  $n \geq 200$  and the bound below it, which is what Table 5 measures; it is not a complexity argument and we do not extrapolate it past  $n = 1000$ , the largest size this corpus carries.

Table 4: Multidimensional benchmark cost/accuracy ladder, all 16,846 held-out instances, Polyak ascent. Time is the median in milliseconds on a serial single-thread timing sample;  $\times$  is that time over GART 2.0’s in the same group. Bold marks every budget at which the bound is both cheaper and more accurate than GART 2.0 in that group, that is, every budget at which GART 2.0 is strictly dominated.

|           | $d \in \{2, 3\}, N = 1,295$ |          |        | $d \in [4, 10], N = 4,534$ |          |              | $d \in [15, 50], N = 5,170$ |          |              | $d = 100, N = 5,847$ |          |              |
|-----------|-----------------------------|----------|--------|----------------------------|----------|--------------|-----------------------------|----------|--------------|----------------------|----------|--------------|
| Row       | ms                          | $\times$ | MAPE   | ms                         | $\times$ | MAPE         | ms                          | $\times$ | MAPE         | ms                   | $\times$ | MAPE         |
| GART 2.0  | 5.87                        | 1.00     | 1.453  | 3.30                       | 1.00     | 0.728        | 3.59                        | 1.00     | 0.340        | 3.26                 | 1.00     | 0.593        |
| $k = 0$   | 2.43                        | 0.24     | 12.190 | 0.13                       | 0.04     | 7.446        | 0.37                        | 0.07     | 3.834        | 0.23                 | 0.07     | 2.133        |
| $k = 10$  | 3.51                        | 0.31     | 7.389  | 0.24                       | 0.08     | 4.500        | 0.36                        | 0.12     | 2.164        | 0.26                 | 0.08     | 1.026        |
| $k = 25$  | 5.18                        | 0.47     | 4.746  | 0.38                       | 0.13     | 3.240        | 0.55                        | 0.17     | 1.461        | 0.37                 | 0.12     | 0.691        |
| $k = 50$  | 7.37                        | 0.69     | 3.177  | 0.62                       | 0.21     | 1.972        | 0.92                        | 0.27     | 0.870        | 0.49                 | 0.15     | <b>0.412</b> |
| $k = 100$ | 12.90                       | 1.27     | 1.388  | 1.21                       | 0.39     | <b>0.692</b> | 1.95                        | 0.56     | <b>0.301</b> | 0.76                 | 0.23     | <b>0.147</b> |
| $k = 200$ | 22.97                       | 2.18     | 0.557  | 2.15                       | 0.69     | <b>0.161</b> | 3.30                        | 0.95     | <b>0.053</b> | 1.19                 | 0.37     | <b>0.025</b> |
| $k = 500$ | 50.80                       | 4.92     | 0.384  | 4.51                       | 1.38     | 0.067        | 6.05                        | 1.75     | 0.011        | 1.12                 | 0.35     | <b>0.004</b> |

Table 5: The multidimensional frontier stratified by instance size as well as by dimension.  $N$  is the number of instances in the cell on which accuracy is measured, every one that both methods score; the cost column is the median of a separate 20-instance sample drawn matched on size across dimensions, so a verdict resting on a few milliseconds should be read as a trade-off. “Dominates” means strictly better on cost and accuracy at once, at the listed ascent budgets. The bound is the Polyak arm.

| $d$ | $n$ band     | $N$   | GART 2.0 MAPE (%) | GART 2.0 (ms) | Verdict against the ladder                                |
|-----|--------------|-------|-------------------|---------------|-----------------------------------------------------------|
| 2   | [20, 100]    | 243   | 1.829             | 2.9           | bound dominates ( $k = 200, 500$ )                        |
|     | [200, 500]   | 108   | 0.663             | 5.7           | <b>GART 2.0 dominates</b> ( $k = 25, 50, 100, 200, 500$ ) |
|     | [600, 1,000] | 135   | 0.429             | 10.0          | <b>GART 2.0 dominates</b> ( $k = 25, 50, 100, 200, 500$ ) |
| 3   | [20, 100]    | 243   | 1.431             | 3.4           | bound dominates ( $k = 100, 200, 500$ )                   |
|     | [200, 500]   | 108   | 0.459             | 12.2          | <b>GART 2.0 dominates</b> ( $k = 100, 200$ )              |
|     | [600, 1,000] | 135   | 0.332             | 25.7          | <b>GART 2.0 dominates</b> ( $k = 25, 50, 100, 200$ )      |
| 4   | [20, 100]    | 243   | 1.015             | 2.4           | bound dominates ( $k = 200$ )                             |
|     | [200, 500]   | 108   | 0.403             | 14.2          | trade-off at every rung                                   |
|     | [600, 1,000] | 135   | 0.251             | 58.2          | <b>GART 2.0 dominates</b> ( $k = 200$ )                   |
| 10  | [20, 100]    | 243   | 0.566             | 2.4           | bound dominates ( $k = 200$ )                             |
|     | [200, 500]   | 108   | 0.204             | 14.7          | trade-off at every rung                                   |
|     | [600, 1,000] | 135   | 0.173             | 100.2         | trade-off at every rung                                   |
| 50  | [20, 100]    | 243   | 0.214             | 2.8           | bound dominates ( $k = 200$ )                             |
|     | [200, 500]   | 108   | 0.095             | 22.7          | trade-off at every rung                                   |
|     | [600, 1,000] | 134   | 0.062             | 114.3         | trade-off at every rung                                   |
| 100 | [20, 100]    | 2,209 | 0.501             | 3.4           | bound dominates ( $k = 100, 200$ )                        |
|     | [200, 500]   | 977   | 0.723             | 26.8          | bound dominates ( $k = 50, 100$ )                         |
|     | [600, 1,000] | 1,185 | 0.824             | 120.4         | bound dominates ( $k = 50, 100$ )                         |



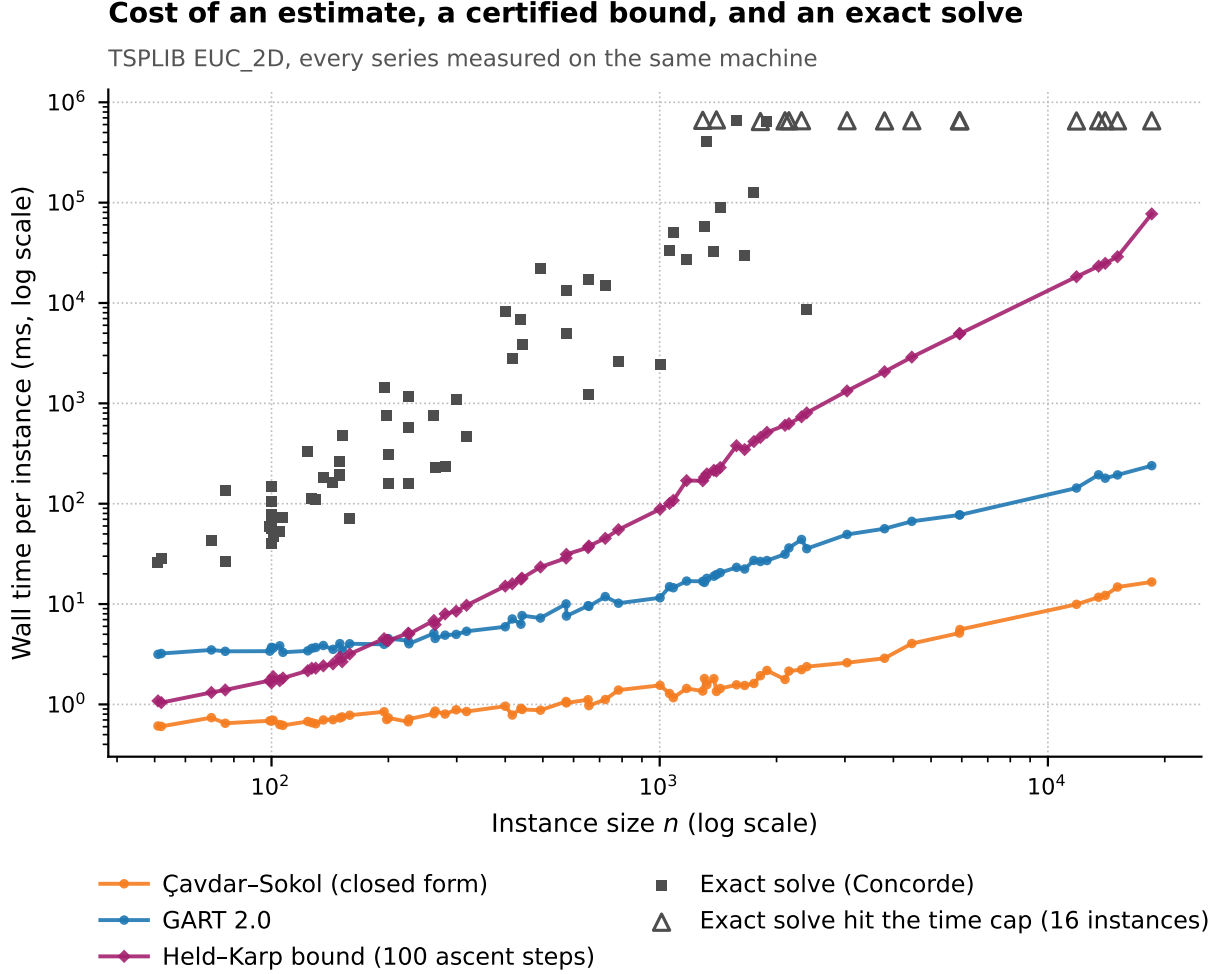


Figure 5: What each family costs per instance on TSPLIB95 EUC\_2D, against instance size, both axes logarithmic. Every series is measured on the same machine, so the comparison needs no cross-hardware correction. The closed form is flat because it reads a handful of summary statistics; GART 2.0 rises gently because its planar extraction is subquadratic; the certified bound rises steeply because it pays  $\Theta(kn^2)$ ; and the exact solve is scattered rather than joined because branch-and-cut time varies by orders of magnitude between instances of the same size. Open triangles are the runs that reached the wall-clock cap without proving optimality, so the exact arm is a lower bound on its own cost. The four families are separated by roughly five orders of magnitude at the largest instance, and only the estimator and the closed form stay inside a real-time budget across the whole range.

#### 5.4 Results on the 2D Diverse and Non-Euclidean Benchmarks

The two subsections above price the bound on the two benchmarks the estimator was designed around. The remaining two carry an accuracy column and no cost column, and the ladders here complete the four. Both are measured on the protocol of Table 23, one estimator per process, with parsing and warm-up outside the clock, and both arms are read off one checkpointed ascent per instance as in Table 3.

Table 6 gives the 2D diverse ladder. GART 2.0 costs 5.00 ms on the typical instance at 2.889% MAPE. The raw certified bound passes it under the Volgenant–Jonker step at a budget of 50, at 0.23 times the cost for 2.702% MAPE and a paired win rate of 54.8%; the calibrated row passes it a rung earlier, at 0.14 times the cost for 2.564%. The reversal is monotone in  $n$  and steeper than the one Section 5.2 reports, because this corpus reaches down to five nodes. On the 360 instances of its smallest size bucket, as Table 22 defines those buckets, the raw bound at a budget of 25 reads 0.439%

MAPE against GART 2.0's 3.212% at 0.07 times the cost, and wins 95.0% of the paired comparisons. On the 610 instances of its largest, nothing on the ladder dominates: the raw bound is more accurate than GART 2.0 only at the top budget, and there it costs 26.08 times as much. This is the rule of Section 5.2 read over a wider range of  $n$ , not a second rule.

Table 7 gives the non-Euclidean ladder, where the estimator needs the MDS embedding of Section 6 and the bound reads the distance matrix directly. Over the 29 instances both methods score, GART 2.0 costs 4.88 ms, of which the embedding is 9.21%, at 3.814% MAPE. The Polyak step at a budget of 500 reaches 0.509% MAPE at 0.69 times that cost and wins all 29 paired comparisons; at a budget of 200 it reads 0.762% for 0.34 times the cost, again on every instance. The cheapest domination anywhere in this study is on this corpus: the Volgenant–Jonker step at a budget of 25 costs 0.065 times GART 2.0 and reads 2.953% against 3.814%. The accuracy factor Section 6 quotes is therefore bought below the estimator's own cost rather than above it.

The two step rules do not order the same way on the two corpora. Volgenant–Jonker leads Polyak at every budget below 500 on the 2D corpus, 3.606% against 7.515% at a budget of 25, and Polyak leads it from a budget of 200 on the non-Euclidean corpus, 0.762% against 1.228%. The assignment of Section 5.1 is therefore not arbitrary, but the crossover is a property of the corpus rather than of the budget, and neither schedule dominates the other across these four benchmarks.

Four checks run over every cell of both tables and none fails. Repeating each timed call returns zero spread across all 18,060 instance–budget pairs of the 2D corpus and all 217 of the non-Euclidean one, so no cost above is read off a single noisy sample. The timed call returns the same bound the accuracy sweep scores, bit-identical on the Volgenant–Jonker arm and within one unit in the last place of a double on the Polyak arm, which is what allows the cost column and the accuracy column of one row to be read together. Every bound printed is at or below the reference cost of its own instance, and every ladder is monotone in  $k$ , over all 2,580 series on the 2D corpus and all 31 on the non-Euclidean one. On the instances small enough to solve outright the bound is checked against an exact dynamic program as well, again with no violation.

Two disclosures bound the cost columns, and both were measured rather than assumed. First, these sessions did not run on a quiet machine, and the two arms are not equally sensitive to that: a paired control against the published quiet-window measurements puts GART 2.0 at 1.48 times its published cost and the bound at 1.10 to 1.14 times, so every cost multiple printed above is between 0.74 and 0.77 of what the same pair would read on a quiet box. That direction runs against the bound rather than for it, and correcting for it costs the bound exactly one of the dominations reported here, the raw Volgenant–Jonker row at a budget of 200 on the 2D corpus, whose 0.79 becomes 1.03. Second, the checkpointed protocol is not flattering the bound either: a control that runs the shipped single-budget call once per rung and shares nothing between rungs costs 0.77 to 0.95 times the checkpointed ladder on the 2D corpus and 0.90 to 1.14 times on the non-Euclidean one, at every budget that contains an ascent.

## 5.5 Complexity Across the Estimator Families

Table 8 states the dominant term of every family in this paper as the code executes it, not as the underlying algorithm is usually quoted, together with whether the row certifies its own output and whether it needs a training corpus. Three entries differ from the textbook reading. The classical closed forms are dominated by a row-deduplication pass rather than by their arithmetic. The convex-hull blow-up this paper cites as a motivation never occurs at runtime, because the implementation caps hull construction at three dimensions and falls back to a bounding box above it. And the 1-tree cannot use the Delaunay shortcut the MST family relies on in the plane: the node potentials break the Euclidean metric, so after the first iteration the perturbed minimum spanning tree is not a subgraph of the triangulation. That is measured rather than assumed. At 144 potential vectors drawn from real ascent trajectories the Delaunay-restricted 1-tree is heavier than the exact one at 141 of them, so the shortcut is unavailable at 141 of the 144, and that is why the bound sits a complexity class above the MST family and not a constant factor above it.

Table 6: 2D diverse benchmark cost/accuracy ladder, all 2,580 instances, both subgradient step rules. Time is the median per-instance time in milliseconds on the solo protocol of Table 23;  $\times$  is that time over GART 2.0’s on the same instances. Raw is the certified bound  $w(\pi_k)$ ; Cal. is  $c_k w(\pi_k)$  at the same  $c_k$  Table 3 uses. Bold marks every cell that is both cheaper and more accurate than GART 2.0, that is, every rung at which GART 2.0 is strictly dominated.

| Row       | Volgenant–Jonker step                  |          |              |              | Polyak step |          |              |              |
|-----------|----------------------------------------|----------|--------------|--------------|-------------|----------|--------------|--------------|
|           | ms                                     | $\times$ | Raw          | Cal.         | ms          | $\times$ | Raw          | Cal.         |
| GART 2.0  | 5.00 ms at 1.00 $\times$ ; 2.889% MAPE |          |              |              |             |          |              |              |
| $k = 0$   | 0.23                                   | 0.046    | 13.938       | 5.654        | 0.31        | 0.062    | 13.938       | 5.654        |
| $k = 10$  | 0.44                                   | 0.087    | 8.352        | 4.257        | 0.49        | 0.098    | 10.572       | 6.872        |
| $k = 25$  | 0.70                                   | 0.14     | 3.606        | <b>2.564</b> | 0.73        | 0.15     | 7.515        | 6.393        |
| $k = 50$  | 1.15                                   | 0.23     | <b>2.702</b> | <b>2.330</b> | 1.12        | 0.22     | 5.526        | 4.978        |
| $k = 100$ | 2.05                                   | 0.41     | <b>2.048</b> | <b>1.763</b> | 1.88        | 0.38     | 3.194        | <b>2.788</b> |
| $k = 200$ | 3.95                                   | 0.79     | <b>1.883</b> | <b>1.618</b> | 3.36        | 0.67     | <b>1.878</b> | <b>1.650</b> |
| $k = 500$ | 9.42                                   | 1.88     | 1.274        | 1.102        | 7.15        | 1.43     | 1.447        | 1.331        |

Table 7: Non-EUC\_2D TSPLIB95 cost/accuracy ladder, the 29 instances both methods score, both subgradient step rules. Columns are as in Table 6; GART 2.0’s time includes the MDS embedding of Section 6, which the bound does not need. The calibration constant  $c_k$  is the planar one, applied here out of its fitted domain, so the Cal. columns are indicative and the Raw columns are the certified result.

| Row       | Volgenant–Jonker step                  |          |              |              | Polyak step |          |              |              |
|-----------|----------------------------------------|----------|--------------|--------------|-------------|----------|--------------|--------------|
|           | ms                                     | $\times$ | Raw          | Cal.         | ms          | $\times$ | Raw          | Cal.         |
| GART 2.0  | 4.88 ms at 1.00 $\times$ ; 3.814% MAPE |          |              |              |             |          |              |              |
| $k = 0$   | 0.03                                   | 0.007    | 15.606       | 6.181        | 0.05        | 0.011    | 15.606       | 6.181        |
| $k = 10$  | 0.17                                   | 0.035    | 8.150        | <b>3.189</b> | 0.16        | 0.033    | 10.087       | 5.481        |
| $k = 25$  | 0.32                                   | 0.065    | <b>2.953</b> | <b>1.784</b> | 0.28        | 0.057    | 6.501        | 5.053        |
| $k = 50$  | 0.56                                   | 0.11     | <b>1.783</b> | <b>1.436</b> | 0.49        | 0.100    | 3.817        | <b>3.129</b> |
| $k = 100$ | 1.10                                   | 0.23     | <b>1.269</b> | <b>1.037</b> | 0.92        | 0.19     | <b>1.990</b> | <b>1.589</b> |
| $k = 200$ | 2.15                                   | 0.44     | <b>1.228</b> | <b>1.018</b> | 1.66        | 0.34     | <b>0.762</b> | <b>0.581</b> |
| $k = 500$ | 5.27                                   | 1.08     | 1.139        | 0.995        | 3.38        | 0.69     | <b>0.509</b> | <b>0.492</b> |

That class difference is the separation that survives everything else in this section, and Figure 5 is what it looks like measured. In the plane GART 2.0 is  $\Theta(n \log n)$  and the bound is  $\Theta(kn^2)$ , and the measured log–log cost exponents in  $n$  agree: 0.975 for GART 2.0 above a thousand nodes against 2.08–2.13 for the bound over the same window. The consequence is a ratio that grows rather than a fixed multiple. GART 2.0 costs a flat 2.0 to 2.3 times the MST-ratio family across four orders of magnitude in  $n$ , which is the signature of a shared complexity class; against the 1-tree at a fixed budget the ratio moves from 7.7 at a thousand nodes to 211 at sixteen thousand. The bound’s advantage at small  $n$  is a statement about constants and the estimator’s advantage at large  $n$  is a statement about growth, and only the second is stable.

The separation is planar, and we do not extend it. On the multidimensional benchmark GART 2.0’s own measured cost exponent in  $n$  is 2.02 at  $d \in [4, 10]$  and 1.82 at  $d \in [15, 50]$ , against 1.785–1.889 for the bound, because the MST construction falls back to a dense all-pairs kernel from  $d = 4$  upward. Only at  $d \in \{2, 3\}$  is that exponent 0.56. On that benchmark both families are quadratic, the separation is a constant, and the constant points the wrong way.

## 5.6 Comparison Against an Exact Solver

This paper’s cost argument has always had an unbounded anchor above it, and that anchor is a floor. We measured Concorde on this machine over the whole EUC\_2D benchmark, on the same box as every other number on the cost axis, so no cross-hardware correction is needed. It proves optimality on most of the corpus in well under a second and

Table 8: Cost class of every estimator family in this paper, as the released code executes it rather than as the algorithm is usually quoted, with whether the row certifies its own output and whether it consumes a training corpus.  $k$  is the 1-tree ascent budget and  $h$  the Hilbert order. Derivations, and the timing sweeps behind the rows that are measured rather than read off the source, are released with the results.

| Family                                   | Dominant term as implemented                                                      | Certifies?                            | Needs training?                |
|------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------|--------------------------------|
| Classical closed forms                   | $\Theta(d n \log n)$ ; a row-deduplication pass, not the arithmetic               | No                                    | Constants fitted in the source |
| Constant multiples of $L_{\text{MST}}$   | $\Theta(n \log n)$ at $d = 2$ (Delaunay); $\Theta(n^2 d + n^2 \log n)$ otherwise  | Lower bound only at $\alpha = 1$      | The two calibrated rows only   |
| Space-filling curve (Hilbert)            | $\Theta(n d h)$ interpreted, plus an $O(n \log n)$ sort                           | Yes, an upper bound: it builds a tour | No                             |
| Learned estimators on this feature block | Feature cost as the MST family above; head $O(1)$ in $n$                          | No                                    | Yes                            |
| GART 1.0                                 | $\Theta(n^2 d)$ time and $\Theta(n^2)$ memory; a dense matrix for two scalars     | No                                    | Yes                            |
| Held–Karp 1-tree, budget $k$             | $\Theta(k n^2 d)$ ; no Delaunay path exists after the first iteration             | Yes, a lower bound                    | No                             |
| Calibrated 1-tree                        | As above                                                                          | No                                    | One scalar per budget          |
| Exact solver                             | $\Theta(n^2 2^n)$ for the Held–Karp DP; branch-and-cut admits no polynomial bound | Yes, exact                            | No                             |

takes minutes on the rest; the largest instances reach the wall-clock cap without proving anything, which makes the exact arm’s cost a lower bound rather than a measurement. Figure 5 draws it beside the estimator, the closed form and the certified bound, and the four families separate by roughly five orders of magnitude at the largest instance. The load-bearing statement is the complexity rather than the ratio: branch-and-cut over an exponential family of subtour constraints admits no polynomial bound, and the Held–Karp dynamic program [Held and Karp, 1962] it falls back to on small instances is  $\Theta(n^2 2^n)$ , so every row of Table 8 is asymptotically cheaper than an exact solve and being cheaper than one distinguishes nothing among them. The comparison that discriminates is the one against the 1-tree, which is polynomial, certified, and in the same cost decade as the estimators.

## 5.7 Label Refutation and Repair

A certified lower bound is also an audit instrument. If  $B$  is a proven lower bound on an instance’s optimum and  $L$  is its stored label, then  $B > L$  by more than the quantisation slack of the metric the solver was handed refutes that label outright, with no tour entering the test. Applying it to all 108,956 labelled instances in this project, with each solver’s own integer scale factor reverse-engineered and verified against the released labels, refutes 40 of them, and 36 of those sit in a set this paper scores. By split they are 35 in the multidimensional test partition, one on TSPLIB, two in training and two in validation, and none at all on the 2D benchmark; the worst overshoots its label by 4.13%. The overlap with Section 3.3 is near-total, since 39 of the 40 lie inside that section’s 184 inconsistent-tour instances, so the certificate found no new population. What it did was upgrade part of the old one from “the stored tour disagrees with the coordinates” to “the stored cost is provably not the optimum”, which is a different and worse defect. Those counts are the pre-repair discovery record and are released as such; none of the 40 survives in a scored table.

The single TSPLIB refutation is a parsing defect rather than a solver defect, and it is the largest single artifact this audit found. The file `linhp318.tsp` declares a fixed-edge section that our parser skips as unrecognised, so the instance is read as a plain tour problem whose parsed coordinates and distance matrix are bit-identical to `lin318`. Its stored label, 41,345, is the fixed-edge Hamiltonian-path optimum; the tour optimum on those coordinates is 42,029, and the 1-tree bound in TSPLIB’s own integer metric, with no slack at all, is 41,802. Every estimator on this paper’s TSPLIB tables was therefore scored against a target no tour on those coordinates can reach. Because the parsed geometry is `lin318`’s,

the tour optimum on it is `lin318`’s published optimum, so this label is repairable exactly rather than approximately, and the tables now use 42,029. A scan of all 111 files found no second unrecognised section keyword and no second duplicated geometry.

Tracing the refutation to its cause exposed a larger population than those 40. The multidimensional corpus was solved twice by two code paths with different distance quantisation, and the second path’s costs were written over the first’s. The generator quantises an edge at  $s(G) \in \{100, 10, 1\}$  for grid  $G \in \{100, 10^3, 10^4\}$ , a fixed resolution of  $G/10^4$ ; the verification path calls  $s_{\text{rob}}(G, n) = s(G) \cdot \min(1, n/500)$ , so for  $n < 500$  the integer matrix handed to the solver is  $500/n$  times coarser and the cost reported back is un-scaled by that same coarse factor. At  $n = 5$  and  $G = 10^4$  one integer unit is 100 grid units. The cross-check that should have caught it computes LKH’s tour length at  $s(G)$  and then divides it by  $s_{\text{rob}}(G, n)$ , inflating it by the same  $500/n$ , so the min always selected the corrupt coarse value. A second, smaller variant used  $s = 1$ , which is neither rule. Reproducing the stored cost of every unquarantined instance from its released coordinates under all three quantisations assigns each of those labels to exactly one path: 15,016 carry the coarse verification scale, 93 the unit scale, and 90,979 the generator’s own resolution. 15,109 labels of 106,272 are therefore wrong, 14.22% of the corpus, 9,662 in training, 2,697 in validation and 2,750 in test.

The defect corrupts the reported *length* and not the chosen *permutation*, which is what makes it repairable. Every repaired label is recomputed from the released coordinates and never rescaled from the stored number. For the 26,557 instances with  $n \leq 10$  we solve exactly with the Held–Karp dynamic program in float64, which is a certificate; the four cases that first exposed the defect reproduce to the digit, `N5_D2_G10000_px_26` at 18,194.600227 against a stored 18,000. Above  $n = 10$  the label becomes the float64 length of the stored tour on the released coordinates, which is an upper bound on the optimum and is exact wherever the stored permutation is; on the  $n \leq 10$  instances, where both quantities exist, the stored tour exceeds the exact optimum by 0.000007% on average on clean rows and 0.0023% on defective ones, and is not optimal on 2.5% of the latter. Where a 1-tree bound closes on the tour the status is upgraded to certified: 2,182 instances. The 2D benchmark needs a different repair, because all 2,580 of its labels are nint sums at a quantum of  $1/G$  and the solver *selects* the tour that minimises the rounded metric, a winner’s curse that pushes the label below the float optimum: mean signed error  $-0.045\%$  overall,  $-0.086\%$  at  $G = 1000$  against  $-0.0031\%$  at  $G = 10^4$ , and 600 of the stored tours are beatable in float64. Those labels are replaced by the better of the stored tour and a float-metric 2-opt descent from it. TSPLIB needs no repair beyond `linhp318`: its published optima are correct for the TSPLIB metric.

The repair has an acceptance test, and it is sharper than the test that found the defect. A quantised label forces the audit to carry the slack  $nq/2$ ; a repaired label is an upper bound on the optimum in the same float64 metric the 1-tree bound is computed in, so  $B \leq \text{OPT} \leq L$  must hold with no slack at all. Across the 145,013 bound checks that six independent sources supply over these instances there are 0 violations, the worst relative excess being  $5.8 \times 10^{-15}$ . **The tables in this paper are the repaired ones.** The pre-repair figures remain retrievable: each results CSV keeps its as-published copy beside it.

What the repair moves is small, and one-directional only in the plane. On the multidimensional test split the label error is symmetric — mean  $+0.0006\%$ , MAPE 0.011%, 1,369 defective labels low against 1,371 high — so correcting it moves GART 2.0 from 0.6201% to 0.6177% and the certified bound at its dominating budget from 0.125% to 0.111%. The 2D bias is systematic and flatters every estimator: GART 2.0’s aggregate moves from 2.9042% to 2.8892% and its near-zero signed bias doubles from  $-0.05$  to  $-0.10$ . On TSPLIB EUC\_2D the single repaired label moves it from 2.5541% to 2.5320%. Across the 14 multidimensional, 25 planar and 20 TSPLIB EUC\_2D rows, *no* estimator changes rank on MAPE or on SDPE. Of the paired tests only one crosses the 5% level, GART 2.0 against a control estimator on the uniform matched domain that this revision no longer reports, and no confidence interval changes which side of zero it lies on. One ordering statistic does move: on TSPLIB close pairs GART 1.0 crosses from above GART 2.0 to below it. Three number families in the supporting studies are still scored against pre-repair labels, because rescoring them would require refitting the models that produced them: the eight row-order permutations of arm A in Section 4.6, the augmentation corpus, and the seed-level bands of the support-arm study. The bound on their movement is the label

error itself, 0.011% on the multidimensional split and 0.050% on the 2D benchmark, against differences those studies report in whole percentage points.

This bears directly on Section 5.2, and the direction does not favour us, which is why it is stated. Repairing 1inhp318 moves GART 2.0’s matched-corpus MAPE from 2.58% to 2.56% and leaves the raw bound’s crossing budget at 50 and its cost multiple at 1.46, but it cuts the margin at that rung from 0.039 to 0.0041 percentage points. Deleting the instance instead would move the crossing to 100 and the multiple to 2.60. The printed crossing budget was never a robust statistic: the interpolated crossing moves only from 48.7 to 49.9 under the repair and to 52.6 under deletion, and the ladder rung is a discretized readout of a margin now four thousandths of a point wide. The domination by the calibrated row is untouched, because its margin is 0.56 percentage points rather than 0.0041.

A coverage gap remains. Seven TSPLIB instances carry no bound at all because the dense perturbed matrix does not fit in memory, five of them inside the scored EUC\_2D set. Nothing in this section is claimed about those instances.

## 5.8 Where Each Method Belongs

The comparison yields a rule keyed to the instance rather than a verdict on either method, and two statements a reader might expect are not among its consequences. GART 2.0 is not cheaper than a 1-tree bound in general: it is cheaper above roughly 400 nodes in the plane and dearer below roughly 150. And it is not uniformly closer to the bound than to the closed forms: that holds against the area-based estimators and fails against the MST-ratio family, from which it is separated by 1.00 percentage points on the full 78-instance EUC\_2D set against 1.35 points from the converged bound on the same set.

A third statement is one this paper made in an earlier form and the evidence does not carry. On the aggregate of every benchmark here the bound is the better instrument, and we report that unchanged. It does not follow that the estimator is superseded, because the multidimensional aggregate is a corpus 62.74% composed of instances of at most 100 nodes, and that is the regime the aggregate describes. Read at fixed size instead (Table 5), the ordering turns over: at  $n \geq 200$  GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and at  $d \in \{2, 3\}$  in the two larger bands it is strictly better than the bound on both axes at every budget we priced. The aggregate and the stratified reading are both correct, and they answer different questions. Only the second is the question a practitioner asks.

The rule that follows has two arguments rather than one. Below roughly 200 nodes, prefer the certified bound at any dimension: it is cheaper, it is more accurate, it needs no training corpus, and it tells the caller how wrong it can be. Above that size prefer the estimator at any dimension it was trained on, most decisively in the plane and at  $d = 3$ , where the ordering not only reverses but keeps reversing, because the bound pays  $\Theta(kn^2)$  for a fixed accuracy in every dimension while the estimator does not. At  $d = 100$ , outside the training grid, prefer the bound at every size; that is the one cell in Table 5 where large  $n$  does not rescue the estimator, and the reason is its own degradation off the grid rather than the bound’s strength. Large instances re-evaluated many times under changing input are the last-mile and partitioning setting of Section 1, they are the regime in which an exact solve is out of reach and a quadratic bound is already expensive, and they are the regime in which GART 2.0’s accuracy is at its best rather than its worst. A practitioner choosing between the two should read Table 5 at their own instance size, and Figure 4 at their own dimension, before reading any aggregate.

## 6 Application: Non-Euclidean Estimation via MDS Embedding

The preceding benchmarks use Euclidean point clouds, and the certified bound of Section 5 needs no more than a symmetric distance matrix. This section asks what the estimator can do on distance-matrix instances, where it needs an embedding and the bound does not. TSPLIB95 contains 33 non-EUC\_2D instances: 4 CEIL\_2D, 2 ATT, 10 GEO, and 17 EXPLICIT. An exhaustive triangle-inequality scan identifies ten EXPLICIT matrices with structural violations (worst direct-edge ratio at least 1.2), which are outside every metric-dependent aggregate in this paper: brg180, brazil58,

gr120, gr48, gr24, gr21, bays29, hk48, dantzig42, and gr17. That leaves 23 instances, seven of them EXPLICIT, and three of those (swiss42, pa561, and fri26) carry mild rounding-level violations, so conclusions are restricted to this screened set rather than to all non-Euclidean TSPs. We evaluate a hybrid pipeline that computes MST features from the original matrix and geometric features from a multidimensional-scaling (MDS) embedding. The paired reference is a constant multiple of the original-distance MST, a fixed  $\alpha = 1.136$  close to the MAPE-minimising constant 1.134 over the full 111-instance TSPLIB set. The MAPE-minimising constant over these 23 instances is 1.1718 and reaches 7.55%, which is the floor any constant multiplier faces on this metric.

## 6.1 MDS preprocessing

CEIL\_2D cases use their native coordinates. For ATT, GEO, and EXPLICIT cases, classical MDS applied to the distance matrix produces an approximate positive-eigenvalue Euclidean representation [Torgerson, 1952, Gower, 1966, Borg and Groenen, 2005, Cox and Cox, 2000]: we double-center  $D^{(2)}$  and take leading positive-eigenvalue components of the resulting Gram matrix [Gower, 1966]. We select the smallest embedding dimension  $\kappa$  retaining at least 99.9% of positive eigenvalue mass, subject to  $\kappa \leq 100$ , and pass  $\kappa$  as the model’s dimension feature. The symbol is distinct from the ascent budget  $k$  of the certified lower bound. The cap prevents five of 19 MDS cases from reaching 99.9%; hence this criterion does not imply distance preservation. Geometric features use the approximate embedding, whereas MST features are computed from the original matrix  $D$  and are therefore unaffected by embedding distortion. Appendix P reports the recovered distance diagnostics.

## 6.2 Results on non-Euclidean TSPLIB95

Table 9: Paired evaluation on the 23 screened non-EUC\_2D TSPLIB95 instances. The ten EXPLICIT matrices with structural triangle-inequality violations are excluded. Count is the size of the stratum;  $N$  is the number of instances the estimator actually scores. GART 2.0 declines si1032, whose greedy-to-MST ratio 1.0260 falls below the in-distribution floor 1.035 the gate applies. That floor is the minimum over the full corpus; the training split’s own minimum is 1.0465, and si1032 is below both, so the decision does not turn on which floor is used. GART 2.0 is therefore scored on 22 instances against 23 for the reference: the Total row is not a like-for-like comparison and the margins are read per stratum. The MAPE-minimising constant over these instances is 1.1718 at 7.55%. MSPE is the mean signed percent error.

| Distance type     | Count     | Estimator              | $N$       | SDPE (%) [95% CI]        | MAPE (%)    | Median  (%) | MSPE (%)     |
|-------------------|-----------|------------------------|-----------|--------------------------|-------------|-------------|--------------|
| ATT               | 2         | GART 2.0               | 2         | 1.42 [—, —]              | 3.66        | 3.66        | -3.66        |
|                   |           | Fixed $\alpha = 1.136$ | 2         | 4.12 [—, —]              | 3.38        | 3.38        | -3.38        |
| CEIL_2D           | 4         | GART 2.0               | 4         | 5.16 [0.11, 6.35]        | 5.77        | 6.05        | 4.04         |
|                   |           | Fixed $\alpha = 1.136$ | 4         | 6.44 [1.08, 7.86]        | 7.54        | 7.77        | 5.95         |
| GEO               | 10        | GART 2.0               | 10        | 2.60 [1.48, 3.23]        | 2.51        | 2.90        | -1.63        |
|                   |           | Fixed $\alpha = 1.136$ | 10        | 9.49 [2.13, 10.93]       | 9.62        | 4.10        | -9.62        |
| EXPLICIT screened | 7         | GART 2.0               | 6         | 3.11 [1.44, 3.87]        | 3.01        | 2.35        | -2.15        |
|                   |           | Fixed $\alpha = 1.136$ | 7         | 9.55 [5.04, 10.88]       | 8.12        | 10.16       | 1.75         |
| <b>Total</b>      | <b>23</b> | <b>GART 2.0</b>        | <b>22</b> | <b>3.89 [2.52, 4.79]</b> | <b>3.34</b> | <b>3.21</b> | <b>-0.93</b> |
|                   |           | Fixed $\alpha = 1.136$ | 23        | 10.48 [6.94, 12.93]      | 8.26        | 6.29        | -2.91        |

Table 9 reports the paired results. On the 22 of the 23 screened instances it accepts, GART 2.0 has aggregate SDPE 3.89% and MAPE 3.34%, against 10.48% and 8.26% for the fixed  $\alpha = 1.136$  multiplier scored on all 23, and 7.55% for the MAPE-minimising constant on this set. The margin over the fixed multiplier therefore measures what the feature set adds to a constant. The Total row is not a like-for-like comparison: GART 2.0 declines si1032 rather than extrapolate below the greedy-to-MST ratio it was trained on, and the  $N$  column marks where the instance sets differ. The dispersion gap is the substantive result:  $\alpha$  ranges from 1.006 to 1.511 across the set, and a single multiplier cannot track that spread without trading one metric against the other. The advantage is not uniform. On the two ATT instances

the fixed constant has the lower MAPE, 3.38% against 3.66%, and on CEIL\_2D the two estimators lie within 1.3 percentage points of SDPE of one another. Groups of two and four instances cannot separate estimators, and we report them only so the aggregate is not read as a uniform result. We do not estimate the relationship between per-instance embedding distortion and prediction error, so this remains a feasibility study for the hybrid pipeline rather than evidence of embedding robustness.

The certified bound needs no embedding here, and that is the whole of the comparison. The Held–Karp relaxation never invokes the triangle inequality, so it is defined on any symmetric matrix and reads these instances directly. On the 29 of this set that both methods score, GART 2.0 obtains 3.81% MAPE against 1.27% for the raw certified bound at an ascent budget of 100 under the Volgenant–Jonker step and 0.51% at a budget of 500 under the Polyak step, a factor of 7.5. The two steps are not interchangeable on this corpus, and Section 5.4 prices both. The bound additionally scores `brg180` and `si1032`, the two instances GART 2.0 declines, one for a structural metric violation and one for falling below the coverage gate. The MDS pipeline is therefore a way to reach these instances with a Euclidean estimator, and not the best way to estimate their tour length.

## 7 Conclusion

We present GART 2.0, a TSP tour-length estimator that predicts the MST-to-tour scaling factor  $\alpha$  from features defined for arbitrary dimension, combining MST edge and topology statistics with centroid and coordinate-range descriptors and computing no convex hull. Across synthetic planar and multidimensional benchmarks and both the Euclidean and the screened non-Euclidean parts of TSPLIB95, it has the lowest aggregate MAPE and SDPE of a seven-baseline roster on every stratum, ahead of a calibrated constant on the multidimensional set, of Çavdar–Sokol on planar i.i.d. uniform instances, and of the asymptotic ratio on TSPLIB EUC\_2D, where the paired difference resolves and the dispersion gap is the wider of the two. The property that makes those results usable is that they improve in the directions that make a solve infeasible. Dispersion falls monotonically across the size range and across the dimension range, including a dimension held out of training entirely, and the same trained model reads non-Euclidean distance matrices through an MDS embedding, well ahead of a fixed multiplier on the instances it accepts.

We then fix the boundary of the method against a certified Held–Karp 1-tree bound, which consumes no training corpus, proves its own output, and needs nothing but a symmetric distance matrix. On the corpus aggregate the bound is the better instrument, more accurate on the multidimensional benchmark at lower cost, and more accurate again on the non-Euclidean instances it reads directly and GART 2.0 reaches only through an embedding. Dimension generality buys reach, not an advantage over a relaxation that is already dimension-general. But that aggregate is mostly small instances, and it is a statement about them: read at fixed size it reverses. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the two larger size bands, in the plane and just above it, it is strictly better than the bound on both axes at every ascent budget we priced. What the estimator owns is scale, because the bound pays a quadratic price in instance size for a fixed accuracy in every dimension while the estimator does not; in the plane the measured cost exponents put the two families in different complexity classes, and an instance many times larger than anything in training is predicted in a fraction of a second. Large instances scored repeatedly are where tour-length estimation earns its place, and below that size, or off the training grid in dimension, the certified bound should be preferred. Both halves of that rule are needed to use either method well.

The same bound used as an audit instrument refuted stored labels, most of them in a scored set, and tracing them to their cause exposed a quantisation defect in a seventh of the multidimensional corpus. The tables in this paper are the repaired ones: labels recomputed from the released coordinates, a residue quarantined as unrecoverable and one TSPLIB parse repaired, verified by zero violations of  $B \leq L$  across every bound check the repair admits, with no estimator changing rank on any stratum as a result (Section 5.7). Three limitations constrain the estimator. Accuracy depends on geometry far more than on scale: error on the near-collinear class is several times what it is on the isotropic class, and Section 4.6 diagnoses that failure as a training-coverage gap and a feature gap at once, with two pre-registered



repair attempts rejected and the released model unchanged. Withholding a whole dimension or the large- $n$  range from training costs a fraction of a MAPE point and appears as a systematic offset rather than as scatter, so the estimator degrades predictably rather than erratically off its training grid, and we demonstrate no correction for that offset. And the repaired labels are not uniformly certificates: some are certified by a 1-tree bound that closes on the tour and some by exact dynamic programming, while the remainder carry an upper bound whose residual on the scored test split is narrow but not zero. Each limitation names its own next step, and one is larger than the others. Both instruments in this paper are scored against reference tours in isolation; the open question is whether the precision advantage, or the certificate, changes the decisions a routing or partitioning algorithm actually makes when either is embedded inside it.

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## A Code and Data Availability

Our implementation, trained model weights, and benchmark scripts are available at <https://github.com/sfeldmanMIG25/Area-and-Distribution-Free-Estimator-for-TSP>. Synthetic instances can be regenerated using the provided scripts with fixed random seeds. The classical estimators and the calibrated constant-ratio baselines are released as separate modules, and the fitted constant-ratio table is frozen as a released artifact rather than refitted at inference. The eight per-benchmark results tables are generated from the scored benchmark outputs and written into the manuscript mechanically rather than typed; a verification pass re-derives all 1,638 table cells from those outputs and reports any that disagrees with the typeset value, and a second pass does the same for the 182 cells of the cost/accuracy ladders of Section 5. The remaining tables, and scalars quoted in the running prose such as the timing split and the audit counts, are transcribed by hand from the same artifacts and are not covered by either check. The reference-tour audit of Section 3.3, the refits of Section 4.5, the MDS diagnostics of Appendix P and the figures of this paper are each released as a standalone reproduction script, and the versions of the Python analysis dependencies are pinned in the repository. For direct audit of target clipping, the two released out-of-interval rows are N1000\_D15\_G100\_pchergphhtgncoe\_59 (training, raw  $\alpha = 0.974823$ ) and N5\_D2\_G100\_oc\_33 (validation, raw  $\alpha = 2.059518$ ).

The GART 2.0 pipeline uses SciPy [Virtanen et al., 2020] for Delaunay triangulation [Delaunay, 1934] and MST construction, Numba [Lam et al., 2015] for centroid-distance computation, NumPy [Harris et al., 2020] symmetric eigendecomposition for classical MDS, LightGBM [Ke et al., 2017] for regression, and scikit-learn [Pedregosa et al., 2011] for metric computation. Estimator timings were collected on one Intel i7 13th Gen machine with 64 GB RAM and Windows 11. Concorde 03.12.19 and LKH-3 3.0.13 generated the stored reference tour costs; the canonical table does not retain solver certificates.

## B Training Data Generation Details

The training dataset was constructed by generating synthetic TSP instances across four distribution classes, with the generator assigned independently per axis:

### B.1 2D benchmark generator design

Four classes (isotropic, biased, geometric, clustered) follow conventions used in the TSP-instance-generation literature, including the variants reported by Cavdar and Sokol [2015] for the distribution-free baseline. The fifth class, Line Noise, samples  $y = mx + b + \epsilon$  with small  $\epsilon$  and then clips out-of-range coordinates to the grid boundary rather than resampling them. Because the slope reaches  $\pm 5$  while  $x$  is drawn across the full grid width, most points fall outside the box and are pinned to a face. The realized geometry is therefore a union of two to four thin segments with dense collinear masses along the boundary, not a single noisy line: at  $n \geq 200$  a median 59.9% of points lie exactly on a face, and  $\alpha$  tracks that fraction closely (Spearman 0.86, rising monotonically from 1.180 to 1.431 across its quartiles) while showing no relationship to the transverse noise width. We describe the class as realized rather than as intended because its difficulty comes from the segment-union structure. The `grid` member of the Geometric Struct. class draws a jittered square lattice, which no product of the four independent per-axis draws of §3.2 can produce, and the training corpus contains no lattice instance.

### B.2 Training corpus

The uniform class draws coordinates from  $\mathcal{U}(0, G)$  where  $G$  is the grid size. The normal class draws from  $\mathcal{N}(G/2, G/6)$  truncated to  $[0, G]$ . The clustered class places  $k \in [3, 15]$  cluster centers uniformly and draws points from  $\mathcal{N}(\mu_k, \sigma_k)$  around each center. The correlated class uses autoregressive structure across dimensions,  $x_{d+1} = \rho \cdot x_d + \epsilon$ , producing axis-aligned anisotropy.

Node counts span  $n \in [5, 1,000]$  with training-visible dimensions  $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ , plus a held-out bucket at  $d = 100$  (5,904 test instances). The pipeline ran Concorde and LKH-3 and retained the shorter tour cost.

The 106,272 solved instances are partitioned into training (69,768, 65.65%), validation (19,584, 18.43%), and test (16,920, 15.92%) using a stratified sample on  $(d, n, \text{grid size})$ . The stratification rule ranks instances within each stratum by a seeded uniform and assigns roughly the first 16% to test, the next 18% to validation, and the remainder to training;  $d = 100$  instances bypass this rule and are locked to the test partition. Table 11 lists the per- $(d, n)$  training-instance count; grid size  $G \in \{100, 1,000, 10,000\}$  is sampled uniformly within each cell.

### B.3 Multidimensional benchmark grid

The full  $(d, n)$  count grid of the 16,920-instance ND benchmark:

Table 10: Multidimensional synthetic benchmark grid: instance count per  $(d, n)$  bucket (16,920 instances total).

| $d \setminus n$     | [5, 10]      | [11, 50]     | [51, 100]    | [101, 200] | [201, 500]   | [501, 1000]  | Total         |
|---------------------|--------------|--------------|--------------|------------|--------------|--------------|---------------|
| $d = 2$             | 162          | 108          | 135          | 27         | 81           | 135          | <b>648</b>    |
| $d = 3\text{--}5$   | 486          | 324          | 405          | 81         | 243          | 405          | <b>1,944</b>  |
| $d = 6\text{--}10$  | 810          | 540          | 675          | 135        | 405          | 675          | <b>3,240</b>  |
| $d = 15\text{--}25$ | 486          | 324          | 405          | 81         | 243          | 405          | <b>1,944</b>  |
| $d = 30\text{--}50$ | 810          | 540          | 675          | 135        | 405          | 675          | <b>3,240</b>  |
| $d = 100$           | 1,476        | 984          | 1,230        | 246        | 738          | 1,230        | <b>5,904</b>  |
| <b>Total</b>        | <b>4,230</b> | <b>2,820</b> | <b>3,525</b> | <b>705</b> | <b>2,115</b> | <b>3,525</b> | <b>16,920</b> |

Table 11: Training-set instance count per  $(d, n)$  cell (69,768 total; grid size  $G \in \{100, 1,000, 10,000\}$  sampled uniformly within each cell; four distribution classes per axis listed in §B).

| $d \setminus n$     | [5, 10]       | [11, 50]      | [51, 100]     | [101, 200]   | [201, 500]   | [501, 1000]   | Total         |
|---------------------|---------------|---------------|---------------|--------------|--------------|---------------|---------------|
| $d = 2$             | 1,026         | 684           | 855           | 171          | 513          | 855           | <b>4,104</b>  |
| $d = 3\text{--}5$   | 3,078         | 2,052         | 2,565         | 513          | 1,539        | 2,565         | <b>12,312</b> |
| $d = 6\text{--}10$  | 5,130         | 3,420         | 4,275         | 855          | 2,565        | 4,275         | <b>20,520</b> |
| $d = 15\text{--}25$ | 3,078         | 2,052         | 2,565         | 513          | 1,539        | 2,565         | <b>12,312</b> |
| $d = 30\text{--}50$ | 5,130         | 3,420         | 4,275         | 855          | 2,565        | 4,275         | <b>20,520</b> |
| <b>Total</b>        | <b>17,442</b> | <b>11,628</b> | <b>14,535</b> | <b>2,907</b> | <b>8,721</b> | <b>14,535</b> | <b>69,768</b> |

## C Detailed Feature Specifications

Table 24 lists the 11 geometric and centroid features, Tables 12 and 13 the 19 MST-derived ones, and Table 14 the single constructive ratio, 31 in all (§3.1). Listed complexities assume the MST has already been built; MST construction itself dominates the asymptotic cost and is analysed in §3.

### C.1 MST Edge Distribution Statistics

The first category collected here captures the local uniformity of the point set through the statistical moments of the MST edge weights. Unlike coordinate-based variance, edge weights are invariant to rotation and translation. High

skewness or kurtosis can indicate a mixture of scales, such as dense clusters separated by voids; these moments are included as candidate predictors of the  $\alpha$  ratio rather than as a tested causal mechanism.

Table 12: MST Edge Statistic Specifications

| Feature Name             | Description & Rationale                                                                           | Complexity    | Origin / Citation                     |
|--------------------------|---------------------------------------------------------------------------------------------------|---------------|---------------------------------------|
| <b>MST Edge Max</b>      | Weight of the longest edge in the MST.                                                            | $O(n)$        | Standard                              |
| <b>MST Quantiles</b>     | 10%, 25%, 50%, 75%, 90% percentiles. Provides a non-parametric distribution profile.              | $O(n \log n)$ | <b>Proposed;</b> via MST [Prim, 1957] |
| <b>MST Edge Mean</b>     | Average edge length in the MST.                                                                   | $O(n)$        | Smith-Miles et al. [2010]             |
| <b>MST Edge Std</b>      | Standard deviation of edge lengths in the MST.                                                    | $O(n)$        | Smith-Miles et al. [2010]             |
| <b>MST Edge Skewness</b> | Asymmetry of edge distribution. High positive skew implies many short edges and few long bridges. | $O(n)$        | Smith-Miles et al. [2010]             |
| <b>MST Edge Kurtosis</b> | Tailedness of distribution. Indicates the presence of extreme outliers.                           | $O(n)$        | Smith-Miles et al. [2010]             |

## C.2 Topological Structure and Clustering Proxies

The second category contains features designed to detect higher-order structures—such as clusters, filaments, and hubs—without performing expensive clustering algorithms. We introduce specific ratios (Dominance, Gap, Normalised Diameter) as  $O(n)$  candidate proxies for these geometric properties. The Leaf Ratio is motivated by the hypothesis that low-leaf trees are more common on linear manifolds than in isotropic clouds; this interpretation is not evaluated independently here.

Table 13: Topological and Clustering Feature Specifications

| Feature Name                  | Description & Rationale                                                                     | Complexity | Origin / Citation                     |
|-------------------------------|---------------------------------------------------------------------------------------------|------------|---------------------------------------|
| <b>MST Dominance Ratio</b>    | Sum of largest $\sqrt{n}$ edges / Total Length. Captures weight of inter-cluster bridges.   | $O(n)$     | <b>Proposed;</b> via MST [Prim, 1957] |
| <b>MST Gap Ratio</b>          | Max edge / Median edge. Measures magnitude of sparsity jumps.                               | $O(1)$     | <b>Proposed;</b> via MST [Prim, 1957] |
| <b>MST Leaf Ratio</b>         | Fraction of nodes with degree $k = 1$ . Topology proxy for dimension.                       | $O(n)$     | Smith-Miles et al. [2010]             |
| <b>MST Degree (Mean, Std)</b> | Branching factor statistics.                                                                | $O(n)$     | Smith-Miles et al. [2010]             |
| <b>MST Degree Max</b>         | Maximum node degree. Identifies "hub" nodes common in High-D spaces.                        | $O(n)$     | Smith-Miles et al. [2010]             |
| <b>MST Diameter</b>           | Sum of edge weights on the longest path between any two leaves, computed via double-BFS.    | $O(n)$     | Graph Theory                          |
| <b>Normalised Diameter</b>    | Diameter / Total Length. Scale-invariant shape proxy ( $= 1.0$ for path-manifolds exactly). | $O(1)$     | <b>Proposed;</b> via double-BFS       |
| <b>Large Edge Count</b>       | Count of edges $> \mu + \sigma$ . Secondary cluster indicator.                              | $O(n)$     | <b>Proposed;</b> via MST [Prim, 1957] |

## C.3 Constructive Ratio

The third and last category holds a single feature. It is the only input GART 2.0 adds to the 30-feature block it inherits, it ranks second by SHAP magnitude (Appendix D), and it is the feature the non-Euclidean coverage gate of Section 6 tests.



Table 14: Constructive Feature Specification (1 feature).

| Feature Name               | Description & Rationale                                                                                                                                                                                                                                                                                                                         | Complexity                                                                               | Origin / Citation                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Greedy-to-MST Ratio</b> | Length of an exact greedy nearest-neighbour tour divided by $L_{\text{MST}}$ . A greedy tour is feasible, so the ratio is a constructed upper bound on the target $\alpha$ . Values outside $[1.035, 2.209]$ , the range spanned by the full 106,272-row corpus, trigger the coverage gate; the training split alone spans $[1.0465, 2.1295]$ . | $O(n^2d)$ dense for $n \leq 3000$ ; $O(nkd)$ with $k = 16$ candidate neighbours above it | <b>Proposed</b> ; greedy nearest-neighbour construction |

## D Feature Importance: SHAP Values

To validate the 31-feature set, we report the SHAP values [Lundberg and Lee, 2017] against the production booster on a 5,000-row sample of the 16,920-row held-out test split. Table 15 ranks all 31 features by mean absolute SHAP contribution. Three observations motivate the final feature set.

Section 3.5 gives the per-feature shares. By group, the nineteen MST-derived features carry 50.7% of total mean absolute SHAP magnitude, the eleven geometric features 25.5%, and the single constructive ratio 23.9%. SHAP magnitude describes model reliance, not a causal effect or proof that a feature group improves generalization.

We used SHAP analysis as a diagnostic during feature reduction, yielding the 31-feature set described in Section 3.1; because this selection used validation feedback, it should not be interpreted as an independent confirmatory test.

Table 15: All 31 GART 2.0 features ranked by mean absolute SHAP magnitude, computed against the production booster on a 5,000-row sample of the 16,920-row held-out test split. The bars show each feature’s share of the total magnitude.

| Rank | Feature                  | Mean  SHAP | Share (%) |                                                                                       |
|------|--------------------------|------------|-----------|---------------------------------------------------------------------------------------|
| 1    | mst_dominance_ratio      | 0.2866     | 26.52     |    |
| 2    | greedy_nn_over_mst       | 0.2578     | 23.86     |    |
| 3    | n_customers              | 0.0922     | 8.53      |    |
| 4    | mst_diameter_normalized  | 0.0886     | 8.20      |    |
| 5    | dimension                | 0.0605     | 5.60      |    |
| 6    | log_bounding_hypervolume | 0.0513     | 4.75      |    |
| 7    | mst_degree_mean          | 0.0414     | 3.84      |    |
| 8    | bounding_hypervolume     | 0.0261     | 2.41      |    |
| 9    | mst_gap_ratio            | 0.0195     | 1.81      |    |
| 10   | centroid_dist_std        | 0.0193     | 1.79      |    |
| 11   | large_edge_count         | 0.0183     | 1.70      |    |
| 12   | mst_leaf_ratio           | 0.0179     | 1.66      |    |
| 13   | mst_diameter             | 0.0154     | 1.42      |    |
| 14   | mst_degree_std           | 0.0119     | 1.10      |    |
| 15   | mst_edge_skew            | 0.0069     | 0.64      |    |
| 16   | aspect_ratio             | 0.0068     | 0.63      |    |
| 17   | mst_degree_max           | 0.0063     | 0.59      |    |
| 18   | centroid_dist_mean       | 0.0059     | 0.55      |    |
| 19   | mst_edge_std             | 0.0051     | 0.47      |    |
| 20   | centroid_dist_iqr        | 0.0046     | 0.42      |    |
| 21   | mst_edge_q25             | 0.0045     | 0.42      |   |
| 22   | mst_edge_max             | 0.0045     | 0.42      |  |
| 23   | mst_edge_kurtosis        | 0.0044     | 0.41      |  |
| 24   | mst_edge_q75             | 0.0039     | 0.36      |  |
| 25   | node_density             | 0.0035     | 0.33      |  |
| 26   | mst_edge_q90             | 0.0035     | 0.33      |  |
| 27   | mst_edge_mean            | 0.0029     | 0.27      |  |
| 28   | mst_edge_q50             | 0.0029     | 0.27      |  |
| 29   | mst_edge_q10             | 0.0027     | 0.25      |  |
| 30   | centroid_dist_max        | 0.0026     | 0.24      |  |
| 31   | log_node_density         | 0.0026     | 0.24      |  |

## E Benchmark-Model Implementation Notes

This appendix records what each scored classical estimator receives, what we repaired, and why three further estimators are surveyed in the literature review and scored nowhere.

**BHH.** For points drawn i.i.d. from a density  $f$  of bounded support,  $L_n/n^{(d-1)/d} \rightarrow \beta_d \int f(x)^{(d-1)/d} dx$  [Beardwood et al., 1959]; for  $f$  uniform on a region of measure  $\mathcal{V}$  the integral is  $\mathcal{V}^{1/d}$ , giving  $\beta_d n^{(d-1)/d} \mathcal{V}^{1/d}$ . The quantity is the measure of the sampling region, and the convex hull of the realized sample is a different, downward-biased statistic that converges to it only as  $n$  grows. Our synthetic generators draw on  $[0, G]^d$ , so  $\mathcal{V} = G^d$  is known exactly, and we report that form on the matched domain. We use  $\beta_2 = 0.7124 \pm 0.0002$  [Johnson et al., 1996] and  $\beta_3 = 0.6979 \pm 0.0002$  [Percus and Martin, 1996]. For  $d \geq 4$  no comparably pinned estimate exists and we substitute the large- $d$  asymptotic  $\sqrt{d}/(2\pi e)$  [Bertsimas and van Ryzin, 1990], a form that source proves for the minimum-spanning-tree and minimum-matching constants and only conjectures for the TSP constant; that substitution, and the fact that  $n \leq 1000$  with  $d \geq 4$  is far from the asymptotic regime, both bound how much the  $d \geq 4$  rows can be read as a test of the theorem.

**Withdrawn: Daganzo, Chien, and Kwon–Golden–Wasil.** Earlier revisions of this paper scored all three. They are withdrawn, and the reason is provenance rather than performance. Their equations reached our code through the literature review in Figliozzi [2009], not through the articles themselves: Daganzo [1984a], which is the Daganzo paper that review carries, Chien [1992] and Kwon et al. [1995] are paywalled, a direct check of each DOI returned no open-access location, and we could not obtain them. Nor is the secondary record self-consistent. For Chien, Çavdar and Sokol [2015] give a Daganzo-form coefficient of 0.69 and Choi [2021] give 0.88, against the 0.67 implied by the  $2.1\bar{r} + 0.67\sqrt{nR}$  we had adopted; for Kwon–Golden–Wasil, Çavdar and Sokol [2015] render the bracket with a plain  $n$  and four-significant-figure coefficients where our transcription carried  $(n+1)$  and two. Choosing between those renderings requires the primaries, which is exactly what we lack, and reporting either would have repeated the defect this appendix exists to disclose. We therefore print no number for any of the three. Section 1.2 still surveys them: they are part of the record this work sits in, and omitting them from the survey would misdescribe the field. Two consequences are worth stating plainly. The strongest classical estimator this paper previously reported was Kwon–Golden–Wasil on its own fitted node range, so withdrawing it removes our most demanding classical baseline; and the surviving pair, BHH and Çavdar–Sokol, are the two whose primary documents we hold.

**Çavdar–Sokol.** This estimator was rebuilt against a primary document we hold: Çavdar [2014], Chapter 4, the same work as Çavdar and Sokol [2015] and by the same author. The area term is the area of the rectangle covering the nodes, and  $\bar{c}_j$  and  $\text{cstdev}_j$  are the mean and standard deviation of the absolute distances to that rectangle’s midpoint line on axis  $j$ , not to the sample mean. The divisor  $\bar{c}_x\bar{c}_y$  sits inside the radical; placing it outside leaves the second term dimensionless. An earlier revision of our code substituted the convex hull for the area and measured dispersion about the sample midrange; both are corrected, and every Çavdar–Sokol row in this paper is the corrected, rotated estimator. Two further features of the published method are now applied. The first is the frame: the estimator is defined on an axis-aligned rectangle and is not rotation invariant, so the implementation rotates each instance into its minimum-area enclosing rectangle by scanning convex-hull edges [Freeman and Shapira, 1975], which is what the source prescribes. The second is the finite- $n$  correction  $0.9325e^{0.00005298n} - 0.2972e^{-0.01452n}$ , Eq. (21) of the dissertation, fitted over  $n \in \{100, 125, \dots, 975\}$  at  $R^2 = 0.9867$ ; the raw estimate is divided by it. Choi [2021] records that the uncorrected estimator underestimates for  $n < 1000$ , which is the direction the correction moves it. That correction is bounded to the range it was fitted over, and the bound binds on this benchmark: 1,320 of the 2,580 2D instances have  $n < 100$ , so for slightly over half the benchmark the ratio is evaluated at the  $n = 100$  endpoint rather than extrapolated below it. The authors’ own figures show the ratio continuing to fall toward 0.4 as  $n$  approaches 10, but publish no fitted form there, so extrapolating would substitute our curve for theirs. We hold the upper end fixed for the same reason: above  $n = 975$  no correction is applied, because Eq. (21) grows without bound ( $E/T = 1.22$  at  $n = 5000$ ) and would contradict the training fit it exists to repair. That leaves a 1.8% step at the upper boundary, where the fitted ratio reads 0.982. Two properties of the source, not of our evaluation, remain: the regression targets Lin–Kernighan tour lengths rather than optima, and every training graph carried a node at each corner of its rectangle, which pins the graph area to the covering rectangle in a way our instances do not.

**Constant-ratio references.** The  $\alpha = 1$  row returns  $L_{\text{MST}}$ . The asymptotic row uses  $\beta_{\text{TSP}}/\beta_{\text{MST}}$  for uniform planar points. The numerator is well established at 0.7124; the denominator is not. We quote 0.6331 following Cortina-Borja and Robinson [2000]. That figure is a simulation estimate and not a theorem, and published estimates of the planar Euclidean MST constant span a range wider than its printed precision, so the ratio 1.1253 should be read as approximate. The calibrated rows take the mean of  $\alpha$  within each dimension, and within each dimension–size cell, over the training split only; the fitted values are frozen and released rather than refitted at inference. At  $d = 100$ , which no training row covers, the table extrapolates from  $d = 50$ .

**Custom Hilbert sort.** We independently min–max normalise every axis to a 16-bit grid, sort by a  $d$ -dimensional Hilbert index [Hilbert, 1891], visit nodes in that order, and close the tour. This is a custom generalization inspired by the planar space-filling-curve heuristic of Bartholdi and Platzman [1982]; its planar performance guarantee does not

transfer to our anisotropically normalised multidimensional ordering. It constructs a tour rather than estimating one, so its error is bounded below by the gap between a space-filling-curve tour and the optimum.

**Sources.** Every constant this paper scores is read from a primary document. The BHH constants come from Johnson et al. [1996] and Percus and Martin [1996]; the Çavdar–Sokol form and both of its constant sets are verified directly against Çavdar [2014], which is open access. No scored constant is transcribed from a secondary source, and where the only available rendering was secondary we withdrew the estimator rather than publish the number.

**Hybrid non-Euclidean builder.** An earlier revision of the hybrid feature builder omitted two of the thirty features that version of the model carried, `log_bounding_hypervolume` and `log_node_density`, and the downstream lookup substituted zero for both on every non-Euclidean instance; it also derived node density from a hypervolume clipped at  $e^{690}$ , which saturates once the embedding dimension is large. Section 6 uses the corrected builder, and the feature lookup now raises on a missing feature rather than defaulting it.

**Convex-hull cap.** Where an estimator receives the convex-hull plug-in we compute the hull only for  $d \leq 3$  and substitute the coordinate bounding box above it, recording which was used. QHull’s output size grows as  $n^{\lfloor d/2 \rfloor}$ , the same blow-up cited in Section 1 as a reason not to build estimators on hull geometry; at  $d = 8$  with  $n = 1000$  a single hull dominates the entire benchmark’s runtime.

## F Classical Estimators: Full Benchmark and Matched Domain

Table 16 reports the two classical estimators twice. The upper panels apply them to the complete 2D and TSPLIB EUC\_2D benchmarks, with the convex hull of each instance standing in for BHH’s region measure, which is the only option on TSPLIB and the fairer option on the degenerate 2D classes. The lower panel restricts to the 210 i.i.d. uniform 2D instances, where the sampling region  $[0, G]^2$  is exact and BHH receives it. Çavdar–Sokol receives the same input in both, the rectangle covering the nodes, so only its instance set changes between the panels.

Table 16: Classical estimators on the full benchmarks and on their matched domains. MSPE is the mean signed percent error; a large MSPE relative to MAPE identifies a systematic offset rather than scatter.

| Domain                             | Estimator                               | $N$   | SDPE (%) [95% CI]    | MAPE (%) | Median  (%) | MSPE (%) |
|------------------------------------|-----------------------------------------|-------|----------------------|----------|-------------|----------|
| Full 2D benchmark                  | BHH                                     | 2,580 | 32.00 [30.00, 33.97] | 23.80    | 15.53       | -13.71   |
|                                    | Çavdar–Sokol                            | 2,580 | 34.61 [31.96, 37.12] | 18.24    | 6.63        | -0.96    |
|                                    | <b>GART 2.0</b>                         | 2,580 | 4.68 [4.42, 4.92]    | 2.89     | 1.49        | -0.10    |
|                                    | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2,580 | 9.38 [8.96, 9.78]    | 5.76     | 2.31        | -0.75    |
| TSPLIB EUC_2D                      | BHH                                     | 78    | 40.75 [26.25, 53.61] | 25.14    | 13.41       | 13.66    |
|                                    | Çavdar–Sokol                            | 78    | 36.34 [24.38, 48.39] | 23.87    | 11.32       | 20.07    |
|                                    | <b>GART 2.0</b>                         | 78    | 2.92 [2.40, 3.38]    | 2.53     | 1.87        | 2.08     |
|                                    | Calibrated MST ratio $\hat{\rho}(d, n)$ | 78    | 4.48 [3.59, 5.32]    | 3.74     | 3.18        | 1.18     |
| 2D random (uniform on $[0, G]^2$ ) | <b>GART 2.0</b>                         | 210   | 2.07 [1.65, 2.48]    | 1.31     | 0.75        | -0.01    |
|                                    | GART 1.0                                | 210   | 4.96 [4.53, 5.35]    | 5.75     | 6.05        | 4.69     |
|                                    | BHH (sampling region)                   | 210   | 7.76 [6.39, 8.98]    | 8.91     | 6.70        | -8.65    |
|                                    | Çavdar–Sokol                            | 210   | 14.28 [11.63, 16.53] | 8.16     | 2.84        | -5.97    |
|                                    | $L_{\text{MST}} (\alpha = 1)$           | 210   | 7.70 [6.39, 8.82]    | 15.60    | 12.52       | -15.60   |
|                                    |                                         |       |                      |          |             |          |

## G Per-Class Detail for the 2D Diverse Benchmark

Table 17 disaggregates the 2D benchmark by the five stress classes of Table 2, with the Geometric Struct. class split into its grid generator and its other two. Two of the six rows carry geometry the training corpus does not contain, Line Noise and grid. Reported inside the 630-instance class aggregate, grid’s signed error is what carries that aggregate to +2.74 MSPE while the two represented generators sit at +0.58; splitting the row is what makes the second unrepresented generator visible.

Table 17: 2D diverse synthetic benchmark by generator class. The Geometric Struct. class of Table 2 is split into its two reported rows: grid, the one Geometric generator with no counterpart in the training corpus, and the two that have one. The split is by training coverage rather than by geometry, so the design taxonomy of Table 2 is unchanged and the six row counts still sum to the 2,580 instances of the benchmark.

| Class                                       | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R^2_{\alpha}$ | Time (ms) |
|---------------------------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|----------------|-----------|
| Isotropic<br>Count=840                      | <b>GART 2.0</b>                         | <b>2.23</b> [2.02, 2.44] | 1.47     | 0.91        | 0.13     | 0.954          | 108       |
|                                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.11 [4.65, 5.53]        | 3.31     | 1.89        | 0.38     | 0.744          | 25.4      |
|                                             | GART 1.0                                | 5.18 [4.94, 5.43]        | 5.38     | 5.29        | 3.82     | 0.701          | 49.5      |
|                                             | $L_{\text{MST}} (\alpha = 1)$           | 7.79 [7.26, 8.29]        | 17.15    | 14.51       | -17.15   | -2.451         | 24.9      |
|                                             | Asymptotic MST ratio                    | 8.77 [8.05, 9.37]        | 7.12     | 3.81        | -6.77    | -0.455         | 22.1      |
|                                             | Custom Hilbert sort                     | 11.50 [10.84, 12.16]     | 30.78    | 35.07       | 30.78    | —              | 0.71      |
| Biased<br>Count=840                         | <b>GART 2.0</b>                         | <b>2.51</b> [2.31, 2.69] | 1.79     | 1.24        | 0.30     | 0.959          | 114       |
|                                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.24 [4.79, 5.66]        | 3.21     | 1.54        | -1.26    | 0.757          | 24.3      |
|                                             | GART 1.0                                | 7.97 [7.58, 8.34]        | 7.93     | 6.19        | 6.70     | 0.487          | 49.7      |
|                                             | $L_{\text{MST}} (\alpha = 1)$           | 8.97 [8.36, 9.53]        | 18.36    | 14.75       | -18.36   | -2.012         | 23.6      |
|                                             | Asymptotic MST ratio                    | 10.09 [9.38, 10.69]      | 8.22     | 4.08        | -8.13    | -0.475         | 21.5      |
|                                             | Custom Hilbert sort                     | 13.14 [12.49, 13.78]     | 32.04    | 35.69       | 32.04    | —              | 0.719     |
| Geometric: Grid <sup>†</sup><br>Count=210   | <b>GART 2.0</b>                         | <b>1.92</b> [1.66, 2.19] | 7.07     | 7.64        | 7.07     | 0.090          | 103       |
|                                             | GART 1.0                                | 4.97 [4.51, 5.43]        | 11.90    | 9.77        | 11.90    | -1.871         | 45.1      |
|                                             | $L_{\text{MST}} (\alpha = 1)$           | 6.01 [4.73, 7.17]        | 4.54     | 2.35        | -4.54    | -0.429         | 27.1      |
|                                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 6.42 [5.38, 7.37]        | 15.99    | 14.82       | 15.99    | -4.371         | 26.6      |
|                                             | Asymptotic MST ratio                    | 6.76 [5.23, 8.04]        | 9.43     | 10.08       | 7.41     | -0.817         | 24.6      |
|                                             | Custom Hilbert sort                     | 6.86 [6.16, 7.52]        | 16.14    | 15.81       | 16.14    | —              | 0.929     |
| Geometric: Boundary, X-Central<br>Count=420 | <b>GART 2.0</b>                         | <b>2.60</b> [2.29, 2.93] | 1.96     | 1.48        | 0.58     | 0.938          | 117       |
|                                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.06 [4.37, 5.71]        | 2.99     | 1.14        | 0.53     | 0.736          | 24.7      |
|                                             | GART 1.0                                | 5.33 [4.94, 5.72]        | 5.50     | 5.28        | 3.81     | 0.700          | 50.8      |
|                                             | $L_{\text{MST}} (\alpha = 1)$           | 7.89 [6.97, 8.68]        | 17.01    | 13.76       | -17.01   | -2.299         | 24.6      |
|                                             | Asymptotic MST ratio                    | 8.87 [7.93, 9.75]        | 6.69     | 2.95        | -6.62    | -0.419         | 22.5      |
|                                             | Custom Hilbert sort                     | 11.61 [10.66, 12.56]     | 31.37    | 35.84       | 31.37    | —              | 0.823     |
| Clustered<br>Count=60                       | <b>GART 2.0</b>                         | <b>2.75</b> [2.06, 3.36] | 2.24     | 1.76        | -1.06    | 0.828          | 148       |
|                                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.36 [4.31, 6.23]        | 7.94     | 7.01        | -7.94    | -0.950         | 33.8      |
|                                             | $L_{\text{MST}} (\alpha = 1)$           | 5.65 [4.69, 6.46]        | 20.46    | 19.41       | -20.46   | -7.816         | 30.7      |
|                                             | Asymptotic MST ratio                    | 6.36 [5.29, 7.29]        | 10.50    | 9.31        | -10.50   | -2.157         | 31.8      |
|                                             | GART 1.0                                | 6.71 [5.46, 7.86]        | 5.41     | 4.06        | -2.53    | -0.071         | 69.2      |
|                                             | Custom Hilbert sort                     | 9.43 [7.18, 11.20]       | 40.97    | 43.87       | 40.97    | —              | 7.14      |
| Line Noise <sup>†</sup><br>Count=210        | <b>GART 2.0</b>                         | <b>6.09</b> [5.67, 6.47] | 10.82    | 11.04       | -10.82   | 0.294          | 106       |
|                                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 8.64 [8.04, 9.18]        | 20.45    | 20.17       | -20.45   | -1.364         | 22.8      |
|                                             | $L_{\text{MST}} (\alpha = 1)$           | 10.67 [10.01, 11.26]     | 34.13    | 38.24       | -34.13   | -5.250         | 24.2      |
|                                             | Asymptotic MST ratio                    | 12.01 [11.26, 12.64]     | 25.88    | 30.51       | -25.88   | -3.154         | 23.3      |
|                                             | GART 1.0                                | 14.90 [13.48, 16.16]     | 12.03    | 10.50       | 2.17     | 0.262          | 47.1      |
|                                             | Custom Hilbert sort                     | 24.64 [23.00, 26.22]     | 38.32    | 42.86       | 38.32    | —              | 0.662     |

Table 17 continued from the previous page

| Class                       | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R^2_{\alpha}$ | Time (ms) |
|-----------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|----------------|-----------|
| <b>Total</b><br>Count=2,580 | <b>GART 2.0</b>                         | <b>4.68</b> [4.42, 4.92] | 2.89     | 1.49        | -0.10    | 0.867          | 113       |
|                             | GART 1.0                                | 7.90 [7.60, 8.16]        | 7.30     | 6.27        | 5.13     | 0.637          | 50.1      |
|                             | Calibrated MST ratio $\hat{\rho}(d, n)$ | 9.38 [8.96, 9.78]        | 5.76     | 2.31        | -0.75    | 0.462          | 25        |
|                             | $L_{\text{MST}} (\alpha = 1)$           | 10.27 [9.91, 10.61]      | 17.95    | 14.61       | -17.95   | -1.621         | 24.9      |
|                             | Asymptotic MST ratio                    | 11.55 [11.15, 11.91]     | 9.20     | 5.34        | -7.68    | -0.378         | 22.6      |
|                             | Custom Hilbert sort                     | 14.19 [13.75, 14.60]     | 30.95    | 34.97       | 30.95    | —              | 0.726     |

## H Rank Agreement

Table 18 reports global rank correlation and close-pair ordering accuracy with  $L_{\text{MST}}$  as the control. Close-pair accuracy is the fraction of instance pairs whose reference costs differ by less than the stated threshold that the estimator orders concordantly with the reference. Pairs whose reference costs are exactly equal carry no reference ordering and are excluded. Every benchmark is enumerated exhaustively, so the statistic is exact and no random seed enters it:  $\binom{2580}{2} = 3,326,910$  pairs on 2D,  $\binom{78}{2} = 3,003$  on TSPLIB EUC\_2D and  $\binom{16846}{2} = 141,885,435$  on the multidimensional set, the 74 quarantined instances of Section 5.7 having no reference cost to order. Of the multidimensional pairs 1,647,719 qualify at the 5% threshold and 3,262,149 at 10%; the corresponding 2D counts are 74,826 and 152,306, and the TSPLIB EUC\_2D counts 54 and 105. The 142 million multidimensional pairs are never held at once: because the relative gap to a larger partner increases with that partner’s cost, sorting the reference costs makes each instance’s qualifying partners a contiguous run whose end binary search locates, so only the qualifying pairs are built. That enumeration was verified to agree pair-for-pair with a direct scan of all 141,885,435 pairs.

Table 18: Rank agreement with reference tour cost. The  $\alpha = 1$  row is the control: any constant multiple of  $L_{\text{MST}}$  induces the same ordering, so the asymptotic-ratio row coincides with it wherever both are defined.

| Benchmark     | Estimator                               | $N$    | Spearman $\rho$ | Kendall $\tau$ | Close pair < 5% (%) | Close pair < 10% (%) |
|---------------|-----------------------------------------|--------|-----------------|----------------|---------------------|----------------------|
| 2D            | <b>GART 2.0</b>                         | 2,580  | 0.9994          | 0.9827         | 74.48               | 83.56                |
|               | GART 1.0                                | 2,580  | 0.9985          | 0.9713         | 69.14               | 76.71                |
|               | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2,580  | 0.9979          | 0.9664         | 68.86               | 76.07                |
|               | Asymptotic MST ratio                    | 2,580  | 0.9961          | 0.9615         | 71.04               | 77.76                |
|               | $L_{\text{MST}} (\alpha = 1)$           | 2,580  | 0.9961          | 0.9615         | 71.04               | 77.76                |
| ND            | <b>GART 2.0</b>                         | 16,846 | 1.0000          | 0.9982         | 92.41               | 96.11                |
|               | Calibrated MST ratio $\hat{\rho}(d, n)$ | 16,846 | 0.9999          | 0.9948         | 81.72               | 89.04                |
|               | $L_{\text{MST}} (\alpha = 1)$           | 16,846 | 0.9996          | 0.9856         | 69.05               | 75.86                |
|               | Custom Hilbert sort                     | 16,846 | 0.9989          | 0.9726         | 58.53               | 64.31                |
| TSPLIB EUC_2D | <b>GART 2.0</b>                         | 78     | 0.9993          | 0.9867         | 72.22               | 80.95                |
|               | GART 1.0                                | 78     | 0.9982          | 0.9773         | 70.37               | 72.38                |
|               | Asymptotic MST ratio                    | 78     | 0.9986          | 0.9793         | 61.11               | 73.33                |
|               | $L_{\text{MST}} (\alpha = 1)$           | 78     | 0.9986          | 0.9793         | 61.11               | 73.33                |

## I Paired Comparisons Against GART 2.0

Aggregate margins are compared on the instances where both estimators return a finite prediction. We report the mean paired difference in absolute percent error, GART 2.0 minus the baseline, with a 95% paired bootstrap interval over 1,000 resamples of instances, and a Wilcoxon signed-rank  $p$ -value. Negative values favour GART 2.0. Table 19 reports the comparisons the body cites; the complete set is released with the replication materials.

Table 19: Paired difference in absolute percent error, GART 2.0 minus baseline.

| Benchmark     | Baseline                                | $N$    | Mean diff. (pp) [95% CI] | Wilcoxon $p$           |
|---------------|-----------------------------------------|--------|--------------------------|------------------------|
| TSPLIB EUC_2D | Asymptotic MST ratio                    | 78     | -1.00 [-1.77, -0.22]     | 0.0049                 |
| TSPLIB EUC_2D | $L_{\text{MST}} (\alpha = 1)$           | 78     | -8.84 [-9.99, -7.63]     | $1.1 \times 10^{-13}$  |
| 2D            | Asymptotic MST ratio                    | 2,580  | -6.31 [-6.68, -5.99]     | $2.5 \times 10^{-289}$ |
| 2D            | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2,580  | -2.87 [-3.07, -2.69]     | $1.3 \times 10^{-159}$ |
| 2D            | $L_{\text{MST}} (\alpha = 1)$           | 2,580  | -15.06 [-15.44, -14.71]  | $< 10^{-300}$          |
| ND            | Calibrated MST ratio $\hat{\rho}(d, n)$ | 16,846 | -1.20 [-1.23, -1.16]     | $< 10^{-300}$          |
| ND            | $L_{\text{MST}} (\alpha = 1)$           | 16,846 | -9.09 [-9.19, -8.98]     | $< 10^{-300}$          |
| 2D uniform    | BHH (sampling region)                   | 210    | -7.60 [-8.56, -6.71]     | $2.0 \times 10^{-35}$  |
| 2D uniform    | Çavdar-Sokol                            | 210    | -6.85 [-8.57, -5.30]     | $7.2 \times 10^{-27}$  |
| 2D uniform    | $L_{\text{MST}} (\alpha = 1)$           | 210    | -14.29 [-15.35, -13.38]  | $3.3 \times 10^{-36}$  |

## J Per-Dimension Detail for the Multidimensional Benchmark

Table 20 reports the multidimensional benchmark broken down by dimension  $d$ .

Table 20: ND benchmark by dimension; same 16,846 scored instances as Table 21.

| Bucket                     | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|----------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| $d = 2$<br>Count=647       | <b>GART 2.0</b>                         | <b>2.37</b> [2.13, 2.60] | 1.57     | 0.91        | 0.28     | 0.966        | 67.9      |
|                            | Calibrated MST ratio $\hat{\rho}(d, n)$ | 6.19 [5.67, 6.71]        | 4.05     | 2.06        | 0.53     | 0.745        | 38.6      |
|                            | $L_{\text{MST}} (\alpha = 1)$           | 9.78 [9.21, 10.35]       | 19.14    | 15.18       | -19.14   | -1.988       | 50.3      |
|                            | Custom Hilbert sort                     | 15.75 [14.80, 16.87]     | 29.74    | 35.22       | 29.74    | —            | 1.47      |
|                            | BHH (sampling region)                   | 30.92 [25.44, 37.70]     | 18.93    | 11.63       | 5.11     | —            | 0.094     |
|                            | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 102 ms    |
| $d = 3-5$<br>Count=1,944   | <b>GART 2.0</b>                         | <b>1.71</b> [1.60, 1.82] | 1.07     | 0.59        | 0.03     | 0.966        | 76        |
|                            | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.05 [3.81, 4.29]        | 2.53     | 1.29        | 0.13     | 0.793        | 41        |
|                            | $L_{\text{MST}} (\alpha = 1)$           | 7.67 [7.39, 7.93]        | 14.96    | 11.73       | -14.96   | -2.285       | 45        |
|                            | Custom Hilbert sort                     | 15.72 [15.24, 16.20]     | 35.70    | 40.14       | 35.70    | —            | 1.62      |
|                            | BHH (sampling region)                   | 24.14 [22.93, 25.27]     | 26.81    | 27.99       | -19.14   | —            | 0.107     |
|                            | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 109 ms    |
| $d = 6-10$<br>Count=3,238  | <b>GART 2.0</b>                         | <b>1.01</b> [0.96, 1.06] | 0.64     | 0.36        | -0.01    | 0.982        | 74.4      |
|                            | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.98 [2.84, 3.11]        | 1.79     | 0.80        | 0.16     | 0.830        | 28.7      |
|                            | $L_{\text{MST}} (\alpha = 1)$           | 6.46 [6.26, 6.64]        | 11.92    | 8.96        | -11.92   | -2.304       | 28.5      |
|                            | BHH (sampling region)                   | 10.28 [9.99, 10.59]      | 27.58    | 27.88       | -27.43   | —            | 0.117     |
|                            | Custom Hilbert sort                     | 13.82 [13.53, 14.11]     | 32.52    | 35.92       | 32.52    | —            | 10.7      |
|                            | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 122 ms    |
| $d = 15-25$<br>Count=1,942 | <b>GART 2.0</b>                         | <b>0.68</b> [0.63, 0.73] | 0.42     | 0.23        | -0.01    | 0.990        | 90.7      |
|                            | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.46 [2.31, 2.61]        | 1.40     | 0.51        | 0.07     | 0.855        | 30.8      |
|                            | $L_{\text{MST}} (\alpha = 1)$           | 5.98 [5.76, 6.20]        | 9.40     | 6.41        | -9.40    | -1.813       | 32.2      |
|                            | BHH (sampling region)                   | 6.17 [5.96, 6.41]        | 26.98    | 26.59       | -26.97   | —            | 0.151     |
|                            | Custom Hilbert sort                     | 11.71 [11.43, 11.98]     | 24.58    | 26.33       | 24.58    | —            | 40.6      |
|                            | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 123 ms    |

Table 20 continued from the previous page

| Bucket                       | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|------------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| $d = 30-50$<br>Count=3,228   | <b>GART 2.0</b>                         | <b>0.48</b> [0.45, 0.50] | 0.29     | 0.15        | -0.03    | 0.995        | 110       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.31 [2.20, 2.41]        | 1.29     | 0.40        | 0.03     | 0.866        | 32.8      |
|                              | BHH (sampling region)                   | 4.34 [4.22, 4.45]        | 28.17    | 27.55       | -28.17   | —            | 0.216     |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 5.94 [5.77, 6.10]        | 8.02     | 4.96        | -8.02    | -1.405       | 33.2      |
|                              | Custom Hilbert sort                     | 7.99 [7.84, 8.14]        | 17.07    | 18.12       | 17.07    | —            | 54.8      |
|                              | Reference-tour generation               | —                        | —        | —           | —        | —            | 136 ms    |
| $d = 100$<br>Count=5,847     | <b>GART 2.0</b>                         | <b>0.44</b> [0.43, 0.45] | 0.59     | 0.63        | 0.51     | 0.991        | 162       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.21 [2.14, 2.28]        | 1.78     | 1.07        | 1.03     | 0.856        | 35.4      |
|                              | BHH (sampling region)                   | 2.76 [2.71, 2.81]        | 29.92    | 29.39       | -29.92   | —            | 0.767     |
|                              | Custom Hilbert sort                     | 4.60 [4.54, 4.66]        | 10.35    | 11.03       | 10.35    | —            | 97.9      |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 5.93 [5.82, 6.06]        | 6.73     | 3.60        | -6.73    | -1.043       | 37.6      |
|                              | Reference-tour generation               | —                        | —        | —           | —        | —            | 120 ms    |
| <b>Total</b><br>Count=16,846 | <b>GART 2.0</b>                         | <b>0.99</b> [0.95, 1.02] | 0.62     | 0.42        | 0.18     | 0.984        | 121       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.94 [2.87, 3.02]        | 1.82     | 0.98        | 0.44     | 0.858        | 33.8      |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 7.26 [7.16, 7.36]        | 9.71     | 7.28        | -9.71    | -1.245       | 36.4      |
|                              | BHH (sampling region)                   | 13.57 [12.85, 14.40]     | 28.01    | 28.56       | -26.18   | —            | 0.248     |
|                              | Custom Hilbert sort                     | 14.40 [14.26, 14.53]     | 21.21    | 16.23       | 21.21    | —            | 51.5      |
|                              | Reference-tour generation               | —                        | —        | —           | —        | —            | 122 ms    |

## K Per-Size Detail for the Multidimensional Benchmark

Table 21 reports the multidimensional benchmark broken down by instance-size bucket.

Table 21: ND benchmark by instance-size bucket.

| Bucket                  | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|-------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| [5, 10]<br>Count=4,229  | <b>GART 2.0</b>                         | <b>1.56</b> [1.47, 1.64] | 0.99     | 0.61        | 0.08     | 0.960        | 53.2      |
|                         | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.42 [5.29, 5.55]        | 4.42     | 4.07        | 0.68     | 0.534        | 3.73      |
|                         | $L_{\text{MST}} (\alpha = 1)$           | 6.05 [5.86, 6.24]        | 19.78    | 18.68       | -19.78   | -5.633       | 3.96      |
|                         | Custom Hilbert sort                     | 7.49 [7.15, 7.82]        | 8.15     | 5.90        | 8.15     | —            | 0.672     |
|                         | BHH (sampling region)                   | 11.23 [10.28, 12.10]     | 33.49    | 33.68       | -32.59   | —            | 0.139     |
|                         | Reference-tour generation               | —                        | —        | —           | —        | —            | 1 ms      |
| [11, 50]<br>Count=2,820 | <b>GART 2.0</b>                         | <b>1.03</b> [0.97, 1.10] | 0.65     | 0.41        | 0.10     | 0.946        | 78.4      |
|                         | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.16 [2.05, 2.26]        | 1.66     | 1.37        | 0.39     | 0.771        | 17.7      |
|                         | $L_{\text{MST}} (\alpha = 1)$           | 4.14 [3.98, 4.32]        | 8.61     | 7.46        | -8.61    | -3.235       | 18.1      |
|                         | Custom Hilbert sort                     | 12.41 [11.94, 12.85]     | 19.59    | 15.81       | 19.59    | —            | 15.9      |
|                         | BHH (sampling region)                   | 14.43 [12.24, 16.94]     | 28.50    | 29.34       | -26.34   | —            | 0.152     |
|                         | Reference-tour generation               | —                        | —        | —           | —        | —            | 108 ms    |



Table 21 continued from the previous page

| Bucket                       | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|------------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| [51, 100]<br>Count=3,520     | <b>GART 2.0</b>                         | <b>0.69</b> [0.66, 0.72] | 0.53     | 0.42        | 0.22     | 0.964        | 108       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 1.21 [1.13, 1.30]        | 0.92     | 0.79        | 0.35     | 0.885        | 32.2      |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 3.50 [3.39, 3.61]        | 6.55     | 5.28        | -6.55    | -2.860       | 34.2      |
|                              | Custom Hilbert sort                     | 13.55 [13.18, 13.92]     | 24.13    | 20.19       | 24.13    | —            | 40        |
|                              | BHH (sampling region)                   | 15.69 [13.62, 18.66]     | 26.49    | 27.69       | -23.59   | —            | 0.184     |
|                              | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 76 ms     |
| [101, 200]<br>Count=705      | <b>GART 2.0</b>                         | <b>0.49</b> [0.45, 0.53] | 0.41     | 0.35        | 0.19     | 0.973        | 145       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 0.89 [0.79, 0.99]        | 0.73     | 0.68        | 0.35     | 0.906        | 56.9      |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 2.93 [2.77, 3.10]        | 5.57     | 4.45        | -5.57    | -3.115       | 57.8      |
|                              | BHH (sampling region)                   | 11.36 [9.62, 12.92]      | 25.24    | 27.21       | -23.65   | —            | 0.383     |
|                              | Custom Hilbert sort                     | 13.12 [12.62, 13.65]     | 26.84    | 23.82       | 26.84    | —            | 73        |
|                              | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 224 ms    |
| [201, 500]<br>Count=2,107    | <b>GART 2.0</b>                         | <b>0.46</b> [0.45, 0.47] | 0.41     | 0.29        | 0.27     | 0.969        | 194       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 0.74 [0.71, 0.78]        | 0.65     | 0.63        | 0.35     | 0.923        | 95.9      |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 2.76 [2.68, 2.84]        | 5.25     | 4.34        | -5.25    | -3.170       | 99.3      |
|                              | BHH (sampling region)                   | 11.70 [10.58, 12.73]     | 25.31    | 27.11       | -23.60   | —            | 1.27      |
|                              | Custom Hilbert sort                     | 12.63 [12.40, 12.84]     | 28.23    | 26.34       | 28.23    | —            | 144       |
|                              | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 886 ms    |
| [501, 1000]<br>Count=3,465   | <b>GART 2.0</b>                         | <b>0.45</b> [0.44, 0.45] | 0.40     | 0.24        | 0.28     | 0.966        | 460       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 0.69 [0.66, 0.71]        | 0.60     | 0.58        | 0.33     | 0.925        | 273       |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 2.61 [2.55, 2.67]        | 5.06     | 4.21        | -5.06    | -3.321       | 326       |
|                              | BHH (sampling region)                   | 11.83 [11.03, 12.69]     | 24.70    | 26.63       | -22.93   | —            | 109       |
|                              | Custom Hilbert sort                     | 12.70 [12.55, 12.87]     | 30.08    | 29.02       | 30.08    | —            | 795       |
|                              | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 3,799 ms  |
| <b>Total</b><br>Count=16,846 | <b>GART 2.0</b>                         | <b>0.99</b> [0.95, 1.02] | 0.62     | 0.42        | 0.18     | 0.984        | 121       |
|                              | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.94 [2.87, 3.02]        | 1.82     | 0.98        | 0.44     | 0.858        | 33.8      |
|                              | $L_{\text{MST}} (\alpha = 1)$           | 7.26 [7.16, 7.36]        | 9.71     | 7.28        | -9.71    | -1.245       | 36.4      |
|                              | BHH (sampling region)                   | 13.57 [12.85, 14.40]     | 28.01    | 28.56       | -26.18   | —            | 0.248     |
|                              | Custom Hilbert sort                     | 14.40 [14.26, 14.53]     | 21.21    | 16.23       | 21.21    | —            | 51.5      |
|                              | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 122 ms    |

## L Per-Size Detail for the 2D Diverse Benchmark

Table 22 reports the 2D diverse benchmark broken down by instance-size bucket.

Table 22: 2D diverse synthetic benchmark by instance-size bucket.

| Bucket                   | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|--------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| [5, 10]<br>Count=360     | <b>GART 2.0</b>                         | <b>4.28</b> [3.92, 4.67] | 3.21     | 2.54        | -0.00    | 0.907        | 58.7      |
|                          | GART 1.0                                | 6.93 [6.27, 7.52]        | 5.27     | 4.08        | 0.85     | 0.784        | 31.1      |
|                          | Custom Hilbert sort                     | 8.84 [8.11, 9.50]        | 8.34     | 5.77        | 8.34     | —            | 0.178     |
|                          | $L_{\text{MST}} (\alpha = 1)$           | 9.29 [8.63, 9.91]        | 33.91    | 34.08       | -33.91   | -6.433       | 10.3      |
|                          | Asymptotic MST ratio                    | 10.46 [9.73, 11.13]      | 25.64    | 25.83       | -25.63   | -3.806       | 10.5      |
|                          | Calibrated MST ratio $\hat{\rho}(d, n)$ | 14.22 [13.24, 15.16]     | 11.69    | 10.19       | 1.14     | -0.003       | 10.6      |
|                          | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 82 ms     |
|                          |                                         |                          |          |             |          |              |           |
| [11, 50]<br>Count=480    | <b>GART 2.0</b>                         | <b>5.83</b> [5.31, 6.35] | 4.07     | 2.66        | -0.81    | 0.708        | 82.6      |
|                          | GART 1.0                                | 6.42 [5.85, 6.95]        | 4.56     | 2.96        | 0.67     | 0.715        | 37.3      |
|                          | $L_{\text{MST}} (\alpha = 1)$           | 8.98 [8.25, 9.68]        | 20.70    | 19.65       | -20.70   | -2.731       | 14.6      |
|                          | Asymptotic MST ratio                    | 10.11 [9.27, 10.90]      | 11.99    | 9.59        | -10.76   | -0.833       | 13.5      |
|                          | Custom Hilbert sort                     | 10.37 [9.73, 11.07]      | 27.17    | 26.96       | 27.17    | —            | 0.346     |
|                          | Calibrated MST ratio $\hat{\rho}(d, n)$ | 11.17 [10.22, 12.09]     | 7.53     | 4.11        | -1.42    | -0.047       | 14.5      |
|                          | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 85 ms     |
|                          |                                         |                          |          |             |          |              |           |
| [51, 100]<br>Count=630   | <b>GART 2.0</b>                         | <b>5.58</b> [4.99, 6.14] | 3.41     | 1.86        | -0.36    | 0.636        | 89        |
|                          | GART 1.0                                | 7.38 [6.79, 8.01]        | 5.39     | 3.52        | 2.33     | 0.494        | 42.9      |
|                          | $L_{\text{MST}} (\alpha = 1)$           | 7.99 [7.24, 8.65]        | 16.93    | 15.69       | -16.93   | -2.370       | 20.7      |
|                          | Asymptotic MST ratio                    | 8.99 [8.14, 9.79]        | 8.09     | 5.98        | -6.53    | -0.425       | 19.5      |
|                          | Calibrated MST ratio $\hat{\rho}(d, n)$ | 9.46 [8.57, 10.30]       | 5.72     | 2.65        | -1.67    | -0.056       | 20.6      |
|                          | Custom Hilbert sort                     | 9.82 [9.11, 10.53]       | 34.21    | 34.06       | 34.21    | —            | 0.605     |
|                          | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 248 ms    |
|                          |                                         |                          |          |             |          |              |           |
| [101, 500]<br>Count=500  | <b>GART 2.0</b>                         | <b>4.25</b> [3.69, 4.75] | 2.39     | 1.09        | 0.18     | 0.593        | 177       |
|                          | GART 1.0                                | 5.81 [5.22, 6.42]        | 9.35     | 8.48        | 8.93     | -0.876       | 74.8      |
|                          | $L_{\text{MST}} (\alpha = 1)$           | 5.88 [5.19, 6.51]        | 13.08    | 12.52       | -13.08   | -3.061       | 37.5      |
|                          | Asymptotic MST ratio                    | 6.61 [5.79, 7.38]        | 4.09     | 2.01        | -2.19    | -0.121       | 34        |
|                          | Calibrated MST ratio $\hat{\rho}(d, n)$ | 6.69 [5.92, 7.44]        | 3.67     | 1.33        | -0.91    | -0.028       | 37.6      |
|                          | Custom Hilbert sort                     | 10.07 [8.92, 11.20]      | 38.03    | 37.89       | 38.03    | —            | 14.6      |
|                          | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 2,965 ms  |
|                          |                                         |                          |          |             |          |              |           |
| [501, 1000]<br>Count=610 | <b>GART 2.0</b>                         | <b>2.75</b> [2.46, 3.04] | 1.64     | 0.79        | 0.45     | 0.650        | 277       |
|                          | $L_{\text{MST}} (\alpha = 1)$           | 4.24 [3.82, 4.64]        | 11.43    | 11.46       | -11.43   | -5.552       | 61.7      |
|                          | Asymptotic MST ratio                    | 4.77 [4.31, 5.21]        | 2.64     | 1.04        | -0.33    | -0.013       | 54.7      |
|                          | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.78 [4.33, 5.21]        | 2.63     | 1.03        | -0.25    | -0.009       | 57.6      |
|                          | GART 1.0                                | 6.27 [5.67, 6.86]        | 10.96    | 9.84        | 10.96    | -5.600       | 108       |
|                          | Custom Hilbert sort                     | 11.07 [9.96, 12.17]      | 38.08    | 38.05       | 38.08    | —            | 50.9      |
|                          | <i>Reference-tour generation</i>        | —                        | —        | —           | —        | —            | 9,032 ms  |
|                          |                                         |                          |          |             |          |              |           |

Table 22 continued from the previous page

| Bucket               | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|----------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| Total<br>Count=2,580 | <b>GART 2.0</b>                         | <b>4.68</b> [4.42, 4.92] | 2.89     | 1.49        | -0.10    | 0.867        | 113       |
|                      | GART 1.0                                | 7.90 [7.60, 8.16]        | 7.30     | 6.27        | 5.13     | 0.637        | 50.1      |
|                      | Calibrated MST ratio $\hat{\rho}(d, n)$ | 9.38 [8.96, 9.78]        | 5.76     | 2.31        | -0.75    | 0.462        | 25        |
|                      | $L_{\text{MST}} (\alpha = 1)$           | 10.27 [9.91, 10.61]      | 17.95    | 14.61       | -17.95   | -1.621       | 24.9      |
|                      | Asymptotic MST ratio                    | 11.55 [11.15, 11.91]     | 9.20     | 5.34        | -7.68    | -0.378       | 22.6      |
|                      | Custom Hilbert sort                     | 14.19 [13.75, 14.60]     | 30.95    | 34.97       | 30.95    | —            | 0.726     |
|                      | Reference-tour generation               | —                        | —        | —           | —        | —            | 459 ms    |

## M Per-Size Detail for the TSPLIB95 EUC\_2D Benchmark

Table 23 reports the TSPLIB95 EUC\_2D benchmark broken down by instance-size bucket.

Table 23: TSPLIB95 EUC\_2D benchmark. Time is median wall clock.

| $n$ range                      | Estimator                               | SDPE (%) [95% CI]        | MAPE (%) | Median  (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|--------------------------------|-----------------------------------------|--------------------------|----------|-------------|----------|--------------|-----------|
| $n \in [51, 150]$<br>Count=23  | <b>GART 2.0</b>                         | <b>2.63</b> [1.34, 3.58] | 2.01     | 1.22        | 1.47     | 0.580        | 3.64      |
|                                | $L_{\text{MST}} (\alpha = 1)$           | 3.77 [2.51, 4.50]        | 13.43    | 12.67       | -13.43   | -9.539       | 1.68      |
|                                | Asymptotic MST ratio                    | 4.24 [2.82, 5.07]        | 3.55     | 2.13        | -2.58    | -0.395       | 1.16      |
|                                | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.68 [3.07, 5.68]        | 4.09     | 3.93        | 1.04     | -0.176       | 1.14      |
|                                | GART 1.0                                | 5.64 [3.58, 7.19]        | 5.44     | 3.92        | 4.07     | -1.307       | 4.05      |
|                                | Custom Hilbert sort                     | 11.46 [8.20, 13.91]      | 39.91    | 38.92       | 39.91    | —            | 0.656     |
|                                |                                         |                          |          |             |          |              |           |
| $n \in [151, 400]$<br>Count=16 | <b>GART 2.0</b>                         | <b>2.86</b> [1.95, 3.51] | 2.38     | 2.50        | 1.51     | 0.706        | 4.51      |
|                                | $L_{\text{MST}} (\alpha = 1)$           | 4.82 [2.13, 6.65]        | 12.38    | 11.56       | -12.38   | -4.801       | 2.33      |
|                                | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.37 [2.49, 7.39]        | 3.73     | 2.43        | 0.01     | 0.049        | 1.91      |
|                                | Asymptotic MST ratio                    | 5.42 [2.39, 7.48]        | 3.39     | 1.27        | -1.41    | -0.088       | 1.84      |
|                                | GART 1.0                                | 5.53 [3.93, 6.54]        | 5.92     | 4.46        | 4.93     | -0.497       | 5.03      |
|                                | Custom Hilbert sort                     | 7.35 [3.56, 10.39]       | 44.24    | 42.42       | 44.24    | —            | 1.44      |
|                                |                                         |                          |          |             |          |              |           |
| $n > 400$<br>Count=39          | <b>GART 2.0</b>                         | <b>3.05</b> [2.32, 3.62] | 2.90     | 1.98        | 2.68     | -0.071       | 20.5      |
|                                | $L_{\text{MST}} (\alpha = 1)$           | 3.58 [2.84, 4.17]        | 9.74     | 9.39        | -9.74    | -6.448       | 9.55      |
|                                | Calibrated MST ratio $\hat{\rho}(d, n)$ | 3.95 [3.13, 4.60]        | 3.54     | 3.24        | 1.75     | -0.122       | 9.52      |
|                                | Asymptotic MST ratio                    | 4.03 [3.20, 4.70]        | 3.57     | 3.15        | 1.57     | -0.132       | 9.27      |
|                                | GART 1.0                                | 7.64 [6.07, 8.92]        | 11.23    | 9.66        | 11.18    | -10.726      | 18.4      |
|                                | Custom Hilbert sort                     | 12.55 [9.60, 15.71]      | 48.66    | 47.03       | 48.66    | —            | 9.14      |
|                                |                                         |                          |          |             |          |              |           |
| Total<br>Count=78              | <b>GART 2.0</b>                         | <b>2.92</b> [2.40, 3.38] | 2.53     | 1.87        | 2.08     | 0.484        | 6.12      |
|                                | $L_{\text{MST}} (\alpha = 1)$           | 4.21 [3.30, 4.99]        | 11.37    | 11.56       | -11.37   | -5.591       | 3.08      |
|                                | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.48 [3.59, 5.32]        | 3.74     | 3.18        | 1.18     | 0.087        | 2.62      |
|                                | Asymptotic MST ratio                    | 4.74 [3.72, 5.62]        | 3.53     | 2.67        | -0.27    | -0.010       | 2.57      |
|                                | GART 1.0                                | 7.45 [6.24, 8.46]        | 8.43     | 6.51        | 7.80     | -3.579       | 6.12      |
|                                | Custom Hilbert sort                     | 11.85 [9.86, 13.59]      | 45.17    | 44.60       | 45.17    | —            | 2.64      |

## N Geometric and Centroid Feature Specifications

Table 24 lists the 11 geometric and centroid features that complement the MST-derived features in Tables 12, 13 and the constructive ratio of Table 14.

Table 24: Geometric and Centroid Feature Specifications (11 features).

| Feature Name                            | Description & Rationale                                                                                                         | Complexity     | Origin / Citation                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------|
| <b>Dimension (<math>d</math>)</b>       | The number of spatial coordinates defining each point.                                                                          | $O(1)$         | Standard                                          |
| <b>Number of Nodes (<math>n</math>)</b> | The total count of nodes in the instance.                                                                                       | $O(1)$         | Standard                                          |
| <b>Aspect Ratio</b>                     | Ratio of the largest dimension range to the smallest. Detects subspace distortion.                                              | $O(n \cdot d)$ | <b>Proposed</b>                                   |
| <b>Bounding Hypervolume</b>             | Product of coordinate ranges: $\prod(\max_d - \min_d)$ . Approximates the sampling-region measure $\mathcal{V}$ for BHH bounds. | $O(n \cdot d)$ | Adapted from Beardwood et al. [1959]              |
| <b>Log Bounding Hypervolume</b>         | Logarithm of hypervolume: $\sum \ln(\max_d - \min_d)$ . Log-space stable for $d \geq 100$ .                                     | $O(n \cdot d)$ | <b>Proposed</b>                                   |
| <b>Node Density</b>                     | Ratio of $n$ to Bounding Hypervolume. Proxy for point intensity $\delta = n/\mathcal{V}$ .                                      | $O(1)$         | <b>Proposed</b> ; cf. Beardwood et al. [1959]     |
| <b>Log Node Density</b>                 | Logarithm of node density: $\ln(n) - \sum \ln(\max_d - \min_d)$ . Log-space stable.                                             | $O(1)$         | <b>Proposed</b>                                   |
| <b>Centroid Dist (Mean, Std)</b>        | Average and deviation of distances to the centroid. Measures central tendency.                                                  | $O(n \cdot d)$ | Adapted from Çavdar and Sokol [2015]              |
| <b>Centroid Dist Max</b>                | Radius of the enclosing sphere.                                                                                                 | $O(n \cdot d)$ | Adapted from Çavdar and Sokol [2015]              |
| <b>Centroid Dist IQR</b>                | Interquartile range of centroid distances. Robust dispersion metric.                                                            | $O(n \cdot d)$ | <b>Proposed</b> ; extends Çavdar and Sokol [2015] |

## O Trained Hyperparameters and Booster Statistics

Table 25 reports the hyperparameters and the resulting booster statistics for the final GART 2.0 model. The hyperparameters are inherited, frozen and not tuned for this model; Section 3.4 reports the search we ran against them and rejected.

Table 25: GART 2.0 hyperparameters and trained-model statistics. The hyperparameter block is inherited and frozen, not tuned for this model; LightGBM uses squared-error regression on the logit-transformed target, early-stopped on validation cost-level MAPE.

| Group                                                 | Parameter                | Value                         |
|-------------------------------------------------------|--------------------------|-------------------------------|
| <i>Hyperparameters (inherited, frozen, not tuned)</i> |                          |                               |
| <i>Tree structure</i>                                 | num leaves               | 148                           |
|                                                       | min child samples        | 23                            |
|                                                       | feature fraction         | 0.548                         |
|                                                       | bagging fraction / freq  | 0.609 / 6                     |
| <i>Optim. &amp; reg.</i>                              | learning rate            | 0.0260                        |
|                                                       | L1 regularization        | $7.76 \times 10^{-2}$         |
|                                                       | L2 regularization        | $1.19 \times 10^{-5}$         |
| <i>Training configuration</i>                         |                          |                               |
| <i>Training setup</i>                                 | boosting type            | gbdt                          |
|                                                       | objective                | regression_l2 (squared error) |
|                                                       | early-stopping patience  | 100                           |
|                                                       | random seed              | 42                            |
| <i>Trained-boosters statistics</i>                    |                          |                               |
| <i>Booster</i>                                        | trees (after early stop) | 1,118                         |
|                                                       | avg leaves / tree        | 147.80                        |
|                                                       | max leaves / tree        | 148                           |
|                                                       | avg tree depth           | 11.83                         |
|                                                       | max tree depth           | 40                            |

Two other learners were fitted on the same feature pipeline before the gradient-boosted ensemble was frozen, and neither displaced it. A residual feed-forward network predicts the same  $\alpha$  from the same MST-and-geometry descriptors, and on the 2D diverse benchmark it costs 137.5 to 146.2 ms per instance against GART 2.0’s 103.3 to 147.8 ms, slower on five of the six generator classes of Table 17 and slower in aggregate. It does not buy accuracy with that time so much as move it: the network reads 2.98% MAPE overall against 2.89% and loses on five of the six classes, but it wins the near-collinear Line Noise class at 9.14% against 10.82% and finishes marginally ahead on aggregate dispersion, 4.61% SDPE against 4.68%. An ordinary least-squares fit is the cheapest of the three at 61.9 to 92.1 ms and pays for it in the tail, reaching 17.9% MAPE on the clustered class against 2.2%, eight times the shipped model’s error, at  $R_\alpha^2 = -9.53$ . Neither is a control on the shipped feature vector and we do not report them as one: the network was fitted on 30 columns and the linear model on 28 against the shipped block’s 31, and both were fitted on an earlier feature table whose reference costs and MST lengths were later revised. What they bound is the model-class question rather than settling it, and nothing in them gave a reason to trade the boosted ensemble away. Their per-instance predictions are released with every other row we score.

## P MDS Embedding for Non-Euclidean Instances

For TSPLIB ATT, GEO (geographic coordinates with great-circle distances), and EXPLICIT (distance matrix only) instances, we apply classical MDS to obtain an approximate positive-eigenvalue Euclidean representation. CEIL\_2D uses native coordinates. The embedding dimension  $\kappa$  is chosen by the rule of §6; this retained-mass rule is a dimension-selection heuristic, not a distance-preservation guarantee.

Given an  $n \times n$  distance matrix  $D$ , classical MDS computes the eigendecomposition of the doubly-centered matrix  $B = -\frac{1}{2}JD^{(2)}J$  where  $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$  and  $D^{(2)}$  is the element-wise squared distance matrix. The embedding coordinates are  $X = V_\kappa \Lambda_\kappa^{1/2}$  where  $V_\kappa$  and  $\Lambda_\kappa$  are the top  $\kappa$  eigenvectors and eigenvalues.

We embed into  $\kappa$  dimensions, selecting the smallest  $\kappa$  that retains  $\geq 99.9\%$  of positive eigenvalue mass (capped at  $\kappa = 100$ ). For GEO instances, the selected dimension ranges from 2 to 61; for screened EXPLICIT instances, from 10 to 100. The MST features (edge weights, degrees, diameter) are computed from the original distance matrix, while geometric features (centroid statistics, bounding volume, aspect ratio) use the approximate MDS coordinates. The dimension  $\kappa$  is passed to the model as the `dimension` feature.

We recomputed these diagnostics directly from the released TSPLIB files. Normalised distance stress is  $\sqrt{\sum_{i<j} (D_{ij} - \hat{D}_{ij})^2 / \sum_{i<j} D_{ij}^2}$ , where  $\hat{D}_{ij}$  is the Euclidean distance induced by the retained MDS coordinates; correlation is Pearson correlation between the same distance vectors. Table 26 reports medians and ranges. CEIL\_2D stress measures only integer-ceiling distortion on a fixed-seed sample of 100,000–1,000,000 pairs; other rows use all pairs. Five capped instances do not reach the 99.9% target: att532, pa561, si175, si535, and si1032.

Table 26: Embedding diagnostics on the 23-instance screened application set. Stress and distance correlation are median [minimum, maximum]. “Below target” counts embeddings retaining less than 99.9% of positive eigenvalue mass because of the  $\kappa = 100$  cap.

| Distance type     | $N$ | $k$ range | Normalised stress       | Distance correlation       | Below target |
|-------------------|-----|-----------|-------------------------|----------------------------|--------------|
| CEIL_2D           | 4   | 2         | 0.0000 [0.0000, 0.0000] | 1.00000 [1.00000, 1.00000] | 0            |
| ATT               | 2   | 6–100     | 0.0028 [0.0007, 0.0048] | 0.99999 [0.99998, 1.00000] | 1            |
| GEO               | 10  | 2–61      | 0.0207 [0.0005, 0.2363] | 0.99951 [0.96944, 1.00000] | 0            |
| EXPLICIT screened | 7   | 10–100    | 0.0198 [0.0041, 0.1208] | 0.99954 [0.99598, 0.99999] | 4            |

The screened application set excludes ten EXPLICIT matrices with structural triangle-inequality violations. Seven EXPLICIT matrices remain, including three with mild rounding-level violations. GART 2.0 declines one of them, si1032, and obtains 3.01% MAPE on the six it accepts. Distortion varies substantially within type (maximum stress 0.2363 for GEO and 0.1208 for screened EXPLICIT), so we do not attribute between-type accuracy differences to the distance label alone.